Embedded Printer
<u>BK-T680/BK-T6112</u>

Programming Manual
V1.7

Shandong New Beiyang Information Technology Co., Ltd.

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Revision History

Date	Version	Remarks	Compiled by
2011-01-28	V1.0	First release	Mr. Xiangang Yang; Mr. Xinping Wang
	V1.1	Update nesting print watermark command and the barcode related	Mr. Xinping Wang
	V1.2	1. Increase the barcode related; 2. Change the Prst control command related.	Mr. Xinping Wang
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2012-8-16	V1.5	Add description of bitmap download command	Mr. Xinping Wang
2012-9-10	V1.6	Add ESC c < m n	Mr. Xinping Wang
2012-11-1	V1.7	Change relevant description of status command.	Mr. Xinping Wang

Contents

1 General Information

1.1 Commands classification

The commands of this printer are classified as below according to their functions:

Print commands: used for printing and feeding paper;

Position commands: to control the print position;

Character commands: to set character's property;

Bitmap commands: to download bitmap and print, including NV and RAM bitmap;

Status commands: used for printer status query;

Barcode commands: used for setting barcode print and property;

2ST commands: to set 2ST printing and print.

Other commands: used for periphery control, micro-definition and initialization;

For the command instruction, please refer to the detailed function of relative commands.

1.2 Key terms

Real-time commands -- The commands that are acted on immediately upon being received by the printer regardless of the printer's busyness or error status.

Page mode -- Under this mode, the printer stores all data in a specified memory and thinks of this as a virtual page. The printer doesn't operate until the printer receives print command either FF or ESC FF.

Standard mode -- Standard mode, that is, line mode is the default mode of printer. The printer prints data and feeds paper only if the printer line buffer is full (data is enough for one print line) or receives the print command like LF.

Marked paper mode – The configuration of the printer is marked paper.

Presenter -- Presenter is the front mechanism of the KIOSK printer. It is used for quick paper-out and retracting the printout. It is abbreviated as PRST in this programming manual.

HRI characters – Human readable interpretation of the barcode.

NV memory– Non-volatile memory in which the data stored does not lose when the power is off.

RAM -- Random Access Memory

Motion unit -- The motion unit has vertical motion unit and horizontal motion unit, which specifies the least distance for vertical motion and horizontal motion.

ASB -- Auto status back

DPI -- Print dots per inch (one inch equals to about 25.4mm). It is used to identify the resolution of the printer. For example, 203DPI means printing 203 dots per inch.

Mini-breadth -- To ensure the PRESENTER to feed the printed tickets out of printer, mini-breadth is allowed before cutting paper, which should be the distance from PRESENTER to cutter.

Font 0 -- A standard character, the size for 203 DPI model Font A is 12 x 24; the size for 300 DPI is 18 x 34;

Font 1 -- Compressed character, the size for 203 DPI model Font B is 9 X 17; the size for 300 DPI is 12 X 24;

Asian character set -- Multibyte code character used in Asian countries, the size for 203DPI is 24 x 24;

the size for 300DPI is 36 x 36.

1.3 Command format explanation

[Function] Describe the function of the commands;

[Format] Explain the format of the commands, such as ASCII, Hex, Decimal and so on;

[Range] The value range of the parameters used in the command;

[Note] Explain the main features and the application notices of commands;

[Default] The initial value used after the printer initializes;

[Relative] Other commands related to the current commend;

[Example] Example used for current or relative commands.

All the command data in the programming Example use HEX. All normal font/characters are data. There is no explanation for the data of command such as 42 43 which is data. The font/character underlined and emphasized is a command such as **1B 40**. All the data inside parentheses after all commands in Example are used to explain the meaning of this command. The parentheses and the data inside it are not the command to be transmitted to the printer.

2 Command Description

2.1 Print command

LF

[Function] Print and feed one line.

[Format] ASCII LF
Hex 0A
Decimal 10

[Notes] Under standard mode, print the data in the print buffer area and feed one line. It only feeds one line when the current buffer area is blank.
Under page mode, the printer executes carriage return and moves the print position to the beginning of the next line.
This command sets the print position to the beginning of the print line.

[Relative] **ESC 2, ESC 3**

FF

[Function] Print and feed paper**.**

[Format] ASCII FF
Hex 0C
Decimal 12

[Description]Under page mode, print the data in the buffer area and return to standard mode.

[Notes]

- Not under Presenter mode 2, but under standard mode, this command has the same function as LF.
- The data in the buffer area is cleared after finishing printing.
- This command sets the print position to the beginning of the line.
- Under Presenter mode 2 and also under standard mode: If the printer under 1st printing mode has already fed paper or printed data, the printer will hold paper (paper will be cut firstly if paper is not cut at this time) when executing this command; if there is data in the buffer area when the printer is under 2st printing mode, the printer will hold paper (paper will be cut firstly if paper is not cut at this time) when executing this command.

[Relative] **ESC FF, ESC L, ESC S**

CR

[Function] Print and carriage return.

[Format] ASCII CR
Hex 0D
Decimal 13

[Description]When this command is enabled, this command functions the same as **LF**; when this command is disabled, this command is ignored.

[Notes] This command is disabled according to the printer configuration.

[Relative] **LF**

ESC FF

[Function] Print in page mode.

[Format]

ASCII	ESC	FF	
Hex	1B	0C	
Decimal	27	12	

[Description] In page mode, print all the data in the buffer area.

[Notes]

- This command is enabled only in page mode.
- After printing, the printer does not clear the data in the buffer area, and does not change the settings of **ESC T** and **ESC W** and the position of the character in the buffer area.
- Don't exit from page mode after finishing printing.

[Relative] **FF, ESC L, ESC S**

ESC J n

[Function] Print and feed paper n lines.

[Format]

ASCII	ESC	J	n
Hex	1B	4A	n
Decimal	27	74	n

[Range] 0 ≤n ≤ 255

[Description] Print the data in the buffer area and feed paper [n × vertical or horizontal motion unit] inches.

[Notes]

- After the printing is completed, this command sets the print position to the beginning of the line.
- The distance of the paper feeding set by this command is not affected by the settings of **ESC 2** or **ESC 3**.
- The horizontal and vertical motion units are set by **GS P**.
- Under standard mode, the printer uses the vertical motion unit (y).
- Under page mode, select vertical motion unit(y) or horizontal motion unit(x) according to the direction of the print area and the starting position of printing:

1) When the starting position is set to the upper left or lower right of the print area using **ESC T**, the vertical motion unit (y) is used.
2) When the starting position is set to the upper right or lower left of the print area using **ESC T**, the horizontal motion unit (x) is used.

- The maximum line spacing is 600mm (23.6 inches) for 300DPI printer and 900mm (35.4 inches) for 203DPI printer. If the setting value exceeds the maximum, it is converted to the maximum automatically.

[Relative] **GS P**

[Example] **<u>1B 40</u>** (initialize printer)

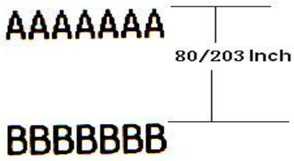
1D 50 CB CB (set the resolution 203×203)

41 41 41 41 41 41 41(data waiting for printing)

1B 4A 50 (print and feed paper 80/203 inches)

42 42 42 42 42 42 42 **0A** (data waiting for printing)

Results:



ESC d n

[Function] Print and feed n lines.

[Format]

ASCII	ESC	d	n
Hex	1B	64	n
Decimal	27	100	n

[Range] 0 ≤n ≤ 255

[Description]Print the data in the buffer area and feed n lines (character line).

[Notes]

- This command sets the print starting position to the beginning of the line.
- This command does not affect the line spacing set by **ESC 2** or **ESC 3**.
- The maximum line spacing is 600mm (23.6 inches) for 300DPI printer and 900mm (35.4 inches) for 203DPI printers. If the setting value exceeds the maximum, it is converted to the maximum automatically.

[Relative] **ESC 2, ESC 3**

2.2 Location command

HT

[Function] Horizontal tab.

[Format]

ASCII	HT
Hex	09
Decimal	9

[Description]Moves the print position to the next horizontal tab position.

[Notes]

- This command is ignored unless the next horizontal tab position has been set.
- If this command is received when the printing position is at the maximum printable width, the printer executes print buffer-full printing of the current line and moves the print position to the beginning of the next line.
- Horizontal tab position is set by **ESC D**.
- When the print buffer is full, the printer acts as follows,

1) Under standard mode, print the data in the current line and move the print position to the beginning of the next line.

2) Under page mode, execute carriage return and move the print position to the beginning of the next line.

The default setting of the horizontal tab position for the paper roll is every 8th ASCII character.

[Relative] **ESC D**

[Example] **0A** (set the print start position to the beginning of the line)

1B 40 (initialize printer)

1B 53(enter standard mode)

33 33 33 33 33 33

1B 44 08 10 1C 00(set the horizontal tab position)

09(move the print position to the next tab)

33 33 33 33

09(the same as above)

33 33 33 33

09(the same as above)

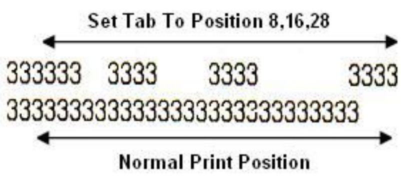
33 33 33 33

0A (print)

33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

0A (print)

Results:



ESC $ nL nH

[Function] Set absolute print position.

[Format]

ASCII	ESC	$	nL	nH
Hex	1B	24	nL	nH
Decimal	27	36	nL	nH

[Range] 0 ≤nL ≤ 255

0 ≤nH ≤ 255

[Description] Set the print position to the beginning of the line with a distance of [(nL + nH × 256) × (vertical or horizontal motion unit)] inches.

[Notes]

- If setting the position outside the specified printable area, this command is ignored.
- The horizontal and vertical motion units are specified by **GS P**.
- In standard mode, the horizontal motion unit (x) is used.
- In page mode, horizontal or vertical motion unit is selected according to the direction and the starting position of the printable area as follows:

1) When the starting position is set to the upper left or lower right of the printable area

using **ESC T**, the horizontal motion unit (x) is used.

2) When the starting position is set to the upper right or lower left of the printable area using **ESC T**, the vertical motion unit (y) is used.

[Relative] **ESC \, GS $, GS \, GS P**

ESC D n1...nk NUL

[Function] Set horizontal tab position.

[Format]

ASCII	ESC	D	n1...nk	NULL
Hex	1B	44	n1...nk	00
Decimal	27	68	n1...nk	0

[Range] 1 ≤ n ≤ 255

0 ≤ k ≤ 32

[Description] Set horizontal tab position.

- n+1 specifies the column number for setting a horizontal tab position from the beginning of the line.
- k indicates the total number of horizontal tab positions to be set.

[Notes]

- The horizontal tab position is stored as a value of [character width × n] measured from the beginning of the line. The character width includes the right-side character spacing, and double-width characters are set with twice the width of normal characters.
- This command cancels the previous horizontal tab settings.
- Up to 32 tab positions (k = 32) can be set. Data exceeding 32 tab positions is processed as normal data.
- Transmit [n] k in ascending order and place a NULL code 0 at the end.
- When [n] k is less than or equal to the preceding value [n] k-1, tab setting is finished and the following data is processed as normal data.
- **ESC D NULL** cancels all horizontal tab position settings.
- The previously specified horizontal tab positions do not change, even if the character width changes.
- Example: When n equals to 8, the print position is the ninth column.

[Default] The default tab positions are at intervals of 8 standard ASCII characters (columns 9, 17, 25).

[Relative] **HT**

ESC T n

[Function] Select print direction in page mode.

[Format]

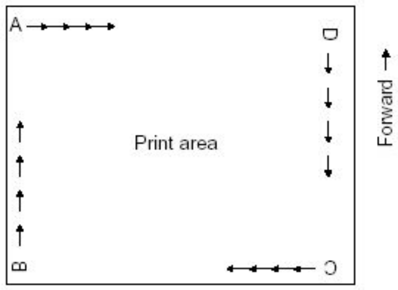
ASCII	ESC	T	n
Hex	1B	54	n
Decimal	27	84	n

[Range] 0 ≤ n ≤ 3

48 ≤n ≤ 51

[Description] Select the print direction and the starting position in page mode. n specifies the print direction and starting position as follows:

n	Print Direction	Starting Position
0, 48	Left to right	Upper left (A in the figure)
1, 49	Bottom to top	Lower left (B in the figure)
2, 50	Right to left	Lower right (C in the figure)
3, 51	Top to bottom	Upper right (D in the figure)



[Notes]

- When the command is input in standard mode, the printer executes only internal flag operation. This command does not affect printing in standard mode.
- This command sets the starting position of the data in the printing area.
- Horizontal or vertical motion unit (x or y) is used according to the starting position of the printing area:

1) If the starting position is in the upper left or lower right of the print area, data is arranged in the direction perpendicular to the paper feed direction:
   Commands using horizontal motion unit: **ESC SP, ESC $, ESC **
   Commands using vertical motion unit: **ESC 3, ESC J, GS $, GS **
2) If the starting position is in the upper right or lower left of the print area, data is arranged in the same direction as the paper feed direction:
   Commands using horizontal motion unit: **ESC 3, ESC J, GS $, GS **
   Commands using vertical motion unit: **ESC SP, ESC $, ESC **

[Default] n = 0

[Relative] **ESC $, ESC L, ESC W, ESC \, GS $, GS P, GS **

[Example] **1B 4C** (enter page mode)

1D 50 CB CB (set printer resolution)
1B 57 20 00 00 00 40 02 90 02 (set the print area in page mode)

1B 54 00(select the print area direction in page mode)

30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30 30
0A (newline)

1B 54 01(select the print area direction in page mode)

31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31
0A (newline)

1B 54 02(select the print area direction in page mode)

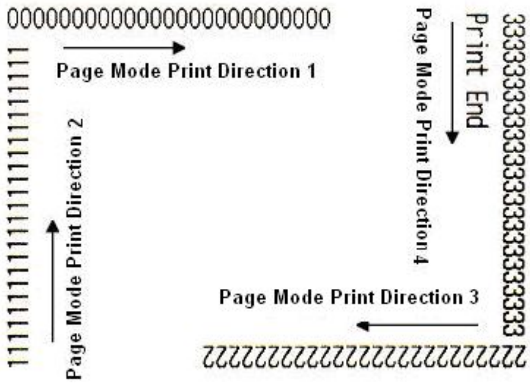
32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32
0A (newline)

1B 54 03 (select the print area direction in page mode)

33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33
0A(print)

50 72 69 6E 74 20 45 6E 64
0C(print)

Results:



ESC W xL xH yL yH dxL dxH dyL dyH

[Function] Set print area in page mode.

[Format]

	ASCII	ESC	W xL xH yL yH dxL dxH dyL dyH
	Hex	1B	57 xL xH yL yH dxL dxH dyL dyH
	Decimal	27	87 xL xH yL yH dxL dxH dyL dyH

[Range] 0 ≤ xL, xH, yL, yH, dxL, dxH, dyL, dyH ≤ 255 (except dxL= dxH=0 or dyL= dyH=0)

[Description] The horizontal starting position, vertical starting position, printing area width, and printing area height are defined as x0, y0, dx (inch), dy (inch), respectively. Each setting for the printing area is calculated as follows:

Horizontal starting position: x0 = [(xL + xH × 256) × (horizontal motion unit)] inches

Vertical starting position: y0 = [(yL + yH × 256) × (vertical motion unit)] inches

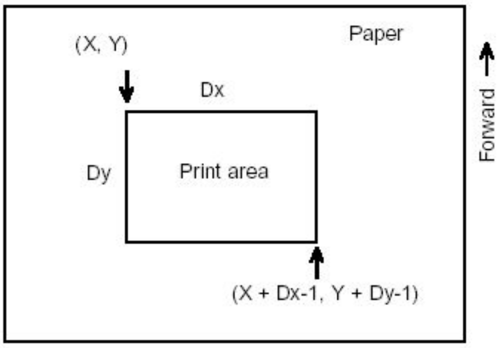
Printing area width: dx = [dxL + dxH × 256] × (horizontal motion unit)] inches

Printing area height: dy = [dyL + dyH × 256] × (vertical motion unit)] inches

[Notes]

- If this command is input in standard mode, the printer executes only internal flag operation. This command does not affect printing in standard mode.
- If the horizontal or vertical starting position is set outside the printable area, the printer stops command processing and processes the following data as normal data.
- If the printing area width or height is set to 0, the printer stops command processing and processes the following data as normal data.
- This command sets the position where data is buffered to the position specified by **ESC T** within the printing area.
- If (horizontal starting position + printing area width) exceeds the printable area, the printing area width is automatically set to (horizontal printable area- horizontal starting position).
- If (vertical starting position + printing area height) exceeds the printable area, the printing area height is automatically set to (vertical printable area – vertical starting position).

- The horizontal and vertical motion units are specified by **GS P**. Changing the horizontal or vertical motion unit does not affect the current printing area.
- Use the horizontal motion unit (x) for setting the horizontal starting position and printing area width, and use the vertical motion unit (y) for setting the vertical starting position and printing area height.
- When the horizontal starting position, vertical starting position, printing area width, and printing area height are defined as X, Y, dx, and dy respectively, the printing area is set as shown in the figure below.



- While using marked paper, set the print area properly to avoid wrong location caused by location error. Recommend setting the print area as whose height of the ticket is -4mm.

[Default] Decided by printer configuration.

[Relative] **CAN, ESC L, ESC T, GS P**

[Example] **0A**

1D 50 CB CB (set printer resolution 203×203)

1B 4C (enter page mode)

1B 57 20 00 00 00 40 01 90 01 (set print area in page mode)

1B 24 00 00(set absolute horizontal starting position to be starting point)

41

1B 24 32 00(set absolute horizontal starting position to be 50/203 inches)

42

1B 24 64 00(set absolute horizontal starting position to be 100/203 inches)

43

0A (newline)

1B 24 00 00(set absolute horizontal starting position to be starting point)

41

1B 5C 32 00(set absolute horizontal starting position to be 50/203 inches)

42

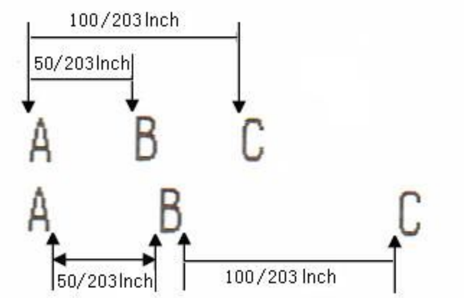
1B 5C 64 00(set relative horizontal starting position to be 100/203 inches)

43

0A (newline)

0C (print in page mode)

Results:



ESC \ nL nH

[Function] Set horizontal relative print position.

[Format]	ASCII	ESC	\	nL	nH
	Hex	1B	5C	nL	nH
	Decimal	27	92	nL	nH

[Range] 0 ≤ nL ≤ 255

0 ≤ nH ≤ 255

[Description] Sets the print starting position based on the current position by using the horizontal or vertical motion unit.

This command sets the distance from the current position to [(nL + nH × 256) ×horizontal or vertical motion unit] inches.

[Notes]

- Any setting that exceeds the printable area is ignored.
- When pitch N is specified to the right, nL+ nH × 256 = N.
- When pitch N is specified to the left (the negative direction): nL+ nH × 256 = 65536 - N.
- The print starting position moves from the current position to [N × horizontal or vertical motion unit]
- The horizontal and vertical motion units are specified by **GS P**.
- In standard mode, the horizontal motion unit is used.
- In page mode, the horizontal or vertical motion unit differs as follows, depending on the starting point of the printing area:

1) When the starting position is set to the upper left or lower right of the printable area using **ESC T**, the horizontal motion unit (x) is used.

2) When the starting position is set to the upper right or lower left of the printable area using **ESC T**, the vertical motion unit (y) is used.

[Relative] **ESC $, GS P**

ESC a n

[Function] Select character alignment mode.

[Format] ASCII ESC a n

Hex 1B 61 n

Decimal 27 97 n

[Range] 0 ≤ n ≤ 2, 48 ≤ n ≤ 50

[Description] Aligns all the data in one line to the specified position

n selects the justification as follows:

n	Justification
0,48	Left justification
1, 49	Centering
2, 50	Right justification

[Notes]

- The command is enabled only when processed at the beginning of the line in standard mode.
- If this command is input in page mode, the printer performs only internal flag operations.
- This command adjusts the blank area according to the **HT, ESC $** or **ESC** \ command.

[Default] n = 0

[Example] **0A**(Entering line mode)

1B 40(Initialization)

1B 61 00(Setting left justification)

41 42 43 **0A**

41 42 43 44 **0A**

41 42 43 44 45 **0A**

1B 61 01(Setting centering)

41 42 43 **0A**

41 42 43 44 **0A**

41 42 43 44 45 **0A**

1B 61 02(Setting right justification)

41 42 43 **0A**

41 42 43 44 **0A**

41 42 43 44 45 **0A**

Results:

Left justification	Centering	Right justification
ABC ABCD ABCDE	ABC ABCD ABCDE	ABC ABCD ABCDE

GS $ nL nH

[Function] Set absolute vertical print position in page mode.

[Format]

ASCII	GS	$	nL	nH
Hex	1D	24	nL	nH
Decimal	29	36	nL	nH

[Range] 0 ≤ nL ≤ 255, 0 ≤ nH ≤ 255

[Description] Set the absolute vertical print starting position for buffer character data in page mode.

This command sets the absolute print position to [(nL + nH × 256) × (vertical or horizontal motion unit)] inches.

[Notes]

- This command is effective only in page mode.
- If the [(nL+ nH × 256) × (vertical or horizontal motion unit)] exceeds the specified printing area, this command is ignored.
- The horizontal starting buffer position does not move after enabling the command.
- The reference starting position is specified by **ESC T**.
- This command operates as follows, depending on the starting position of the printing area specified by **ESC T**:

1) When the starting position is set to the upper left or lower right, this command sets the absolute position in the vertical direction.
2) When the starting position is set to the upper right or lower left, this command sets the absolute position in the horizontal direction.

- The horizontal and vertical motion units are specified by **GS P**.

[Relative] **ESC $, ESC T, ESC W, ESC \, GS P, GS **

GS L nL nH

[Function] Set left margin.

[Format]

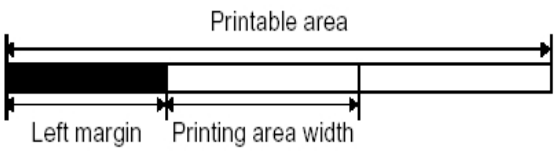
ASCII	GS	L	nL	nH
Hex	1D	4C	nL	nH
Decimal	29	76	nL	nH

[Range] 0 ≤ nL ≤ 255

0 ≤ nH ≤ 255

[Description] Sets the left margin using nL and nH.

The left margin is set to [(nL + nH × 256) × horizontal motion unit)] inches.



[Notes]

- This command is effective only processed at the beginning of the line in standard mode.
- If this command is input in page mode, the printer performs only internal flag

operations.

- This command does not affect printing in page mode.
- If the setting exceeds the printable area, this command is ignored.
- The horizontal and vertical motion units are specified by **GS P.** Changing the horizontal and vertical motion unit does not affect the current left margin.

[Default] nL = 0, nH = 0

[Relative] **GS P, GS W**

[Example] **0A** (Set printing position at the beginning of the line)

1B 40 (Initialization)

30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 **0A**

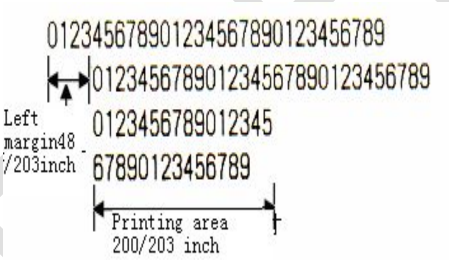
1D 4C 30 00 (Set left margin to 48/203 inch)

30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 **0A**

1D 57 C8 00 (Sets printing width to 200/203 inch)

30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 36 37 38 39 **0A**

Results:



GS P xL xH yL yH

[Function] Set horizontal and vertical motion units.

[Format]	ASCII	GS	P	xL	xH	yL	yH
	Hex	1D	50	xL	xH	yL	yH
	Decimal	29	80	xL	xH	yL	yH

[Range] x=xL+xHx256,y=yL+yHx256

0 ≤ x ≤ 360 ,0 ≤ y ≤ 360

0 ≤ y ≤360

[Description] Set the horizontal and vertical motion units to approximately 25.4/ x mm {1/ x inches} and approximately 25.4/ y mm {1/ y inches} respectively. When x and y are set to 0, the default setting of each value is used.

[Notes]

- The horizontal direction is perpendicular to the paper feed direction and the vertical direction is the paper feed direction.
- In standard mode, the following commands use x or y, regardless of character rotation

(upside-down or 90° clockwise rotation):

1) Commands using x: **ESC SP, ESC $, ESC \, FS S, GS L, GS W**
2) Commands using y: **ESC 3, ESC J, GS V**

- In page mode, the following command use x or y, depending on character orientation:

1) When the print starting position is set to the upper left or lower right of the printing area using **ESC T** (data is buffered in the direction perpendicular to the paper feed direction):
   Commands using x: **ESC SP, ESC $, ESC W, ESC \, FS S**
   Commands using y: **ESC 3, ESC J, ESC W, GS $, GS \, GS V**
2) When the print starting position is set to the upper right or lower left of the printing area using **ESC T** (data is buffered in the paper feed direction):
   Commands using x: **ESC 3, ESC J, ESC W, GS $, GS **
   Commands using y: **ESC SP, ESC $, ESC W, ESC \,FS S, GS V.**

- If the related commands are not executed after changing motion unit, relative settings shall not be changed.
- The minimum motion unit is a combined result of this command and other commands.

[Default] 300 DPI model: x = 305, y = 305; 203 DPI model: x = 203, y = 203

[Relative] **ESC SP, ESC $, ESC 3, ESC J, ESC W, ESC \, GS $, GS L, GS V, GS W, GS **

GS W nL nH

[Function] Set printing area width.

[Format]

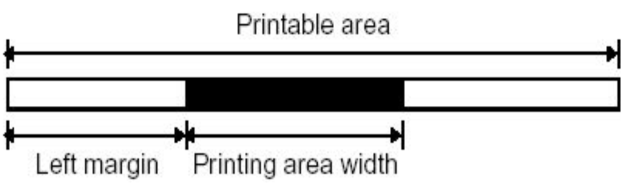
ASCII	GS	W	nL	nH
Hex	1D	57	nL	nH
Decimal	29	87	nL	nH

[Range] $0 \le nL \le 255$

$0 \le nH \le 255$

[Description] Sets the printing area width to the area specified by nL and nH.

The printing area width is set to [(nL + nH × 256) × horizontal motion unit]] inches.



[Notes]

- This command is effective only processed at the beginning of the line in page mode.
- This command does not affect printing in page mode.
- If the [left margin + printing area width] exceeds the printable area, [printable area width - left margin) is used.
- The horizontal and vertical motion units are specified by **GS P**. Changing the horizontal and vertical motion units does not affect the current left margin and print area width.
- The horizontal motion unit (x) is used for calculating the printing area width.

[Relative] **GS L, GS P**

GS \ nL nH

[Function] Set relative vertical print position in page mode.

[Format]

ASCII	GS	\	nL	nH
Hex	1D	5C	nL	nH
Decimal	29	92	nL	nH

[Range] $0 \leq nL \leq 255$

$0 \leq nH \leq 255$

[Description] Set the absolute vertical print starting position from the current position in page mode.

This command sets the distance from the current position to [(nL + nH × 256) × vertical or horizontal motion unit] inches.

[Notes]

- This command is ignored unless page mode is selected.
- When pitch N is specified to the movement downward:

  nL + nH × 256 = N

  When pitch N is specified to the movement upward (the negative direction).

  nL + nH × 256 = 65536 – N
- Any setting that exceeds the specified printing area is ignored.
- This command function as follows specify motion unit, depending on the print starting position set by **ESC T**:

1) When the starting position is set to the upper left or lower right of the printing, the vertical motion unit (y) is used.
2) When the starting position is set to the upper right or lower left of the printing area, the horizontal motion unit (x) is used.

- The horizontal and vertical motion units are specified by **GS P**.

[Relative] **ESC $, ESC T, ESC W, ESC \, GS $, GS P**

GS T n

[Function] Set print position to the beginning of print line.

[Format]

ASCII	GS	T	n
Hex	1D	54	n
Decimal	29	84	n

[Range] n = 0, 1, 48, 49

[Description] Set the print position to the beginning of print line.

• n specifies the data processing in the print buffer:

n	Printing position
0, 48	Set the print position to the beginning of print line after deleting all data in the print buffer.
1, 49	Set the print position to the beginning of print line after printing all data in the print buffer.

2.3 Character command

CAN

[Function] Cancel print data in page mode.

[Format] ASCII CAN

Hex 18

Decimal 24

[Description] In page mode, deletes all print data in the print buffer.

[Notes]

- This command is enabled only in page mode.
- If data that existed in the previously specified printing area also exists in the currently specified printing area, it is deleted.

[Relative] ESC L, ESC W

[Example] **1B 40** (Initialization)

1D 50 CB CB (Setting resolution 203×203)

1B 4C (Entering page mode)

1B 57 00 00 00 00 20 02 E8 00 (Setting printing width and height in page mode)

31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 7A 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79 31 32 33 34 35 36 37 38 39 30 61 62 63 64 65 64 66 67 68 69 6A 6B 6C 6D 6E 6F 70 71 72 73 74 75 76 77 78 79

1B 57 44 00 10 00 7C 01 AA 00(Setting the size of page needing to be deleted)

18(Delete data in page buffer)

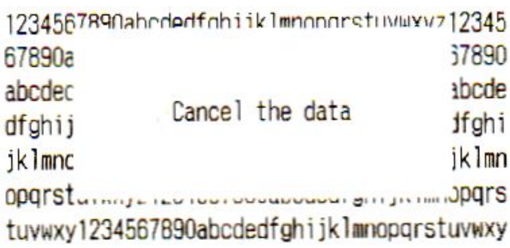
1B 24 64 00(Setting absolute horizontal print position as 100 dots)

1D 24 60 00 (Setting absolute vertical print position as 96 dots)

43 61 6E 63 65 6C 20 74 68 65 20 64 61 74 61 20

0A 0C (Printing)

Results:



ESC SP n

[Function] Set character right-side spacing.

[Format]	ASCII	ESC	SP	n
	Hex	1B	20	n
	Decimal	27	32	n

[Range] 0 ≤ n≤255

[Description] Set the character spacing for the right side of the character to [n × horizontal or vertical motion units] inches.

[Notes]

- The horizontal and vertical motion units are specified by **GS P**.
- Without setting of right character spacing, only changing the horizontal or vertical motion unit does not affect the current right-side spacing.
- The right-side character spacing for double-width mode is twice the normal value. When characters are enlarged, the right-side character spacing is enlarged accordingly.
- This command sets values independently in each mode (standard and page modes).
- In standard mode, the horizontal motion unit is used.
- In page mode, the horizontal or vertical motion unit is specified depending on starting position and direction of the printable area as follows:

1) When the starting position is set to the upper left or lower right of the printable area using **ESC T**, the horizontal motion unit (x) is used.
2) When the starting position is set to the upper right or lower left of the printable area using **ESC T**, the vertical motion unit (y) is used.

- For 300DPI printers, the maximum right-side spacing is 288/305 inches. Any setting exceeding the maximum is converted to the maximum automatically.
- For 203DPI printers, the maximum right-side spacing is 288/203 inches. Any setting exceeding the maximum is converted to the maximum automatically.

[Default] n = 0

[Relative] **GS P**

[Example] **1B 40**

1B 20 00(Set right-side character spacing as 0)

41 41 41 41 41 **0A**

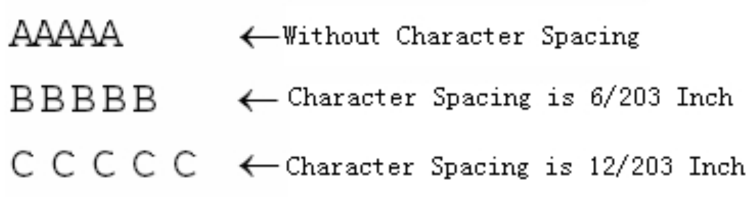
1B 20 06(Set character spacing as 6/203 inch)

42 42 42 42 42 **0A**

1B 20 0C(Set character spacing as 12/203 inch)

43 43 43 43 43 **0A**

Results:



ESC ! n

[Function] Select character print mode(s)

[Format] ASCII ESC ! n

Hex 1B 21 n

Decimal 27 33 n

[Range] 0 ≤ n ≤ 255

[Description] Select character print mode(s) using n as follows:

Bit	1/0	HEX	Decimal	Function
0	0	00	0	Standard ASCII Font (12 × 24)
	1	01	1	Compress ASCII Font (9 × 17)
1,2				Undefined
3	0	00	0	Emphasized mode not selected
	1	08	8	Emphasized mode selected
4	0	00	0	Double-height mode not selected
	1	10	16	Double-height mode selected
5	0	00	0	Double-width mode not selected
	1	20	32	Double-width mode selected
6				Undefined
7	0	00	0	Underline mode not selected
	1	80	128	Underline mode selected

[Notes]

- When both double-height and double-width modes are selected, quadruple size characters are printed.
- The printer can underline all characters, but can not underline the space set by **HT** or 90° clockwise rotated characters.
- The thickness of the underline is that selected by **ESC -**, regardless of the character size.
- When some characters in a line are double or more height, all the characters on the line are aligned at the baseline.
- **ESC E** can also turn on or off emphasized mode. However, the setting of the last received command is effective.
- **ESC –** can also turn on or off underline mode. However, the setting of the last received command is effective.
- **GS !** can also select character size. However, the setting of the last received command is effective.

[Default] n = 0

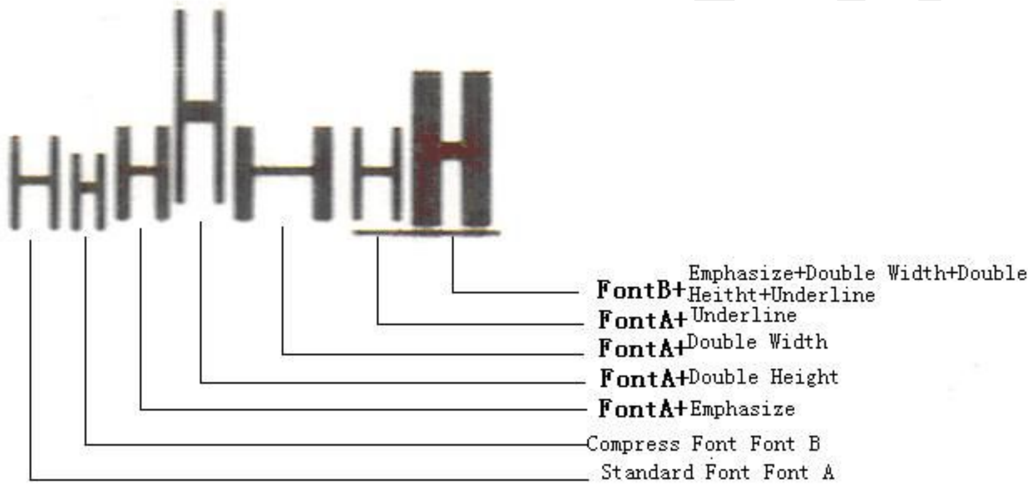
[Relative] **ESC -, ESC E, GS !**

[Example] 1B 40(Initialization)

1B 21 00(Select normal print mode)

48

1B 21 01(Select compress font mode)
48
1B 21 08(Select emphasized mode)
48
1B 21 10(Select double-height mode)
48
1B 21 20(Select double-width mode)
48
1B 21 80(Select underline mode)
48
1B 21 B9 (Select compress, emphasized, double-width, double-height and underline mode)
48 **0A**
Results:



ESC % n

[Function] Select/cancel user-defined character set.

[Format]

ASCII	ESC	%	n
Hex	1B	25	n
Decimal	27	37	n

[Range] 0 ≤ n ≤255

[Description] Select or cancel the user-defined character set.

When the LSB of n is 0, the user-defined character set is canceled.

When the LSB of n is 1, the user-defined character set is selected.

[Notes]

- When the user-defined character set is canceled, the internal character set is automatically selected.
- n is available only for the least significant bit.

[Default] n = 0

[Relative] **ESC &, ESC ?**

ESC & y c1 c2 [x1 d1...d(y × x1)]...[xk d1...d(y × xk)]

[Function] Define user-defined characters.

[Format]

ASCII	ESC	&	y c1 c2 [x1 d1...d(y × x1)]...[xk d1...d(y × xk)]
Hex	1B	26	y c1 c2 [x1 d1...d(y × x1)]...[xk d1...d(y × xk)]
Decimal	27	38	y c1 c2 [x1 d1...d(y × x1)]...[xk d1...d(y × xk)]

[Range] 203DPI model:

$y = 3$

$32 \leq c1 \leq c2 \leq 127$

$0 \leq x \leq 12$ Standard ASCII font

$0 \leq x \leq 9$ Compressed ASCII font

300DPI model:

$y = 5$ Standard ASCII font

$y = 3$ Compressed font

$32 \leq c1 \leq c2 \leq 127$

$0 \leq x \leq 18$ Standard ASCII font

$0 \leq x \leq 13$ Compressed ASCII font

$0 \leq d1 \ldots d(y \times xk) \leq 255$

[Description] Defines user-defined characters.

- y specifies the number of bytes in the vertical direction.
- c1 specifies the beginning character code for the definition, and c2 specifies the final code.
- x specifies the number of dots in the horizontal direction.

[Notes]

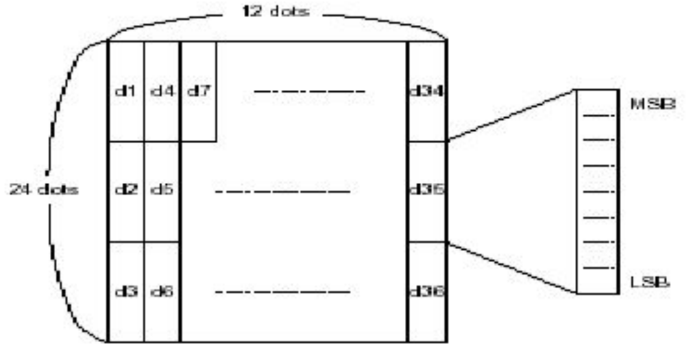
- The allowable character code range is from ASCII code <20>H to <7F>H (96 characters).
- It is possible to define multiple characters for consecutive character codes. If only one character is desired, use c1 = c2.
- d is the data for the downloaded characters. The dot pattern is in the horizontal direction from the left side.
- The data to define a user-defined character is (y × x) bytes.
- Set a corresponding bit to 1 to print a dot or 0 to not print a dot.
- The user-defined character definition is cleared when:

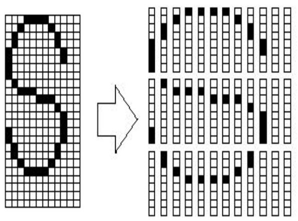
1) ESC is executed.
2) The power is turned off.

All user-defined Characters defined with this command will be cleared when the GS * command is executed.

[Example] 203 DPI model

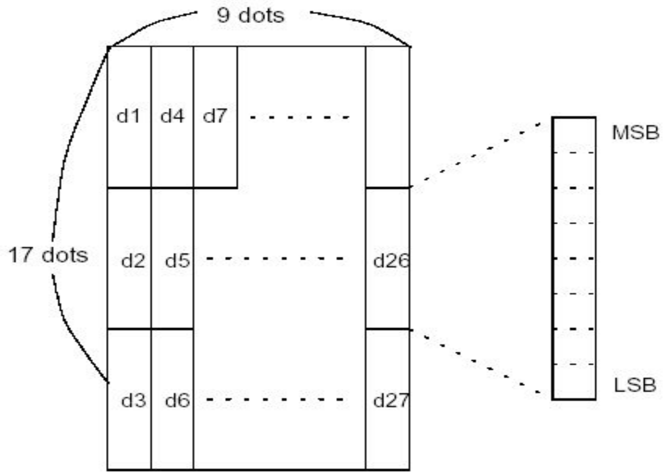
When standard ASCII font (12 × 24) is selected

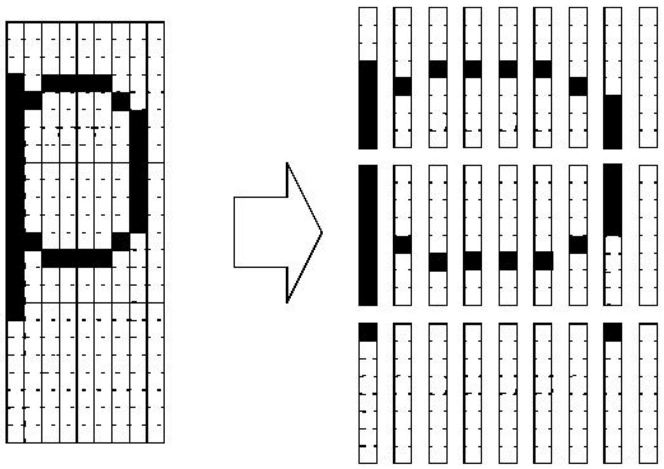




d1= <0F>H d4 = <30>H d7 = <40>H . . . .
d2 = <03>H d5 = <80>H d8 = <40>H . . . .
d3 = <00>H d6 = <00>H d9 = <20>H . . . .

When compressed ASCII font (9×17) is selected





d1 = <1F>H d4 = <08>H d7 = <10>H...
d2 = <FF>H d5 = <08>H d8 = <04>H...
d3 = <80>H d6 = <00>H d9 = <00>H...

[Relative] **ESC %, ESC**

ESC – n

[Function] Turn underline mode on/off

[Format] ASCII ESC - n
Hex 1B 2D n
Decimal 27 45 n

[Range] 0 ≤ n ≤2, 48 ≤n ≤ 50

[Description] Turns underline mode on or off, based on the following values of n:

n	Function
0, 48	Turns off underline mode
1, 49	Turns on underline mode (1-dot thick)
2, 50	Turns on underline mode (2-dots thick)

[Notes]

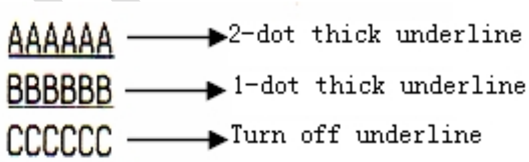
- The printer can underline all characters (including right-side character spacing), but cannot underline the space set by **HT**.
- The printer cannot underline 90° clockwise rotated characters and white/black inverted characters.
- Changing the character size does not affect the current underline thickness.
- Underline mode can also be turned on or off by using **ESC !**. Note, however, that the last received command is effective.

[Default] n = 0

[Relative] **HT, ESC !**

[Example] **1B 40**
1B 2D 02(2-dot thick underline)
41 41 41 41 41 41 **0A**
1B 2D 01(1-dot thick underline)
42 42 42 42 42 42 **0A**
1B 2D 00(Turn off underline)
43 43 43 43 43 43 **0A**

Results:



ESC ? n

[Function] Cancel user-defined characters.

[Format] ASCII ESC ? n
Hex 1B 3F n
Decimal 27 63 n

[Range] 32 ≤n ≤127

[Description]Cancel user-defined characters.

[Notes]

- This command cancels the pattern defined for the character code specified by n. After the user-defined characters are canceled, the corresponding pattern for the internal character is printed.
- If a user-defined character has not been defined for the specified character code, the

printer ignores this command.

[Relative] **ESC &, ESC %**

ESC E n

[Function] Turn emphasized mode on/off.

[Format]

ASCII	ESC	E	n
Hex	1B	45	n
Decimal	27	69	n

[Range] 0 ≤ n ≤ 255

[Description] Turn emphasized mode on or off.

- When the LSB of n is 0, emphasized mode is turned off.
- When the LSB of n is 1, emphasized mode is turned on.

[Notes]

- Only the least significant bit of n is effective.
- Printer output is the same in double-strike mode and in emphasized mode.
- **ESC !** turn on and off emphasized mode in the same way, the last received command is available.

[Default] n = 0

[Relative] **ESC ! ESC G**

[Example] 1B 40

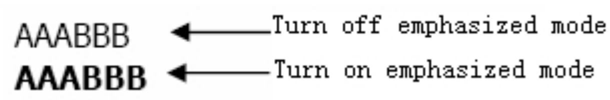
1B 45 01(Emphasized mode is selected)

41 41 41 41 41 **0A**

1B 45 00(Emphasized mode is not selected)

42 42 42 42 42 **0A**

Results:



ESC G n

[Function] Turn on/off double-strike mode.

[Format]

ASCII	ESC	G	n
Hex	1B	47	n
Decimal	27	71	n

[Range] 0 ≤ n ≤ 255

[Description] Turns double-strike mode on or off.

- When the LSB of n is 0, double-strike mode is turned off.
- When the LSB of n is 1, double-strike mode is turned on.

[Notes]

- Only the least significant bit of n is effective.
- Printer output is the same in double-strike mode and in emphasized mode.

[Default] n = 0

[Relative] **ESC E, ESC !**

[Example] See **ESC E**.

ESC M n

[Function] Select character font.

[Format] ASCII ESC M n

Hex 1B 4D n

Decimal 27 77 n

[Range] n = 0, 1, 2,3,48, 49,50,51

[Description] Select character fonts.

n	Function
0,48	Select standard ASCII font
1,49	Select compressed ASCII font
2,50	Select user defined character
3,51	Select Asian font, include: 1) Simplified Chinese(GB2312) 2) Traditional Chinese(BIG5) 3) Japanese (JIS/SJIS/EUC) 4) Korean (KSC5601/EUC) The printer will print selected Chinese font [simplified Chinese, traditional Chinese, Japanese (EUC) or Korean (EUC)] according to its configuration.

[Example] 1B 40

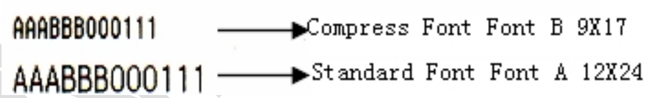
1B 4D 01(Compressed font selected)

41 41 41 42 42 42 30 30 30 31 31 31 **0A**

1B 4D 00(Standard font selected)

41 41 41 42 42 42 30 30 30 31 31 31 **0A**

Results:



ESC R n

[Function] Select international character set.

[Format] ASCII ESC R n

Hex 1B 52 n

Decimal 27 82 n

[Range] 0 ≤ n ≤ 13

[Description]Select an international character set n from the following table,see below:

<table>
<tr><th rowspan="2">n</th><th rowspan="2">Country</th><th colspan="12">ASCII Code (Hex)</th></tr>
<tr><th>23</th><th>24</th><th>40</th><th>5B</th><th>5C</th><th>5D</th><th>5E</th><th>60</th><th>7B</th><th>7C</th><th>7D</th><th>7E</th></tr>
<tr><td>0</td><td>U.S.A</td><td>#</td><td>$</td><td>@</td><td>[</td><td>\</td><td>]</td><td>^</td><td>`</td><td>{</td><td>|</td><td>}</td><td>~</td></tr>
<tr><td>1</td><td>France</td><td>#</td><td>$</td><td>à</td><td>°</td><td>ç</td><td>§</td><td>^</td><td>`</td><td>é</td><td>ù</td><td>è</td><td>¨</td></tr>
<tr><td>2</td><td>Germany</td><td>#</td><td>$</td><td>§</td><td>Ä</td><td>Ö</td><td>Ü</td><td>^</td><td>`</td><td>ä</td><td>ö</td><td>ü</td><td>ß</td></tr>
<tr><td>3</td><td>U.K.</td><td>£</td><td>$</td><td>@</td><td>[</td><td>\</td><td>]</td><td>^</td><td>`</td><td>{</td><td>|</td><td>}</td><td>~</td></tr>
<tr><td>4</td><td>Denmark I</td><td>#</td><td>$</td><td>@</td><td>Æ</td><td>Ø</td><td>Å</td><td>^</td><td>`</td><td>æ</td><td>ø</td><td>å</td><td>~</td></tr>
<tr><td>5</td><td>Sweden</td><td>#</td><td>¤</td><td>É</td><td>Ä</td><td>Ö</td><td>Å</td><td>Ü</td><td>é</td><td>ä</td><td>ö</td><td>å</td><td>ü</td></tr>
<tr><td>6</td><td>Italy</td><td>#</td><td>$</td><td>@</td><td>°</td><td>\</td><td>é</td><td>^</td><td>ù</td><td>à</td><td>ò</td><td>è</td><td>ì</td></tr>
<tr><td>7</td><td>Spain I</td><td>Pt</td><td>$</td><td>@</td><td>¡</td><td>Ñ</td><td>¿</td><td>^</td><td>`</td><td>¨</td><td>ñ</td><td>}</td><td>~</td></tr>
<tr><td>8</td><td>Japan</td><td>#</td><td>$</td><td>@</td><td>[</td><td>¥</td><td>]</td><td>^</td><td>`</td><td>{</td><td>|</td><td>}</td><td>~</td></tr>
<tr><td>9</td><td>Norway</td><td>#</td><td>¤</td><td>É</td><td>Æ</td><td>Ø</td><td>Å</td><td>Ü</td><td>é</td><td>æ</td><td>ø</td><td>å</td><td>ü</td></tr>
<tr><td>10</td><td>Denmark II</td><td>#</td><td>$</td><td>É</td><td>Æ</td><td>Ø</td><td>Å</td><td>Ü</td><td>é</td><td>æ</td><td>ø</td><td>å</td><td>ü</td></tr>
<tr><td>11</td><td>Spain II</td><td>#</td><td>$</td><td>á</td><td>¡</td><td>Ñ</td><td>¿</td><td>é</td><td>`</td><td>í</td><td>ñ</td><td>ó</td><td>ú</td></tr>
<tr><td>12</td><td>Latin America</td><td>#</td><td>$</td><td>á</td><td>¡</td><td>Ñ</td><td>¿</td><td>é</td><td>ü</td><td>í</td><td>ñ</td><td>ó</td><td>ú</td></tr>
<tr><td>13</td><td>Korea</td><td>#</td><td>$</td><td>@</td><td>[</td><td>W</td><td>]</td><td>^</td><td>`</td><td>{</td><td>|</td><td>}</td><td>~</td></tr>
</table>

[Default] n = 0

ESC V n

[Function] Turn 90° clockwise rotation mode on/off.

[Format] ASCII ESC V n

Hex 1B 56 n

Decimal 27 86 n

[Range] 0 ≤ n ≤ 1, 48 ≤ n ≤ 49

[Description] Turn 90° clockwise rotation mode on/off. n is used as follows:

n	Function
0, 48	Turns off 90° clockwise rotation mode
1, 49	Turns on 90° clockwise rotation mode

[Notes]

- This command only affects printing in standard mode.
- When underline mode is turned on, the printer does not underline 90° clockwise rotation.
- Double-width and double-height commands in 90° rotation mode enlarge characters in the opposite directions from double-height and double- width commands in normal mode.

[Default] n = 0

[Relative] **ESC !, ESC -**

[Example] 1B 40

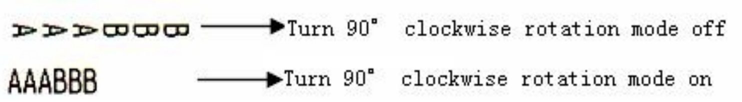
1B 56 01(Turn 90° clockwise rotation mode on)

41 41 41 42 42 42 **0A**

1B 56 00(Turn 90° clockwise rotation mode off)

41 41 41 42 42 42 **0A**

Results:



ESC t n

[Function] Select codepage.

[Format] ASCII ESC t n

Hex 1B 74 n

Decimal 27 116 n

[Range] 0 ≤ n ≤ 5, 16 ≤ n ≤ 19,

[Description] Select a page n from the codepage table:

n	Page
0	PC437
1	Katakana
2	PC850
3	PC860
4	PC863
5	PC865
16	WPC1252
17	PC866
18	PC852
19	PC858

[Notes]

- If printer font is Chinese character font, use **ESC M n** command to select English font first, then select page code with this command.
- Details see appendix 2 (Character code table)

[Default] n = 0

[Relative] **ESC M n, FS**

ESC { n

[Function] Turn on/off upside-down printing mode.

[Format] ASCII ESC { n

Hex 1B 7B n

Decimal 27 123 n

[Range] 0 ≤ n ≤ 255

[Description] Turns upside-down printing mode on or off.

- When the LSB of n is 0, upside-down printing mode is turned off.
- When the LSB of n is 1, upside-down printing mode is turned on.

[Notes]

- Only the LSB of n is valid.
- This command is enabled only when processed at the beginning of a line in standard mode.
- When this command is input in page mode, the printer performs only internal flag operations.
- This command does not affect printing in page mode.

[Default] n = 0

[Example] 1B 40

1B 7B 01(Turn on upside-down printing mode)

41 42 43 44 45 46 **0A**

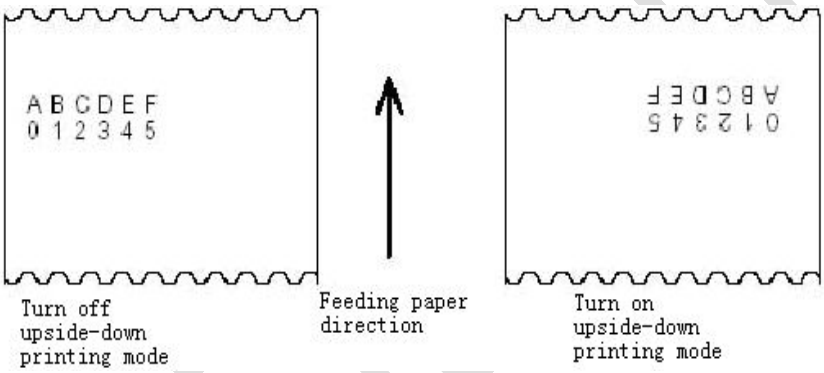
30 31 32 33 34 35 **0A**

1B 7B 00(Turn off upside-down printing mode)

41 42 43 44 45 46 **0A**

30 31 32 33 34 35 **0A**

Results:



GS ! n

[Function] Select character size.

[Format]

ASCII	GS	!	n
Hex	1D	21	n
Decimal	29	33	n

[Range] 0 ≤ n ≤ 255

(1 ≤ vertical number of times ≤ 6, 1 ≤ horizontal number of times ≤ 6)

[Description]Selects the character height using bits 0 to 2 and selects the character width using bits 4 to 7 as follows:

Bit	Off/On	Hex	Decimal	Function
0-3	Character height selection. See Table 2.			
4-7	Character width selection. See Table 1.			

Table 1 Character Width Selection			Table 2 Character Height Selection		
Hex	Decimal	Width	Hex	Decimal	Height
00	00	0(Normal)	00	0	1 (Normal)
10	16	2(double Width)	01	1	2(Double height)
20	32	3	02	2	3
30	48	4	03	3	4
40	64	5	04	4	5
50	80	6	05	5	6
60	96	6	06	6	6
70	112	6	07	7	6

[Notes]

- This command is effective to all characters (alphanumeric and Kanji) except for HRI characters.
- If n is outside of the defined range, this command will be ignored.
- In standard mode, the vertical direction is the paper feed direction, and the horizontal direction is perpendicular to the paper feed direction. However, when character orientation changes in 90° clockwise-rotation, the relationship between vertical and horizontal directions is reversed.
- In page mode, vertical and horizontal directions are based on the character orientation.
- When characters are enlarged with different sizes on one line, all the characters on the line are aligned at the baseline.
- The **ESC !** command can also turn double-width and double-height modes on or off. However, the setting of the last received command is effective.

[Default] n = 0

[Relative] **ESC !**

GS B n

[Function] Select white/black reverse printing mode.

[Format]

ASCII	GS	B	n
Hex	1D	42	n
Decimal	29	66	n

[Range] 0 ≤ n ≤ 255

[Description] Turn on or off white/black reverse printing mode.

- When the LSB of n is 0, white/black reverse mode is turned off.
- When the LSB of n is 1, white/black reverse mode is turned on.

[Notes]

- Only the LSB of n is valid.
- This command is available for all built-in characters and user-defined characters except

HRI characters.

- When white/black reverse printing mode is on, it also applied to character spacing set by **ESC SP**.
- This command does not affect bit-image, user-defined bit-image, barcode, HRI characters, and spacing skipped by **HT, ESC $, and ESC **.
- This command does not affect the space between lines.
- Underline mode is disabled when White/black reverse mode is selected. Otherwise it is enabled when white/black reverse mode is turned off.

[Default] n = 0

[Example] **1B 40**

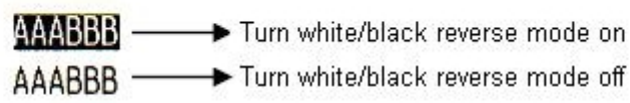
1D 42 01(Turn white/black reverse mode on)

41 41 41 42 42 42 **0A**

1D 42 00(Turn white/black reverse mode off)

41 41 41 42 42 42 **0A**

Results:



FS ! n

[Function] Set print mode(s) for Kanji characters.

[Format]

ASCII	FS	!	n
Hex	1C	21	n
Decimal	28	33	n

[Range] 0 ≤n ≤ 255

[Description] Set the print mode for Chinese character, using n as follows:

Bit	0/1	Hex	Decimal	Function
0, 1	--	--	--	Undefined
2	0	00	0	Double-width mode is off
	1	04	4	Double-width mode is on
3	0	00	0	Double-height mode is off
	1	08	8	Double-height mode is on
4-6	--	--	--	Undefined
7	0	00	0	Underline mode is off
	1	80	128	Underline mode is on

[Notes]

- When double-width and double-height mode are set (including right and left side character spacing), the width and height are doubled at the same time.
- The printer can underling all the characters (including right and left side character spacing), except the space by HT and 90 degree clockwise-rotated characters.
- The thickness of underline is specified by FS, regardless with the character size

- When some of the characters on a line are not in the same height, all the characters on the line are aligned at the baseline.
- It is possible to emphasize the Chinese character by **FS W** or **GS!**. The last received command is effective.
- It is possible to choose underline mode by **FS –**, the last received command is effective.

[Default] n = 0

[Relative] **FS - , FS W,GS !**

FS &

[Function] Select Kanji character mode.

[Format]

ASCII	FS	&
Hex	1C	26
Decimal	28	38

[Description] Select Kanji character mode.

[Notes]

- When the Chinese character mode is selected, the printer checks the code is for Chinese character or not, then process the first byte and the second byte if the code is for Chinese. When use Chinese GB18030 character set, the printer process the third and the forth byte if the Chinese character is forth-byte code.
- The printer select Chinese character mode automatically when the power is turned on.

[Relative] **FS . , FS C**

FS - n

[Function] Turn underline mode on/off for Kanji characters.

[Format]

ASCII	FS	-	n
Hex	1C	2D	n
Decimal	28	45	n

[Range] $0 \le n \le 2$, $48 \le n \le 50$

[Description]Turn underling mode off/on for Chinese character based on the value of the n:

n	Function
0, 48	Turn underline mode off for Chinese character
1, 49	Turn underline mode on for Chinese character(1 dot thick)
2, 50	Turn underline mode on for Chinese character(2 dot thick)

[Notes]

- The printer can underline all the characters(including right and left side character spacing),except the space by HT and 90 degree clockwise-rotated characters.
- When the underline mode for Chinese character is turned off, underline printing is no longer executed, but the previously specified thickness is not changed, the default thickness is 1 dot.
- The thickness of underling will not change even the size of character changes.
- It is possible to turn off the underling mode for Chinese character by FS!, the last

received command is effective.

[Default] n = 0

[Relative] **FS !**

FS .

[Function] Cancel Kanji character mode.

[Format]

ASCII	FS	.
Hex	1C	2E
Decimal	28	46

[Description] Cancel Kanji character mode.

[Notes]

- When Chinese character mode is canceled, all the characters are processed one byte for one time as ASCII code.
- The printer select Chinese character mode automatically when the power is turned on.

[Relative] **FS &, FS C**

FS 2 c1 c2 d1...dk

[Function] Define user-defined Kanji characters.

[Format]

ASCII	FS	2	c1	c2	d1...dk
Hex	1C	32	c1	c2	d1...dk
Decimal	28	50	c1	c2	d1...dk

[Range] c1 and c2 indicate character codes for the defined characters.

c1 = FEH,

A1H ≤ c2 ≤ FEH

0 ≤ d ≤ 255

k = 72

[Description] Define user-defined Chinese character specified by c1 and c2.

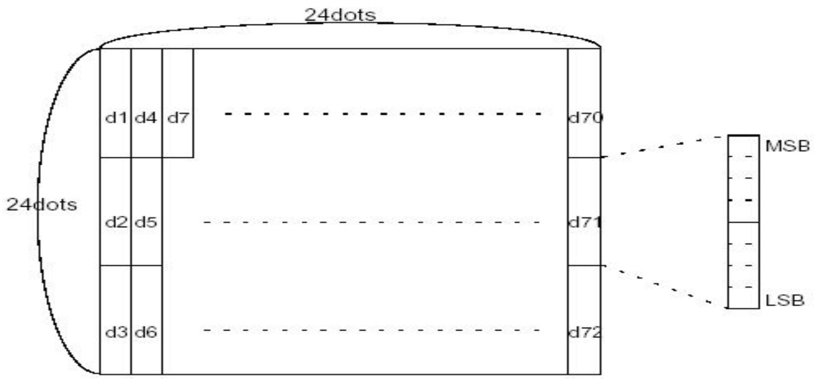
[Notes]

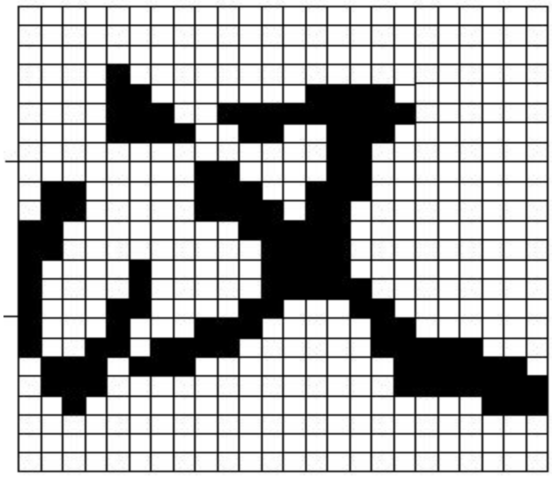
- c1, c2 indicate character code for the defined character, c1 specifies first byte, and c2 specifies the second byte.
- d indicates data. 1 indicates to print 1 dot, 0 indicates not print a dot.

[Default] No user defined Chinese Kanji.

[Relative] **FS C**

The relationship of user-defined Chinese character and the data is described as following:





D1=00H, D4=00H, D7=00H, D10=00H. . . . . .
D2=1FH, D5=78H, D8=60H, D11=00H. . . . . . .
D3=C0H, D6=30H, D9=38H, D12=70H. . . . . . .

FS C n

[Function] Select Japanese character mode.

[Format] ASCII FS C n

Hex 1C 43 n

Decimal 28 67 n

[Range] 0 ≤ n ≤ 1, 48 ≤ n ≤ 49

[Notes] Select Japanese character mode according to the value of n:

n	Code mode
0,48	JIS code
1,49	SHIFT JIS code

- This command is valid only under mode of Japanese fonts.
- Under JIS code mode, the following codes are valid:
  1st byte: <21>H to <7E>H
  2nd byte: <21>H to <7E>H
- Under SHIFT JIS code mode, the following codes are valid:
  1st byte: <81>H to <9F>H 和 <E0>H to <EF>H
  2nd byte: <40>H to <7E>H 和<80>H to <FC>H

[Default] n = 0

FS S n1 n2

[Function] Set the right and left character spacing.

[Format]

ASCII	FS	S	n1	n2
Hex	1C	53	n1	n2
Decimal	28	83	n1	n2

[Range] 0 ≤ n1 ≤ 255

0 ≤ n2 ≤ 255

[Description] Set the right spacing and the left spacing to n1 and n2 respectively.

The left spacing is [n1 × horizontal or vertical motion unit] inch, and the right spacing is [n2 ×horizontal or vertical motion unit] inch.

[Notes]

- When double-width is set, the right and the left spacing are doubled.
- In standard mode, the horizontal motion unit is used.
- In page mode, it is depending on starting position of the printable area as follows:

(1) When the starting position is set to the upper left or lower right, the horizontal motion unit is used.

(2) When the starting position is set to the upper right or lower left, the vertical motion unit is used.

(3) The maximum spacing of Chinese character is 36 mm, all setting exceeding the maximum is converted to the maximum automatically.

[Default] n1 = 0, n2 = 0

FS W n

[Function] Turn quadruple-size mode on/off for Kanji characters.

[Format]

ASCII	FS	W	n
Hex	1C	57	n
Decimal	28	87	n

[Range] 0 ≤ n ≤ 255

[Description] Select/cancel quadruple-size mode of Chinese character.

When the LSB of n is 0, quadruple-size mode of Chinese character is turned off.

When the LSB of n is 1, quadruple-size mode of Chinese character is turned on.

[Notes]

- Only the LSB of n is effective.
- When quadruple-size mode is turned on, the size of Chinese character is same as when double-width mode and double-height mode are both turned on.
- When quadruple-size mode is turned off, the characters are printed in normal size.
- All the characters are aligned at the baseline.
- **FS !** or **GS !** also can select or cancel quadruple-size mode of Chinese character, the last received command is effective..

[Default] n = 0

[Relative] **FS !, GS !**

2.4 Bitmap command

ESC * m nL nH d1... dk

[Function] Download and print bit-image.

[Format] ASCII ESC * m nL nH d1...dk

Hex 1B 2A m nL nH d1...dk

Decimal 27 42 m nL nH d1...dk

[Range] m = 0, 1, 32, 33, 35

0 ≤ nL ≤ 255

0 ≤ nH ≤ 3

0 ≤ d ≤255

[Description]

Selects a bit-image mode with m for the number of dots specified by nL and nH, d1...dk is data, of which number is decided by horizontal dots number and selected bit-image mode as follows:

203DPI

m	Bit-Image Mode	Vertical Direction		Horizontal Direction	
		Number of Dots	Dot Density	Dot Density	Number of Data (K)
0	8-dot single-density	8	67 DPI	101 DPI	nL + nH × 256
1	8-dot double-density	8	67 DPI	203 DPI	nL + nH × 256
32	24-dot single-density	24	203 DPI	101 DPI	(nL + nH × 256) × 3
33,35	24-dot double-density	24	203 DP	203 DPI	(nL + nH × 256) × 3

300DPI

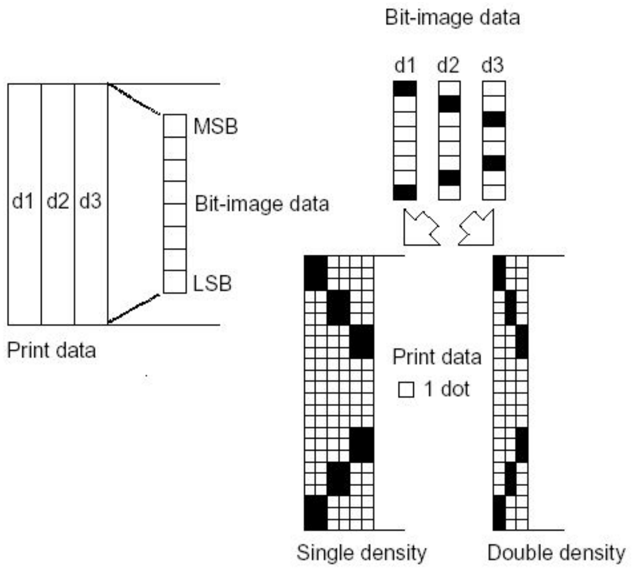
m	Mode	Vertical Direction		Horizontal Direction	
		Number of Dots	Dot Density	Dot Density	Number of Data (K)
0	8-dot single-density	8	100 DPI	150 DPI	nL + nH × 256
1	8-dot double-density	8	100 DPI	300 DPI	nL + nH × 256
32	24-dot single-density	24	300 DPI	150 DPI	(nL + nH × 256) × 3
33,35	24-dot double-density	24	300 DPI	300 DPI	(nL + nH × 256) × 3

[Notes]

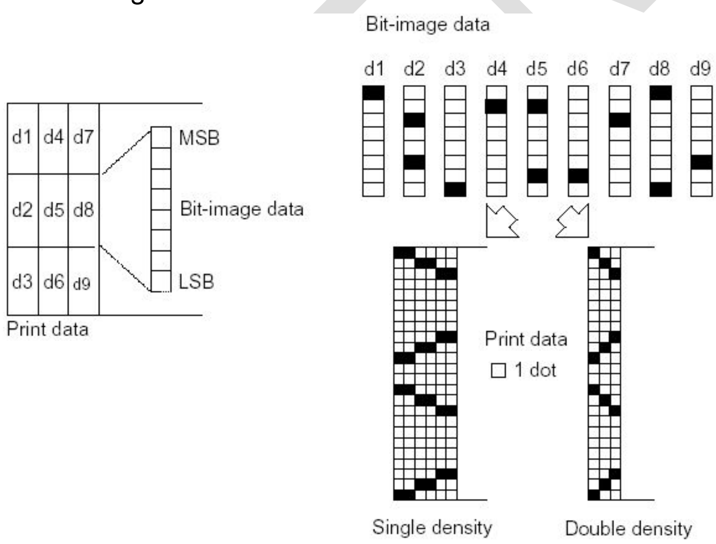
- If the value of m is out of the specified range, nL and data following are processed as normal data.
- If the bit-image data input exceeds the number of dots to be printed on a line, the excess data is ignored.
- d indicates the bit-image data. Set a corresponding bit to 1 to print a dot or to 0 to not print a dot.
- After printing a bit-image, the printer returns to normal data processing mode.

- This command is not affected by print modes (emphasized, double-strike, underline, character size or white/black reverse printing), except upside-down printing mode.
- The relationship between the image data and the dots to be printed is as follows:

When 8-dot bit-image is selected:



When 24-dot bit-image is selected:



GS # n

[Function] Specify a number for the bit-image to be downloaded.

[Format] ASCII GS # n

Hex 1D 23 n

Decimal 29 35 n

[Range] $0 \leq n \leq 255$

[Description] Specify a number for the bit-image to be downloaded. This number is to be used when downloading and printing this bit-image.

[Notes] The command is only enabled for bit-images in RAM and the settings are erased when the printer is turned off.

ESC (+ *.BMP file data)

[Function] Download BMP bit-image.

[Format]

ASCII	ESC		(+ *.BMP file data)
Hex	1B		(+ *.BMP file data)
Decimal	27		(+ *.BMP file data)

[Range] The horizontal width of bit-image (Max.) is 576, and the vertical dots (Max.) is 512.

[Description] If adding 1b command before BMP bit-image, then it can be downloaded into RAM.

[Notes]

- This command is only for the bit-image downloaded into RAM, and the data set will lose after the power is not connected.
- The storing place of this bit-image is the same with 1D 2A RAM, and you can use 1D 23 n command to select bit-image number.
- This bit-image can be printed through 1d 2f n command.
- Only support single color (binary) bit-image.

GS * x y d1...d(x × y × 8)

[Function] Download RAM bit-image.

[Format]

ASCII	GS	*	x y d1...d(x × y × 8)
Hex	1D	2A	x y d1...d(x × y × 8)
Decimal	29	42	x y d1...d(x × y × 8)

[Range] $1 \le x \le 255$, $1 \le y \le 255$

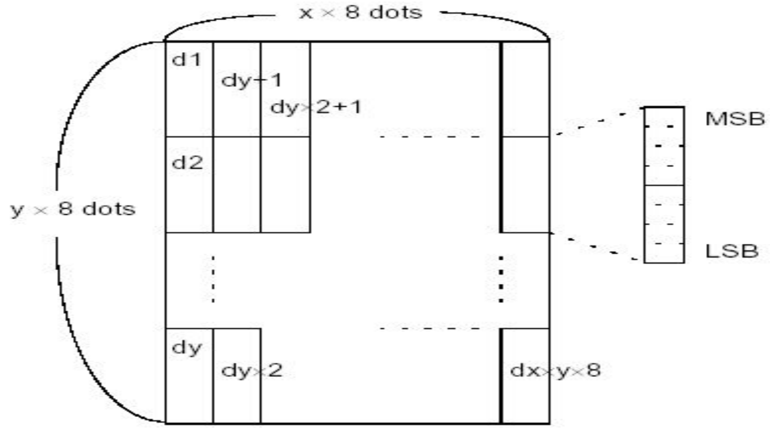
$0 \le d \le 255$

[Description] Define a downloaded bit-image using the number of bytes specified by x and y

- x specifies the number of dots in the horizontal direction.
- y specifies the number of dots in the vertical direction.

[Notes]

- The number of dots in the horizontal direction is x × 8, in the vertical direction it is y × 8.
- If x × y is out of the specified range, this command is disabled.
- The maximum capacity for bit-images in RAM is 128K bytes.
- The execution of ESC & command will erase the bit-images in RAM downloaded with this command.
- The d indicates bit-image data. Data (d) specifies a bit printed to 1 and not printed to 0.
- The downloaded bit-image is cleared when the power is turned off.
- The downloaded bit-image is cleared when executing ESC @ command.
- The number for the bit-image downloaded is defined by **GS #** command.
- The following figure shows the relationship between the downloaded bit-image and the printed data.



[Relative] GS /

GS / m

[Function] Print downloaded RAM bit-image.

[Format]

ASCII	GS	/	m
Hex	1D	2F	m
Decimal	29	47	m

[Range] 0 ≤ m ≤ 3, 48 ≤ m ≤ 51

[Description] Print a downloaded bit-image using the mode specified by m.

m selects a mode from the table below:

203DPI:

m	Mode	Vertical Dot Density (DPI)	Horizontal Dot Density (DPI)
0, 48	Normal	203	203
1, 49	Double-width	203	101
2, 50	Double-height	101	203
3, 51	Quadruple	101	101

300DPI

m	Mode	Vertical Dot Density (DPI)	Horizontal Dot Density (DPI)
0, 48	Normal	300	300
1, 49	Double-width	300	150
2, 50	Double-height	150	300
3, 51	Quadruple	150	150

[Notes]

- This command is ignored if a downloaded bit-image has not been defined.
- This command has no effect in the print modes (emphasized, double-strike, underline, character size, or white/black reverse printing), except for upside-down printing mode.
- The command sets the position after printing to the beginning of next line.
- If the downloaded bit-image to be printed exceeds the printable area, the excess data is not printed.
- The command prints bit-images in RAM and not that in NV memory. The number for the bit-image to be printed is defined by GS **#** command.

[Relative] **GS *, GS #**

GS v 0 m xL xH yL yH d1....dk

[Function] Print raster bit-image.

[Format]

ASCII	GS	v	0	m xL xH yL yH d1...dk
Hex	1D	76	30	m xL xH yL yH d1...dk
Decimal	29	118	48	m xL xH yL yH d1...dk

[Range] $0 \le m \le 3$, $48 \le m \le 51$

$0 \le xL \le 255$

$0 \le xH \le 255$

$0 \le yL \le 255$

$0 \le d \le 255$

$k = (xL + xH \times 256) \times (yL + yH \times 256)$ ($k \ne 0$)

[Description] Select raster bit-image mode. The value of m selects the mode, as follows:

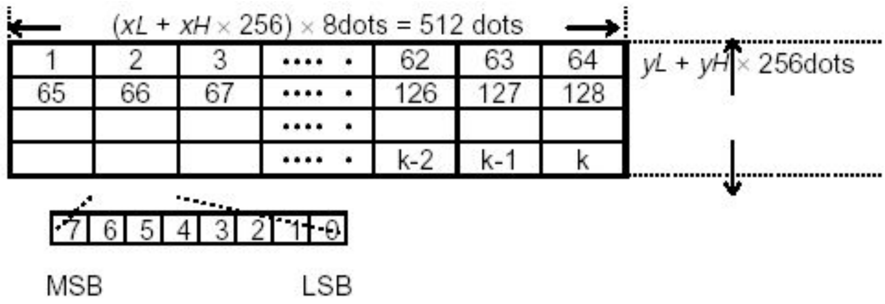
m	Mode	Vertical Resolution (DPI)		Horizontal Resolution (DPI)	
		203 DPI	300 DPI	203 DPI	300 DPI
0, 48	Normal	203 DPI	305DPI	203 DPI	300 DPI
1, 49	Double-width	203 DPI	305DPI	101 DPI	150 DPI
2, 50	Double-height	101DPI	152DPI	203 DPI	300 DPI
3, 51	Quadruple	101DPI	152DPI	101 DPI	150 DPI

- xL, xH indicate the number of data bytes (xL+ xH × 256) in the horizontal direction of the bit-image.
- yL, yH indicate the number of data bytes (yL+ yH × 256) in the vertical direction of the bit-image.

[Notes]

- In standard mode, this command is effective only when there is no data in the print buffer.
- This command has no effect in all print modes (character size, emphasized, double-strike, upside-down, underline, white/black reverse printing, etc.) for raster bit-image.
- Data outside the printing area is discarded.
- The **ESC a** (Select justification) setting is also effective on raster bit-images.
- When this command is received during macro definition, the printer ends macro definition, and begins performing this command. The definition of this command should be cleared.
- d indicates the bit-image data. Set a bit to 1 prints a dot and setting it to 0 does not print a dot.

[Example] When xL+ xH × 256 = 64



FS p n m

[Function] Print NV bit-image.

[Format]

ASCII	FS	p	n	m
Hex	1C	70	n	m
Decimal	28	112	n	m

[Range] 1 ≤ n ≤ 255

0 ≤ m ≤ 3, 48 ≤ m ≤51

[Description] Print a NV bit-image n using the mode specified by m.

203DPI:

m	Mode	Vertical Dot Density (DPI)	Horizontal Dot Density (DPI)
0.48	Normal	203	203
1.49	Double-width	203	101
2.50	Double-height	101	203
3.51	Quadruple	101	101

- n is the number of the NV bit-image (defined using the **FS q** command).
- m specifies the bit-image mode.

[Notes]

- NV bit-image means a bit-image which is defined in a non-volatile memory by **FS q** and printed by **FS p**.
- This command is not effective when the specified NV bit-image has not been defined.
- This command is not affected by print modes (emphasized, double-strike, underline, character size, white/black reverse printing, or 90° rotated characters, etc.), except upside-down printing mode.
- If the downloaded bit-image to be printed exceeds one line, the excess data is not printed.

[Relatives] **ESC *, FS q, GS /, GS v 0**

FS q n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n

[Function] Define NV bit-image.

[Format]

ASCII	FS	q	n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n
Hex	1C	71	n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n
Decimal	28	113	n [xL xH yL yH d1...dk]1...[xL xH yL yH d1...dk]n

[Range] 1 ≤ n ≤ 255

$0 \le xL \le 255$

$1 \le (xL + xH \times 256) \le 1023$

$1 \le (yL + yH \times 256) \le 8191$

$0 \le d \le 255$

$k = (xL + xH \times 256) \times (yL + yH \times 256) \times 8$

Total defined data area (MAX.) = 512K bits (64K bytes)

[Description] Define the NV bit-image specified by n.

- n specifies the number of the defined NV bit-image.
- xL, xH specifies (xL + xH × 256) × 8 dots in the horizontal direction for the NV bit-image you are defining.
- yL, yH specifies (yL + yH × 256) × 8 dots in the vertical direction for the NV bit-image you are defining.

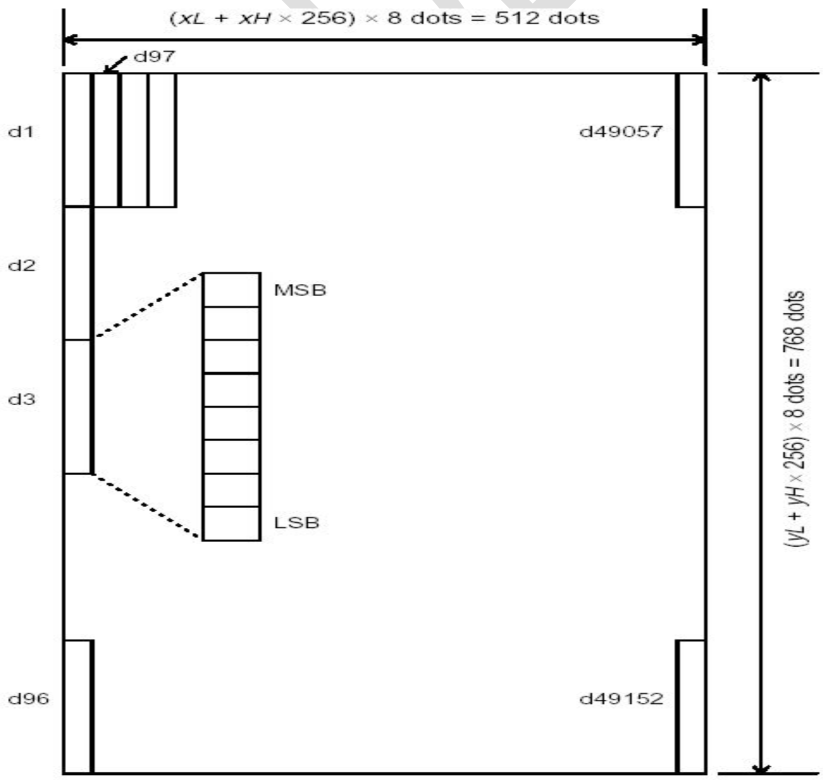
[Notes]

- Frequent write command execution may cause damage to the NV memory. Therefore, it is recommended to write the NV memory 10 times or less a day.
- This command cancels all NV bit-images that have already been defined by this command. The printer cannot redefine only one of several data definitions previously defined. In this case, all data needs to be sent again.
- During the processing of this command, the printer is busy to write data to NV memory and stops receiving any more commands. Therefore, before the ending of the processing of this command, do not send any other commands even real-time commands to the printer.
- NV bit-image means a bit-image which is defined in a non-volatile memory by **FS q** and printed by **FS p**.
- In standard mode, this command is effective only when processed at the beginning of the line.
- This command commences effective when 7 bytes <FS~yH> is processed as a normal value.
- When the amount of data exceeds the capacity left in the range defined by xL, xH, yL, yH, the printer processes the defined range by xL, xH, yL, yH.
- In the first group of NV bit-images, when any of the parameters xL, xH, yL, yH is out of the definition range, this command is disabled.
- In groups of NV bit-images other than the first one, when the printer processes xL, xH, yL, yH out of the defined range, it stops processing this command and starts writing into the NV images. At this time, NV bit-images that haven't been defined are disabled (undefined), but any NV bit-images before that are enabled.
- The d indicates the definition data. In data (d) a 1 bit specifies a dot to be printed and a 0 bit specifies a dot not to be printed.
- This command defines n as the number of a NV bit-image. Numbers rise in order from NV bit-image 01H. Therefore, the first data group [xL xH yL yH d1...dk] is NV bit-image

01H, and the last data group [xL xH yL yH d1...dk] is NV bit-image n. The total agrees with the number of NV bit-images specified by command FS p.

- A definition data of a NV bit-image consists of [xL xH yL yH d1...dk]. Therefore, when only one NV bit-image is defined n=1, the printer processes a data group [xL xH yL yH d1...dk] once. The printer uses ([data: (xL + xH × 256) × (yL + yH × 256) × 8] + [header:4]) bytes of NV memory.
- The definition area in this printer is a maximum of 512K bits (64K bytes). This command can define several NV bit-images, but cannot define a bit-image data whose total capacity [bit-image data + header] exceeds 512K bytes (64K bytes).(Different printer downloads different space, please refer to printer configuration).
- During the processing of this command, the printer does not transmit status and does not respond to status inquiry as well.
- When this command is received during macro definition, the printer ends macro definition, and begins performing this command.
- Once a NV bit-image is defined, it is not erased by performing ESC @, reset, and power off.
- Printer initialization is performed automatically after bitmap download (1b 40).
- This command performs only definition of a NV bit-image and does not perform printing. Printing of the NV bit-image is performed by the FS p command.

[Example] When xL = 64, xH = 0, yL = 96, yH = 0



[Relative] **FS p**

2.5 Status command

DLE EOT n

[Name] Real-time status transmission.

[Format]

ASCII	DLE	EOT	n
Hex	10	04	n
Decimal	16	4	n

[Range] 2 ≤ n≤7, 9 ≤ n≤11

[Description] Transmits the selected printer status specified by n in real-time, according to the following parameters:

n = 2: Transmit peripheral equipment status.

n = 3: Transmit error status.

n = 4: Transmit paper roll sensor status.

n = 5: Transmit printer status (Equal to first byte data of ASB).

n = 6: Transmit printer status (Equal to second byte data of ASB).

n = 7: Transmit printer status (Equal to third byte data of ASB).

n = 9: Transmit PRST status.

n = 10: Reset printer.

n = 11: Clear printer errors.

n = 12: Equal to fourth byte data of ASB

n = 13: Equal to fifth byte data of ASB

n = 14: Equal to sixth byte data of ASB

n = 15: Equal to seventh byte data of ASB

n = 16: Equal to eighth byte data of ASB

n = 17: Prst type, printer type and double-path configuration or not

[Notes] When n is out of range, the return value is invalid.

- The printer transmits specified status immediately upon receiving this command, when the command is enabled.
- Even though the printer is not selected using **ESC =** (select peripheral device), this command is still effective.
- The printer transmits the status without confirming whether the host computer can receive data.
- This command is only valid for serial and USB interfaces. If printer has both serial and USB interface, data return to serial and USB interface at the same time.
- The reset command is used to clear the error status and the print buffer as well. Make sure to cancel the print duty in the application when using this command.

n = 2: Transmits printer status defined as below:

Bit	Off/On	Hex.	Description
0	0	0	Fixed to 0
1	1	2	Fixed to 1
2	0/1	00/04	Top cover closed/ Top cover open
3	0/1	00/08	No FEED button action / FEED button action
4	1	10	Fixed to 1
5	0/1	0/20	Paper adequate/ Paper end
6	0/1	00/40	Normal / Error
7	0	00	Fixed to 0

n=3, Transmits error status defined as below:

Bit	Off/On	Hex.	Description
0	0	00	Fixed to 0
1	1	02	Fixed to 1
2	0	00	Fixed to 0
3	0/1	00/08	Cutter normal/Cutter error
4	1	10	Fixed to 1
5	0/1	00/20	Top cover closed/ Top cover open
6	0/1	00/40	Print head temperature normal/ Print head overheated
7	0	00	Fixed to 0

n=4, Transmits paper roll sensor status defined as below:

Bit	Off/On	Hex.	Description
0	0	00	Fixed to 0
1	1	02	Fixed to 1
2	0/1	00/04	Paper adequate / Paper near end
3	0/1	00/08	Paper adequate / Paper near end
4	1	10	Fixed to 1
5	0/1	00/20	Paper adequate / Paper end
6	0/1	00/40	Paper adequate / Paper end
7	0	00	Fixed to 0

n = 5: Transmits printer status defined as below:

Bit	Off/On	Hex.	Description
0	0/1	00/01	Paper adequate/Paper near end
1	1/1	00/02	Top cover closed/ Top cover open
2	0/1	00/04	Paper adequate/Paper end
3	0/1	00/08	Printhead normal/overheated
4	0/1	00/10	Cutter normal/Cutter error
5	0/1	00/20	Normal/Paper jam
6	0/1	00	Fixed to 0
7	0	00/80	Normal printer voltage/Abnormal printer voltage

Note: bit5 is valid under the configuration of having anti-jam mode.

n = 6: Transmits printer status defined as below:

Bit	Off/On	Hex.	Description
0	0/1	00/01	PRST without paper / PRST with paper
1	0/1	00/02	Normal/Paper jam at printing site
2	0/1	00/04	Normal/Paper jam on PRST
3	0/1	00/08	Normal/Black mark missing
4	0/1	00/10	Auto paper loading normal/ error
5	0/1	00/20	Buffer not full/buffer full
6	0/1	0/40	System normal/system error
7	0	0	Fixed to 0

- The receive buffer size of the printer is 4096 byte, receiving and processing at the same time, “buffer full” is only a real time status of the buffer, not a error status.

When using USB interface:

1) When there are 2560 bytes data in receive buffer, printer enters into “buffer full” status, and in this case the printer will close port and the program can not send data to the printer.
2) In “buffer full” status, when data in receive buffer reduce to 2303 bytes from 2560 bytes, the receive buffer status will turn to “buffer not full”, and the program can still send data to the printer.

When using serial interface:

1) When there are 3840 bytes data in receive buffer, printer enters into “buffer full” status, and in this case the printer will close port and the program can not send data to the printer.
2) In “buffer full” status, when data in receive buffer reduce to 3583 bytes from 3840 bytes, the receive buffer status will turn to “buffer not full”, and the program can still send data to the printer.

- System error described in the above table is just an extended function and reserved error type, currently the printer do not have this error type.

n = 7: Transmits printer course status:

Bit	Off/On	Hex.	Description	Error
0~4	00000/10000	00/10	Printer idle status Normal/Error	Top cover open, print head overheated, paper end
	00001/10001	01/11	Printing status Normal/ Error	Top cover open, print head overheated, paper end, paper jam
	00010/10010	02/12	Cutting paper status Normal/ Error	Top cover open, print head overheated, paper end, cutter error, paper jam
	00011/10011	03/13	PRST feeding paper status Normal/ Error	Top cover open, print head overheated, paper end, paper jam
	00100/10100	04/14	PRST waiting status Normal/Error	Top cover open, print head overheated, paper end
	00101/10101	05/15	PRST retraction status Normal/Error	Top cover open, print head overheated, paper end, paper jam
	00110/10110	06/16	PRST ejection status Error/Normal	Top cover open, print head overheated, paper end, paper jam
	00111/10111	07/17	Auto paper loading status Normal/Error	Top cover open, print head overheated, paper end
5	0/1	00/20	PrstIn without paper /PrstIn with paper	
6	0/1	00/40	No FEED button action/FEED button action	
7	0/1	00/80	Paper retract normally / user take the paper away in the process of retracting	

- The seventh byte will be cleared up after the next cut.
- PrstIn indicates the state of PrstIn sensor before or after retraction, with paper means that the paper holder is full of paper or some paper is absorbed on the channels.
- When PrstIn sensor is in with paper state before retraction, the printer cannot judge weather the paper was taken away by user in the retraction process.

n = 9, Transmit PRST status:

Bit	Off/On	Hex.	Description
0	0/1	00/01	PRST without paper / PRST with paper
1-3	0	00	Fixed to 0
4	1	02	Fixed to 1
5-7	0	00	Fixed to 0

n = 10, reset printer

- Printer shall clear error information and buffer area and execute half-auto feeding course with this command.
- Printer status recovers with the command after manually cancel error.

n = 11, clear printer error

- When executes this command, printer shall clear errors automatically according to error status. If the operation succeeds, it enters normal status, otherwise only via manual operation.

n = 12:

Bit	Status	Hex	Function
0	0/1	00/01	Paper cabinet not full/full
1	0/1	00/02	Return 4-byte ASB /return 8-byte ASB
2~3	00	00	Fixed to 0
4~7	1111	f0	Fixed to 1

n = 13:

Bit	Status	Hex	Function
0	0	00	Fixed to 0
1	0/1	00/02	Fixed to 0
2	0/1	00/04	Fixed to 0
3	0/1	00/08	Fixed to 0
4	0/1	00/10	Fixed to 0
5~6	00	0	Fixed to 0
	01	20	Fixed to 0
	10	40	Fixed to 0
	11	60	Fixed to 0
7	0	00	Fixed to 0

n = 14:

Bit	Status	Hex	Function
0	0	00	Fixed to 0
1	0/1	00/02	Normal status in one ticket control mode/completing clearing disabled printing status
2	0/1	00/04	Printing not disabled in one ticket control mode, printing enabled/printing disabled
3	0/1	00/08	Pulling paper/normal
4	0/1	00/10	Reserved, fixed to 0
5	0/1	00/20	Path 1 weekend paper near end
6	0/1	00/40	Path 2 weekend paper near end
7	0	00	Fixed to 0

Note:

- BIT5 indicates weekend paper near end status when the printer is not configured with double path module.
- The printer does not stop printing when paper near end status occurs by default. Users can set the printer to stopping printing when paper near end status occurs by using KIOSKUtility.

n = 15:

Bit	Status	Hex	Function
0~7	0	00	Reserved, fixed to 0

n = 16:

Bit	Status	Hex	Function
0~1	0	00	Fixed to 0

n = 17:

First byte:

Bit	Status	Hex	Function
0~7	0	0~255	0：No Prst 1：Standard DC motor Prst 2：Anti-jam and anti-pull module 3： Standard DC motor Prst, and anti-jam and anti-pull module Other: Not support at the moment

Second byte:

Bit	Status	Hex	Function
0~1	1	1	1ST lower print head printer
	2	2	1ST upper print head printer
	3	3	2ST printer
2	0/1	00/04	1ST paper/ 2ST paper
3	0/1	00/08	No double path/Double path
4~7	0	0	Reserved, fixed to 0

Note: When using USB interface, the transmitted command format please refer to 3.4 Data return format of USB interface.

GS a n

[Function] Enable/Disable Automatic Status Back (ASB).

[Format] ASCII GS a n

Hex 1D 61 n

Decimal 29 97 n

[Range] 0 ≤ n ≤ 255

[Description] Enables or disables Automatic Status Back function.

When the LSB of n = 0, the ASB function is disabled.

When the LSB of n = 1, the ASB function is enabled.

[Notes]

- Only the LSB of n is valid.
- If printer has both serial and USB interface, data return to serial and USB interface at the same time.
- When using serial interface and sending this command enables the ASB functions, the printer automatically transmits a four -bytes data of current status.
- When using USB interface, the ASB function is enabled by default, and the application program can read eight-byte/four-byte printer status information at any time.
- When using USB interface and the printer work under class mode, plus the ASB function enabled, program can read printer status at any time, the format are as follows:

0x05 (auto status data length +1) + four-byte ASB data;

- When using serial interface, the ASB function is disabled as default configuration.

First byte (printer information)

Bit	Off/On	Hex.	Status for ASB
0	0/1	00	Paper adequate/ Paper near end
1	0/1	02	Top cover closed / Top cover open
2	0/1	00/04	Paper adequate/ Paper end
3	0/1	00/08	Print head temperature normal/ Print head overheated
4	0/1	00/10	Cutter normal/ Cutter error
5	0/1	00	Normal/paper jam
6	0/1	00	Reserve
7	0	00	Printer voltage normal/abnormal

Note: bit5 is valid under the configuration of having anti-jam mode.

Second byte (printer information)

Bit	Off/On	Hex.	Description
0	0/1	00/01	PRST without paper / PRST with paper
1	0/1	00/02	Normal/Paper jam on printing part
2	0/1	00/04	Normal/Paper jam on PRST
3	0/1	00/08	Normal/Black mark missing
4	0/1	00/10	Normal/Auto paper loading error
5	0/1	00/20	Buffer not full/buffer full
6	0/1	00/40	System normal/error
7	0	00	Fixed to 0

Third byte (printer information)

Bit	Off/On	Hex.	Description	Possible error*
0-4	00000/10000	00/10	Printer idle status Normal/Error	Top cover open, Printhead overheated, paper end/ near end
	00001/10001	01/11	Printing status Normal/ Error	Top cover open, Printhead overheated, paper end/ near end ,Paper jam
	00010/10010	02/12	Cutting paper status Normal/ Error	Top cover open, Printhead overheated, paper end / near end , cutter error Paper jam
	00011/10011	03/13	PRST feeding paper status Normal/ Error	Top cover open, Printhead overheated, paper end / near end , paper jam
	00100/10100	04/14	PRST waiting status Normal/Error	Top cover open, Printhead overheated, paper end / near end
	00101/10101	05/15	PRST retraction status Normal/Error	Top cover open, Printhead overheated, paper end / near end , paper jam
	00110/10110	06/16	PRST ejection status Error/Normal	Top cover open, Printhead overheated, paper end / near end , paper jam
	00111/10111	07/17	Auto paper loading status Normal/Error	Top cover open, Printhead overheated, paper end / near end
5	0/1	00/20	PrstIn without paper / PrstIn with paper	

6	0/1	00/40	No FEED button action/FEED button action	
7	0/1	00/80	Paper retract normally / user take the paper away in the process of retracting	

- The function of the fifth and the seventh byte are optional (by enable/disable the states of PrstIn sensor), the fifth and the seventh byte set to 0 when the function is effective.
- The fifth and the seventh byte both indicates states and do not effect the action of printer.
- The seventh byte will be cleared up after the next cut.
- PrstIn indicates the state of PrstIn sensor before or after retraction, with paper means that the paper holder is full of paper or some paper is absorbed on the PRE path.
- When PrstIn sensor is in with paper state before retraction, the printer cannot judge weather the paper was taken away by user in the retraction process.

Bits of the fourth byte data are defined as below:

Bit	Status	Hex	Function
0	0/1	00/01	Paper cabinet not full/full
1	0/1	00/02	Return 4-byte ASB/return 8-byte ASB
2~3	00	00	Fixed to 0
4~7	1111	f0	Fixed to 1

Bits of the fifth byte data are defined as below:

Bit	Status	Hex	Function
0	0	00	Fixed to 0
1	0/1	00/02	Fixed to 0
2	0/1	00/04	Fixed to 0
3	0/1	00/08	Fixed to 0
4	0/1	00/10	Fixed to 0
5~6	00	0	Fixed to 0
	01	20	Fixed to 0
	10	40	Fixed to 0
	11	60	Fixed to 0
7	0	00	Fixed to 0

Bits of the sixth byte data are defined as below:

Bit	Status	Hex	Function
0	0	00	Fixed to 0
1	0/1	00/02	Fixed to 0
2	0/1	00/04	Fixed to 0
3	0/1	00/08	Normal/pulling paper
4	0/1	00/10	Reserved, fixed to 0
5	0/1	00/20	Path 1 weekend paper near end
6	0/1	00/40	Path 2 weekend paper near end
7	0	00/80	Paper near end and stop printing

Note:

- BIT5 indicates weekend paper near end status when the printer is not configured with double path module.
- The printer does not stop printing when paper near end status occurs by default. Users can set the printer to stopping printing when paper near end status occurs by using KIOSKUtility.

Bits of the seventh byte data are defined as below:

Bit	Status	Hex	Function
0~7	0	00	Reserved, fixed to 0

Bits of the eighth byte data are defined as below:

Bit	Status	Hex	Function
0~1	0	00	Fixed to 0
2	1	04	Fixed to 1
3	0	00	Fixed to 0
4~7	1	0F	Fixed to 1

[Relative] **DLE EOT n**

GS r n

[Function] Transmit status.

[Format] ASCII GS r n

Hex 1D 72 n

Decimal 29 114 n

[Range] n = 1, 49

[Description] Transmits the status specified by n as follows:

N	Function
1, 49	Transmits paper sensor status and paper end sensor status

[Notes]

- This command is valid for serial model and /or USB model only. If printer has both serial and USB interface, data return to serial and USB interface at the same time.
- The status types to be transmitted are shown below:

Paper sensor status (n = 1, 49):

Bit	Off/On	Hex	Decimal	Sensor	Status for ASB
0, 1	Off	00	0	Paper near end sensor	paper adequate
	On	03	3		paper near end
2, 3	Off	00	0	Paper presence sensor	paper adequate
	On	0c	12		paper end
4	Off	00	0		Not used. Fixed to 0.
5,6	Off	00	0		Undefined
7	Off	00	0		Not used. Fixed to 0.

- When using USB interface, the transmitted command format please refer to 3.4 Data return format of USB interface.

[Relative] **DLE EOT**

2.6 Barcode command

GS H n

[Function] Select printing position for HRI characters.

[Format] ASCII GS H n

Hex 1D 48 n

Decimal 29 72 n

[Range] 0 ≤ n ≤ 3, 48 ≤ n ≤ 51

[Description] Select the printing position of HRI characters when printing a barcode.

n selects the printing position as follows:

n	Printing position
0, 48	Not printed
1, 49	Above the barcode
2, 50	Below the barcode
3, 51	Both above and below the barcode

- HRI indicates Human Readable Interpretation.
- The commands (include white/black reverse, emphasized, character size, underline, 90° clockwise-rotation) don't affect HRI.

[Notes] HRI characters are printed using the font specified by **GS f**.

[Default] n = 0

[Relative] **GS f, GS k**

GS f n

[Function] Select font for Human Readable Interpretation (HRI) characters.

[Format] ASCII GS f n

Hex 1D 66 n

Decimal 29 102 n

[Range] n = 0, 1, 48, 49

[Description] Select a font for the HRI characters used when printing a barcode.

n selects a font from the following table:

n	Font
0,48	Standard ASCII characters
1,49	Compressed ASCII characters

[Notes]

- HRI indicates Human Readable Interpretation.
- HRI characters are printed at the position specified by **GS H**.

[Default] n = 0

[Relative] **GS H, GS k**

GS h n

[Function] Select barcode height.

[Format]

ASCII	GS	h	n
Hex	1D	68	n
Decimal	29	104	n

[Range] 1 ≤ n ≤ 255

[Description] n specifies the number of dots in the vertical direction.

[Default] n = 216

[Relative] **GS k**

①GS k m d1...dk NUL②GS k m n d1...dn

[Function] Select a barcode type and print barcode.

[Format]

①ASCII	GS	k	m	d1...d k	NUL	
Hex	1D	6B	m	d1...d k	00	
Decimal	29	107	m	d1...d k	0	
②ASCII	GS	k	m	n	d1... dn	
Hex	1D	6B	m	n	d1... dn	
Decimal	29	107	m	n	d1... dn	

[Range] ①0 ≤ m ≤ 6 (m and d depends on the barcode system used)

②65 ≤ m ≤ 73 (m and d depends on the barcode system used)

[Description] Select a barcode type and print the barcode.

m selects a barcode type as follows:

m		Type	Number	d
①	0	UPC-A	11 ≤ k ≤ 12	48 ≤ d ≤ 57
	1	UPC-E	11 ≤ k ≤ 12	48 ≤ d ≤ 57,d1=48
	2	JAN13 (EAN13)	12 ≤ k ≤ 13	48 ≤ d ≤ 57
	3	JAN 8 (EAN8)	7 ≤ k ≤ 8	48 ≤ d ≤ 57
	4	CODE39	1 ≤ k ≤ 255	45 ≤ d ≤ 57, 65 ≤ d ≤ 90, 32, 36, 37,43
	5	ITF	1 ≤ k ≤ 255	48 ≤ d ≤ 57
	6	CODABAR	1 ≤ k ≤ 255	48 ≤ d ≤ 57, 65 ≤ d ≤ 68, 36, 43, 45,46,47,58

	10	PDF417	1 ≤ k ≤ 255	0 ≤ d ≤ 255
	11	QRCODE	1 ≤ k ≤ 928	0 < d ≤ 255
	12	MAXICODE	1 ≤ k ≤ 84	48 ≤ d ≤ 57,65 ≤ d ≤ 90, 97 ≤ d ≤ 122
②	65	UPC-A	11 ≤ n ≤ 12	48 ≤ d ≤ 57
	66	UPC-E	11 ≤ n ≤ 12	48 ≤ d ≤ 57,d1=48
	67	JAN13 (EAN13)	12 ≤n ≤ 13	48 ≤ d ≤ 57
	68	JAN 8 (EAN8)	7 ≤n ≤ 8	48 ≤ d ≤ 57
	69	CODE39	1 ≤ n ≤ 255	45 ≤ d ≤ 57, 65 ≤ d ≤ 90, 32, 36, 37,43
	70	ITF	1 ≤ n ≤ 255	48 ≤ d ≤ 57
	71	CODABAR	1 ≤ n ≤ 255	48 ≤ d ≤ 57 65 ≤ d ≤ 68, 36, 43,45,46,47 58
	72	CODE93	1 ≤ n ≤ 255	0 ≤ d ≤ 127
	73	CODE128	2 ≤ n ≤ 255	0 ≤ d ≤ 127
	75	PDF417	1 ≤ n ≤ 255	0 ≤ d ≤ 255
	76	QRCODE	1 ≤ n ≤ 255	0 ≤ d ≤ 255
	77	MAXICODE	1 ≤ n ≤ 138	48 ≤ d ≤ 57,65 ≤ d ≤ 90
	80	GS1 DataBar Omnidirectional	n=13	48 ≤ d ≤ 57
	81	GS1 DataBar Truncated	n=13	48 ≤ d ≤ 57
	82	GS1 DataBar Limited	n=13	48 ≤ d ≤ 57
	83	GS1 DataBar Expanded	2 ≤ n ≤ 255	32 ≤ *d* ≤ 34, 37 ≤ *d* ≤ 63, 65 ≤ *d* ≤ 90, *d* = 95, 97 ≤ *d* ≤ 122, *d* = 123 [*d1* = 40, 48 ≤ *d2* ≤ 57, 48 ≤ *d3* ≤ 57, or 48 ≤ *d1* ≤ 57, 48 ≤ *d2* ≤ 57]
	84	GS1-128	2 ≤ n ≤ 255	0 ≤ d ≤ 127

[Notes ①]

- This command ends with a NULL code.
- When UPC-A or UPC-E barcode system is selected, if more than 12 bytes barcode data is received, the printer will process the excess data as normal data.
- When JAN13 (EAN13) barcode system is selected, if more than 13 barcodes data is received, the printer will process the excess data as normal data.
- When JAN8 (EAN8) barcode system is selected, if more than 8 bytes barcodes data is

received, the printer will process the excess data as normal data.

- The number of data for ITF barcode must be even numbers. When an odd number of data is input, the printer ignores the last received data.
- The beginning code and the ending code of CODEBAR barcode must be one of A, B, C and D. The ending codes can be replaced with T, E, * and N.
- When QRCODE is selected, d1...d k (d1…dn)consist of 5 parts and the format is shown as below:

1) Daabbcc,

   D: Link structure mode, input specific mark symbol”D”. This mode is optional and the following three parameters and separator should be assigned if this mode is selected.

   aa: The position of the specific symbol; input 2 bytes decimal data

   bb: The total number of the symbols; input 2 bytes decimal data

   cc: The even and odd data; input 2bytes hexadecimal data

   , : Are fixed separator symbol.

2) E: Error correction grade Range: L,M,Q,H. The correction grade is increasing from L to H.
3) M: Mask image reference. Range: Default as automatic mask.
4) M: Data input mode Range: A or M, A means automatic mode (Recommended).M means manual input mode. If A is selected, the character mode is not necessary to be assigned; If M selected, the character mode must be assigned. The default is A mode.
5) <Character mode><DATA1>,

   < Character mode ><DATA2>,

   < Character mode ><DATA3>,

   ………………………

   < Character mode ><DATAn>

   Note: n>=200

   Character input mode<N,A,B,K>

   N: Numbers(0~9)

   A: Mixed by alphabet and numbers(0~9)(A~Z)(SP, $, %, *, +, -, ., /,:)

   B: 8 Bit byte mode(0x00~0xFF)

   K: JIS

   The legal width of the bar: The ratio of the bar is not changeable.

   Example:

   1D 6B 0B 51 41 2C 30 31 32 33 34 35 36 37 38 39 41 42 43 44 20 32 44 20 63 6F 64 65 00(Automatic mode is recommended and the character symbol A can be omitted)

   1D 6B 4c 12 48 4D 2C 4E 31 32 33 34 35 36 37 38 39 31 32 33 34 35

   1D 6B 0B 4D 4D 2C 41 41 43 2D 34 32 00

1D 6B 0B 4C 4D 2C 4E 30 31 32 33 34 35 36 37 38 39 30 31 32 33 34 35 2C 41 41 42 43 2C 42 30 30 30 36 71 72 63 6F 64 65 00

1D 6B 0B 46 2C 4C 4D 2C 4E 30 31 32 33 34 35 36 37 38 39 2C 41 31 32 41 41 42 42 2C 42 30 30 30 36 71 72 63 6F 64 65 00

• Maxicode supports different code mode and compression mode, and its capacity is different for different character sets. Maxicode has maximum 138 numbers, other characters do not have so many numbers. Maxicode has five parts in following formats:

1) The basic postal code in 5 numbers;
2) The second postal code in 4 numbers;
3) The country code in 3 numbers;
4) The service class in 3 numbers;
5) The character strings

Legal character: alphabet and numbers;

Length of variable: changeable;

Legal length of the bar: The ratio of the bar is not changeable.

Example:

1D 6B 0C 33 32 37 38 39 35 35 35 35 38 34 30 36 36 36 54 48 49 53 20 50 41 43 4B 41 47 45 49 53 20 47 4F 49 4E 47 20 54 4F 20 44 41 54 41 4D 41 58 43 4F 52 50 2E 00

[Notes ②]

- n indicates the number of barcode data, and the printer processes n bytes from the next character data as barcode data.
- If n is outside of the specified range, the printer stops command processing and processes the following data as normal data.

[Notes (standard mode)]

- If d is outside of the specified range, the command is ineffective.
- If the horizontal size exceeds printing area, excess data is printed.
- This command feeds as much paper as is required to print the barcode, regardless of the line spacing specified by **ESC 2** or **ESC 3**.
- This command is enabled only when no data exists in the print buffer. When data exists in the print buffer, the printer processes the data following m as normal data.
- After printing barcode, this command sets the print position to the beginning of the line.
- This command is not affected by print modes (emphasized, double-strike, underline, character size, white/black reverse printing, or 90° rotated character, etc.), except for upside-down printing mode.

[Notes (page mode)]

- This command develops barcode data in the print buffer, but does not print it. After processing barcode data, this command moves the print position to the right side dot of the barcode.

- If d is out of the specified range, the command is ignored.
- If barcode width exceeds the printing area, the printer does not print the barcode
- When using the CODE 128 (m = 73) in this printer, take the following points into account for data transmission:

1) The top of the barcode data string must be code set selection character (any of CODE A, CODE B or CODE C) which selects
2) Special characters are defined by combining two characters "{" and one character. The ASCII character "{" is defined by transmitting "{" twice consecutively.

Specific character	Transmit data		
	ASCII	Hex	Decimal
SHIFT	{S	7B, 53	123,83
CODE A	{A	7B, 41	123, 65
CODE B	{B	7B, 42	123, 66
CODE C	{C	7B, 43	123, 67
FNC1	{1	7B, 31	123, 49
FNC2	{2	7B, 32	123, 50
FNC3	{3	7B, 33	123, 51
FNC4	{4	7B, 34	123, 52
"{"	{{	7B, 7B	123, 123

[Example] Example data for printing "No. 123456"
In this example, the printer first prints "No." using CODE B, then prints the following numbers using CODE C.
GS k 73 10 123 66 78 111 46 123 67 12 34 56

No.123456

- If the top of the barcode data is not the code set selection character, the printer stops command processing and processes the following data as normal data.
- If combination of "{" and the following character does not apply any special character, the printer stops command processing and ignore the following data.
- If the printer receives characters that cannot be used in the special code set, the printer stops command processing and ignore the following data.
- The printer does not print HRI characters that correspond to the shift characters or code set selection characters.
- HRI character for the function character is space.
- HRI characters for the control character (<00>H to <1F>H and <7F>H) are not printed.
- Be sure to keep spaces on both right and left sides of a barcode. (Spaces are different depending on the types of the barcode. Spaces mean blank space).

[Relative] **GS H, GS f, GS h, GS w**.

[Notes] <u>**1B 40**</u>(Initialize printer)

4A 41 4E 31 33 **0A**
1D 48 01(Set the width of the barcode unit 1)
1D 66 01(HRI character use condensed character)
1D 77 01(HRI character print above the barcode)
1D 68 40(Barcode height is 64/203 inch)
1D 6B 02 30 31 32 33 34 35 36 37 38 39 30 35 39 00 0A
1D 48 02(Set the width of the barcode unit 2)
1D 66 01(HRI character use condensed character)
1D 77 02(HRI character print under the barcode)
1D 68 80(Barcode height is 128/203 inch)
1D 6B 02 30 31 32 33 34 35 36 37 38 39 30 35 39 00 0A
1D 48 03(Set the width of the barcode unit 3)
1D 66 00(HRI character use standard character)
1D 77 03(HRI character print both above and under the barcode)
1D 68 C8(Barcode height is 162/203 inch)
1D 6B 02 30 31 32 33 34 35 36 37 38 39 30 35 39 00 0A
Result:

JAN13
0123456789050
Barcode unit width 1
Barcode height 64/203 inch
HRI condensed character
on the above

Barcode unit width 2
Barcode height 128/203 inch
HRI condensed character on
the below
0 1 2 3 4 5 6 7 8 9 0 5 0

0 1 2 3 4 5 6 7 8 9 0 5 0
Barcode unit width 2
Barcode height 200/203 inch
HRI condensed character on
both above and below
0 1 2 3 4 5 6 7 8 9 0 5 0

GS o n

[Function] Set parameters of QRCODE barcode

[Format]

ASCII	GS	o	m nA nB nC
Hex	1D	6F	m nA nB nC
Decimal	29	111	m nA nB nC

[Range] m = 0 , 1≤ nA ≤255 ,0≤nB≤1,1≤nC ≤2

The meaning of parameter n is shown as below:

Parameter	Meaning
nA	Basic element width
nB	Language mode 0:Chinese; 1:Japanese
nC	Symbol type 1:Original type; 2:Enhanced type(Recommended)

[Notes] When the value of parameter is outside the specified range, the command is not valid.

[Default] nA = 3, nB=0, nC=2.

GS q n

[Function] Set correction grade of barcode PDF417.

[Format]

ASCII	GS	q	n
Hex	1D	71	n
Decimal	29	113	n

[Range] 0≤ n ≤8

[Notes] Set correction grade of PDF417 code, the higher the correction grade is, the bigger the capacity of the barcode is.

GS w n

[Function] Set barcode width.

[Format]

ASCII	GS	w	n
Hex	1D	77	n
Decimal	29	119	n

[Range] 2 ≤ n ≤ 6

[Description] Set the horizontal size of the barcode.

n specifies the barcode width as follows:

n	Module Width (dots) for Multi-level Barcode	Binary-level Barcode	
		Thin element width (dots)	Thick element width (dots)
2	2	2	5
3	3	3	8
4	4	4	10
5	5	5	13
6	6	6	15

- Single-level barcodes are as follows:
  UPC-A, UPC-E, JAN13 (EAN13), JAN8 (EAN8), CODE93, CODE128
- Binary-level barcodes are as follows:CODE39, ITF, CODABAR

[Default] n = 2

[Relative] **GS k**

GS (k pL pH cn fn n (cn = 51, fn = 67)

[Function] Set the module width of 2-dimensional GS1 DataBar.

[Format]

ASCII	GS	(	k	pL	pH	cn	fn	n
Hex	1D	28	6B	pL	pH	cn	fn	n
Decimal	29	40	107	pL	pH	cn	fn	n

[Range] (pL + pH × 256) = 3,(pL = 3, pH = 0)

cn = 51

fn = 67

2 ≤ n ≤ 8

[Notes] Set the width of one module of 2-dimensional GS1 DataBar to n dots.

[Default] n=2

GS (k pL pH cn fn nL nH (cn = 51, fn = 71)

[Function] Set the maximum width of GS1 DataBar: 2-dimensional GS1 DataBar Expanded Stacked.

[Format] ASCII GS (k pL pH cn fn nL nH

Hex 1D 28 6B pL pH cn fn nL nH

Decimal 29 40 107 p pH cn fn nL nH

[Range] (pL + pH × 256) = 4, (pL =4, pH = 0)

cn = 51

fn = 71

106 ≤ (nL + nH × 256) ≤ 3952, (nL + nH × 256) = 0 (0 ≤ nL ≤ 255, 0 ≤ nH ≤ 15)

[Notes] Set the maximum width of GS1 DataBar Expanded Stacked (2-dimensional GS1 DataBar) to (nL + nH 256) dots.

[Default] (nL + nH × 256) = 141 (nL = 141, nH = 0)

GS (k pL pH cn fn m n d1...dk (cn = 51, fn = 80)

[Function] Define and print GS1 Databar.

[Format] ASCII GS (k pL pH cn fn m n d1...dk

Hex 1D 28 6B pL pH cn fn m n d1...dk

Decimal 29 40 107 pL pH cn fn m n d1...dk

[Range] 6≤(pL + pH × 256) ≤259 (0 ≤ pL ≤ 255, pH = 0, 1)

cn = 51

fn = 80

m = 48

n = 72, 73, 76

0 ≤ d ≤ 255

k = (pL + pH × 256) - 4

[Notes] n is used to select the defined GS1 DataBar type as following:

n	Barcode type
72	GS1 DataBar Stacked
73	GS1 DataBar Stacked Omnidirectional
74	GS1 DataBar Expanded Stacked

GS (k pL pH cn fn n (cn = 52, fn = 67)

[Function] Set module width of composite symbology.

[Format] ASCII GS (k pL pH cn fn n

Hex 1D 28 6B pL pH cn fn n

Decimal 29 40 107 pL pH cn fn n

[Range] (pL + pH × 256) = 3 (pL = 3, pH = 0)

cn = 52

fn = 67

2 ≤ n ≤ 8

[Notes] Set one module width of composite symbology to *n* dots.

[Default] n=2

GS (k pL pH cn fn nL nH (cn = 52, fn = 71)

[Function] Set the maximum width of GS1 DataBar expanded stacked of composite symbology.

[Format]

ASCII	GS	(	k	pL	pH	cn	fn	nL	nH
Hex	1D	28	6B	pL	pH	cn	fn	nL	nH
Decimal	29	40	107	pL	pH	cn	fn	nL	nH

[Range] (pL + pH × 256) = 4 (pL= 4, pH = 0)

cn = 52

fn = 71

106 ≤ (pL + pH × 256) ≤ 3952, (nL + nH × 256) = 0 (0 ≤ nL ≤ 255, 0 ≤ nH ≤ 15)

[Notes] Sets the maximum width of GS1 DataBar Expanded Stacked to (pL + pH 256).

[Default] (nL + nH × 256) = 141 (nL = 141, nH = 0)

GS (k pL pH cn fn n (cn = 52, fn = 72)

[Function] Set a font for HRI character when print barcode.

[Format]

ASCII	GS	(	k	pL	pH	cn	fn	n
Hex	1D	28	6B	pL	pH	cn	fn	n
Decimal	29	40	107	pL	pH	cn	fn	n

[Range] (pL + pH × 256) = 3 (pL = 3, pH = 0)

cn = 52

fn = 72

0 ≤ n ≤ 2, 48 ≤ n ≤ 50

0 ≤ n ≤ 2, 48 ≤ n ≤ 50, n = 97, 98

[Notes] Selects whether or not to use the font of HRI character, and selects a font by n as following:

n	Font
0,48	Does not use HRI character.
1,49	Standard ASCII character (12 × 24)
2,50	Compressed ASCII character (9 × 17)
97	Special standard ASCII character (12 × 24)
98	Special compressed ASCII character (9 × 17)

Notes: Following characters do not support HRI character.

GS1 DataBar Stacked

GS1 DataBar Stacked Omnidirectional

GS1 DataBar Expanded Stacked

[Default] n=0

GS (k pL pH cn fn m a b d1...dk (cn = 52, fn = 80)

[Function] Define and print GS1 composite symbology.

[Format] ASCII GS (k pL pH cn fn m a b d1...dk

	Hex 1D 28 6B pL pH cn fn m a b d1...dk
	Decimal 29 40 107 pL pH cn fn m a b d1...dk

[Range] 7 ≤ (pL + pH × 256) ≤ 2366 (0 ≤ pL ≤ 255, 0 ≤ pH ≤ 9) [when a = 48]

8 ≤ (pL + pH × 256) ≤ 2366 (0 ≤ pL ≤ 255, 0 ≤ pH ≤ 9) [when a = 49]

cn = 52

fn = 80

m = 48

a = 48, 49

65 ≤ b ≤ 77 [when a = 48]

b=65, 66 [when a = 49]

0 ≤ d ≤ 255

k = (pL + pH × 256) – 5

[Notes] Define and print GS1 composite symbology.

- When a=48, b specifies the type of defined complex code as following:

b	Barcode type
65	EAN8
66	EAN13
67	UPC-A
68	UPC-E (6-digit version (0 excluded))
69	UPC-E (11-digit version (0 included))
70	GS1 DataBar Omnidirectional
71	GS1 DataBar Truncated
72	GS1 DataBar Stacked
73	GS1 DataBar Stacked Omnidirectional
74	GS1 DataBar Limited
75	GS1 DataBar Expanded
76	GS1 DataBar Expanded Stacked
77	GS1-128

- When a=49, b specifies the type of element as following:

b	Element type
65	CC-A, CC-B, CC-C
66	Fixed to CC-C

2.7 Two-color command

ESC r n

[Function] Enter/Exit two-color print mode.

[Format] ASCII ESC r n

Hex 1B 72 n

Decimal 27 114 n

[Range] 0 ≤n ≤ 1

[Notes] • n=0, exit two-color mode.

• n=1, enter two-color mode.

[Default] n = 1

[Relative] ESC C

[Example] **1B 40**

1B 72 01 (enter two-color print mode)

1D 21 11 (set double-height, double-width)

1B 43 01 (select color 2)

41

1B 43 00 (select color 1)

41

1B 43 01 (select color 2)

41

1B 43 00 (select color 1)

41

0A (print)

1B 72 01 (exit two-color print mode)

Result:

ESC C n

[Function] Select print color.

[Format]

ASCII	ESC	C	n
Hex	1B	43	n
Decimal	27	67	n

[Range] 0 ≤n ≤ 1

[Notes] • n=0, select color 1.

• n=1, select color 2.

[Relative] **ESC r**

GS (N pL pH fn a

[Function] Two-color print setting command, set print color and entering/exiting two-color mode.

[Format] ASCII GS (N pL pH fn a

Hex 1D 28 4E pL pH fn a

Decimal 27 67 n

[Range] pL = 2, pH = 0, fn = 48, a = 48, 49, 50

[Notes] • a=48, exit two-color print mode.

• a=49, enter two-color print mode, and select color 1.

• a=50, enter two-color print mode, and select color 2.

[Relative] **ESC r, ESC C**

2.8 Upside-down print command

GS (z nL nH 0 S

[Function] Enter upside-down print mode, start incepting upside-down data.

[Format] ASCII GS (z nL nH 0 S

Hex 1D 28 7A nL nH 30 53

Decimal 29 40 122 nL nH 48 83

[Range] nL = 2 nH = 0

[Notes]

- The difference between upside-down command and **ESC { n** upside-down printing: this upside-down printing command can print the note upside down, while **ESC { n** can only print the character line upside down.
- This command is used at the beginning of the upside-down page. The part behind the command is to be printed. It cannot be printed out immediately, but it is stored in the buffer. When the printer incepts the command to cut paper (**GS V**)or to exit upside-down print mode, the printer print upside down.
- This command can only be used at the beginning of the line, otherwise it will be ignored.
- This command should cooperate with cut paper command or exit upside-down print mode command; otherwise it will not be able to print upside down.
- The data to be printed in upside-down mode must be smaller than the command buffer (the capacity of buffer can be checked by printing self-test pages), for pages larger than command buffer:

1) Printer ignores print data;
2) If ending with exiting upside-down printing mode, the printer will enter normal print mode with no action;
3) If ending with cut paper command, the printer will enter normal print mode after the action of cutting paper.

- Forbid command

The following commands are not supported under upside-down printing mode.If sending the following commands under upside-down printing mode, the printer may not perform the expected result.

Command	Function
GS :	Start / end macro definition
GS ^	Perform macro definition
ESC D	Set horizontal tab position
FS q	Define NV bitmap
ESC =	Select printer
GS (A	Perform testing print
ESC c 7	Grayscale print function

Note: Although **FS q** command is not supported under upside-down printing mode, **FS p** command is supported. If NV bitmap is to be printed, enter upside-down printing mode by command. Before entering upside-down printing mode, first send **FS q** command to define NV bitmap. Example is as follows:

1C 71 01 (define NV bitmap,……..stand for bitmap data)

1D 28 7A 02 00 30 53 (enter upside-down printing mode)

1C 70 01 00(print the bitmap downloaded in FLASH)

1D 28 7A 02 00 30 45(print NV bitmap and exit upside-down printing mode)

- Perform the command immediately

In upside-down printing mode, this kind of command will be performed before printing. Details are as follows:

Command	Function
GS a	Automatically back status.
DLE ENQ n	Real-time request.
DLE DC4	Real-time produce cash drawer pulse.
GS r	Return back status.
ESC p	Produce cash drawer control pulse

Steps to enter upside-down printing mode through command:

1) Send command of entering upside-down printing mode;
2) Send page;
3) Send command of exiting upside-down printing mode or cutting paper and print sample page.

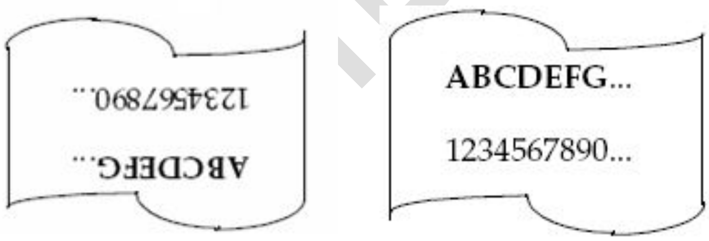
[Example] **1D 28 7A 02 00 30 53** (enter upside-down printing mode)

41 42 43 44 45 46 47 2E 2E 2E **0A 0A**

31 32 33 34 35 36 37 38 2E 2E 2E **0A** (print sample page)

1D 56 42 00 (cut paper and exit upside-down printing mode)

Normal print mode and upside-down print:



GS (z nL nH 0 E

[Function] Print page data, exit upside-down printing mode and enter normal print mode.

[Format]

ASCII	GS	(	z	nL	nH	0	E
Hex	1D	28	7A	nL	nH	30	45
Decimal	29	40	122	nL	nH	48	69

[Range] nL = 2

nH = 0

[Notes]

- This command should be used at the end of the upside-down page. After receiving the command, the page data will be printed upside down.
- This command can only be used at the beginning of each line, otherwise it will be ignored.

- This command should be used together with the command to enter upside-down printing mode, otherwise upside-down printing will not be able to perform.

2.9 Watermark command

GS { w f n1 n2 n3 n4 n5

[Function] Set parameters of watermark bitmap, and enter watermark mode.

[Format] ASCII GS { w f n1 n2 n3 n4 n5

Hex 1D 7B 77 02 n1 n2 n3 n4 n5

Decimal 29 123 119 02 n1 n2 n3 n4 n5

[Range] 0 ≤ n1 ≤ 1

0 ≤ n2 ≤ 2

1 ≤ n3≤ 255

0 ≤ n4 ≤ 255

1 ≤ n5 ≤ 255

[Notes]

- n1 indicates watermark print mode:
  n1 = 0: print watermark bitmap when feeding paper
  n1 = 1: print watermark bitmap when printing
- n2 indicates align mode of watermark printing
  n2 = 0: left justification
  n2 = 1: centering
  n2 = 2: right justification
- n3 indicates enlargement mode selection of watermark printing. 0-3 bits select height, and 4-7 bits select width, the values of which are as follows:

Width selection			Height selection		
Hex	Decimal	Horizontal enlargement	Hex	Decimal	Vertical enlargement
10	16	1 (normal)	01	1	1 (normal)
20	32	2 (double-width)	02	2	2 (double-height)
30	48	3	03	3	3
40	64	4	04	4	4
…	…	…	…	…	…
F0	240	15	0F	15	15

- n4 indicates watermark brightness adjustment, and the recommended value is 0x30.
- n5 indicates bitmap number (defined by command **FS q**).
- This command is enabled only when processed at the beginning of the line.
- This command is only enabled in line mode.
- Define Flash bitmap using command **FS q** first before using this command.

[Example] 1D 7B 77 02 01 00 22 40 01

Explanation:

n1=0x01: indicating watermark bitmap is printed only when there is a print task.

n2=0x00: indicating left justification of watermark bitmap.

n3=0x22: indicating watermark bitmap is enlarged twice in horizontal and vertical directions respectively.

n4=0x40: watermark brightness is 0x40.

n1=0x01: select the Flash bitmap whose number is 1 as watermark bitmap.

GS { w n

[Function] Enter\Exit watermark mode.

[Format]

ASCII	GS	{	w	n
Hex	1D	7B	77	n
Decimal	29	123	119	n

[Range] 0 ≤ n ≤ 1

[Notes]

- n = 0: exit watermark mode.
- n = 1: enter watermark mode.
- This command is enabled only when processed at the beginning of the line.
- Before using this command, set watermark parameters using watermark setting command first.
- Printer returns to normal print mode after using this command to exit watermark mode.

GS . B E . . 3 .

[Function] Set parameters of watermark bitmap, enter watermark mode.

[Format]

ASCII	GS	.	B	E	.	.	E	3	..	n1	n2	n3	n4	n5	n6	n7
Hex	1D	99	42	45	92	9a	45	30	02	n1	n2	n3	n4	n5	n6	n7
Decimal	29	153	66	69	146	154	69	48	2	n1	n2	n3	n4	n5	n6	n7

[Range]

0 ≤ n1 ≤ 1

0 ≤ n2 ≤ 2

0 ≤ n3≤ 1

1 ≤ n4≤ 255

0 ≤ n5 ≤ 255

1 ≤ n6 ≤ 255

1 ≤ n7 ≤ 255

[Notes]

- n1 indicates watermark print mode:

  n1 = 0: print watermark bitmap when feeding paper

  n1 = 1: print watermark bitmap when printing

- n2 indicates align mode of watermark printing

  n2 = 0: left justification

n2 = 1: centering

n2 = 2: right justification

- n3 indicates upside down print mode.
  n3 = 0: disable upside-down.
  n3 = 1: enable upside-down.
- n4 indicates enlargement mode selection of watermark printing. 0-3 bits select height, and 4-7 bits select width, the values of which are as follows:

Width			Height		
Hex	**Decimal**	**Enlarge in landscape**	**Hex**	**Decimal**	**Enlarge in vertical**
10	16	1(normal)	01	1	1 (normal)
20	32	2 (double-width)	02	2	2 (double -height)
30	48	3	03	3	3
40	64	4	04	4	4
…	…	…	…	…	…
F0	240	15	0F	15	15

- n5 indicates watermark brightness adjustment, and the recommended value is 0x30.
- n6 indicates bitmap number (defined by command **FS q**).
- n7 indicates watermark number, and the obverse/reverse side can select different watermark numbers.
- Define Flash bitmap using command **FS q** first before using this command.

GS . B E . . 3

[Function] Enable/Disable obverse side/reverse side watermark print mode.

[Format] ASCII GS . B E . . E 3 n1 n2 n3

Hex 1D 99 42 45 92 9a 45 30 n1 n2 n3

Decimal 29 153 66 69 146 154 69 48 n1 n2 n3

[Range] 0 ≤ n1 ≤ 1

1 ≤ n2≤ 255

1 ≤ n3 ≤ 255

[Notes]

- n1 indicates watermark printing:
  n1 = 0: indicates watermark printing is disabled
  n1 = 1: indicates watermark printing is enabled
- n2 indicates watermark number. (Note: This watermark number should be corresponding with the watermark number set.)
- n3 indicates side number
  n3 = 0: indicates to print on obverse side.
  n3 = 1: indicates to print on reverse side.
- This command is enabled only when processed at the beginning of the line.

- Before using this command, set watermark parameters using watermark setting command first.
- Printer returns to normal print mode after using this command to exit watermark mode.

[Example] 1F FE 00 00 01(cancel the watermark print 01 on obverse side)

2.10 Nesting print command

US STX STX n1 n2 n3 n4 n5

[Function] Set bitmap parameters, and enter nesting print mode.

[Format] ASCII GS . B E . . E 0 .. n1 n2 n3 n4 n5

Hex 1D 99 42 45 92 9a 45 31 02 n1 n2 n3 n4 n5

Decimal 29 153 66 69 146 154 69 49 2 n1 n2 n3 n4 n5

[Range] 0 ≤ n1 ≤ 1

0 ≤ n2 ≤ 2

0 ≤ n3≤ 1

1 ≤ n4≤ 255

0 ≤ n5 ≤ 255

1 ≤ n6 ≤ 255

1 ≤ n7 ≤ 255

[Notes]

- n1 indicates nesting print mode:
  n1= 0: left justification
  n1 = 1: centering
  n1 = 2: right justification
- n2 indicates upside-down mode:
  n2= 0: upside-down mode is disabled
  n2 = 1: upside-down mode is enabled
- n3 indicates enlargement mode selection of nesting print. 0-3 bits select height, and 4-7 bits select width, the values of which are as follows:

Width			Height		
Hex	Decimal	Enlarge in landscape	Hex	Decimal	Enlarge in vertical
10	16	1(normal)	01	1	1 (normal)
20	32	2(double-width)	02	2	2(double-height)
30	48	3	03	3	3
40	64	4	04	4	4
…	…	…	…	…	…
F0	240	15	0F	15	15

- n4 indicates bitmap number (defined by command **FS q**).
- n5 indicates nesting print number, and the obverse/reverse side can select different nesting print numbers.

- Define Flash bitmap using command **FS q** first before using this command.

US n1 n2 n3

[Function] Enable/Disable obverse side/reverse side nesting print mode.

[Format]

ASCII	GS	.	B	E	.	.	E	1	n1	n2	n3
Hex	1D	99	42	45	92	9a	45	31	n1	n2	n3
Decimal	29	153	66	69	146	154	69	49	n1	n2	n3

[Range] $0 \le n1 \le 1$

$1 \le n2 \le 255$

$1 \le n3 \le 255$

[Notes]

- n1 indicates nesting print:
  n1 = 0: nesting print is disabled.
  n1 = 1: nesting print is enabled.
- n2 indicates nesting print number.
- n3 indicates side number.
  n3 = 0: nesting print on obverse side.
  n3 = 1: nesting print on reverse side.
- This command is enabled only when processed at the beginning of the line.
- Before using this command, set nesting print parameters using nesting print setting command first.
- Printer returns to normal print mode after using this command to exit nesting print mode.

[Example] 1F FE 02 00 00 01(Cancel nesting print 01 on obverse side)

2.11 Gray scale command

FS r n xl xh yl yh zl zh d1 d2 d3...d(k)

[Function] Download Flash gray scale bitmap.

[Format]

ASCII	FS	r	n	xL	xH	yL	yH	zL	zH	d1	d2	d3 ...d(k)
Hex	1C	72	n	xL	xH	yL	yH	zL	zH	d1	d2	d3 ...d(k)
Decimal	28	114	n	xL	xH	yL	yH	zL	zH	d1	d2	d3 ...d(k)

[Range] $1 \le n \le 255$

xL = 1,xH = 0

$1 \le (yL + yH \times 256) \le 65536$

$1 \le (zL + zH \times 256) \le 8190$

$0 \le d \le 255$

$k = (yL + yH \times 256) \times (zL + zH \times 256) \times 8$

[Notes]

- The maximum capacity of Flash depends on printer configuration. The Flash download capacity supported by printer can be checked by printing self-test page. The size of Flash bitmap should not exceed the maximum value of FLASH download capacity; otherwise it will cause download failure.

- n specifies the number of the Flash bitmaps.
- yL, yH specify the number of dots in the horizontal direction for Flash bitmap (yL + yH × 256) × 8.
- zL, zH specify the number of dots in the vertical direction for the Flash bitmap (zL + zH × 256) × 8.
- This command is disabled in upside-down mode.
- Frequent execution of this command may cause damage to the Flash memory. Therefore, it is recommended to write the Flash 10 times or less a day.
- This command cancels all Flash bitmaps that have already been defined by this command. The printer can not redefine one of the several bitmaps previously defined, and in this case, all data needs to be sent again.
- Because printer is in busy status when processing this command, it writes data to Flash and stops receiving other commands. Thus, when executing this command, sending other commands including immediate commands is forbidden.
- Flash gray scale bitmap is the bitmap which is defined by **FS r**, stored in a Flash memory and printed by command **FS p.**
- In standard mode, this command is enabled only when processed at the beginning of the line.
- The 9 bytes <from FS~ zH> are processed as command data, not data of graphics.
- In the first group of Flash bitmaps, when any one parameter of yL, yH, zL, zH exceeds the defined range, this command is disabled.
- When downloading several bitmaps, if the printer processes yL, yH, zL, zH out of the defined range, it stops processing this command. At this time, bitmaps that haven't been defined are disabled (undefined), but bitmaps before that are enabled.
- d indicates the defined bitmap data. Set a corresponding bit to be 1 for printing a dot, or to be 0 for not printing a dot.
- This command defines n as the number of Flash bitmaps. Numbers rise in order from Flash bitmap 1. Therefore, the first data group [yL yH zL zH d1...dk] is Flash bitmap 1, and the last data group [yLyH zL zHd1...dk] is Flash bitmap n. It is the same when printing bitmaps using command **FS p**.
- Define a Flash bitmap consisting of [yL yH zL zH d1...dk]. Therefore, when only one Flash bitmap, n=1. The printer uses number of bytes of Flash memory as follows: ([byte number of bitmap data: (yL + yH × 256)× (zL+ zH × 256)× 8] + [header: 4]).
- When processing this command, printer does not process other commands.
- Once a Flash bitmap is defined, it can not be erased by executing **ESC @**, reset, and power off commands.
- Printer initialization is performed automatically after bitmap download (1b 40).
- This command only defines Flash bitmap and does not execute printing. Flash bitmap printing is executed by the **FS p** command.

- Format of the gray scale bitmap: every dot line of gray scale image is indicated by four dot lines of data. The four dot lines of data form different levels of gray scale bitmaps. The table below shows the gray scale level of a dot and the data of this dot in the four dot lines of data. The corresponding relation is as below:

Actual gray scale level	Data of the first dot line	Data of the second dot line	Data of the third dot line	Data of the fourth dot line
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

[Relative] FS p

ESC c 6 n yl yh zl zh d1 d2 d3 ...d(k)

[Function] Download RAM gray scale bitmap.

[Format]

ASCII	ESC	c	6	n	yl	yh	zl	zh	d1	d2	d3 ...d(k)
Hex	1B	63	36	n	yl	yh	zl	zh	d1	d2	d3 ...d(k)
Decimal	27	9	54	n	yl	yh	zl	zh	d1	d2	d3 ...d(k)

[Range]
$0 \le n \le 7$
$0 \le d \le 255$
$(yL + yJ \times 256) > 0$
$(zL+ zJ \times 256) > 0$
$k = (yL + yH \times 256) \times (zL+ zH \times 256) \times 8$
$k > 0$

[Notes]

- n specifies the bitmap number of RAM gray scale bitmap.
- yL, yH specify the number of dots in the horizontal direction for the Flash bitmap $(yL + yH \times 256) \times 8$.
- zL, zH specify the number of dots in the vertical direction for the Flash bitmap $(zL + zH \times 256) \times 8$.
- This command is disabled in upside-down mode.

- If the related parameter of this command exceeds the defined range, this command is disabled.
- If the RAM gray scale bitmap defined by this command exceeds the maximum download capacity of RAM 128kB, this command is disabled.
- The downloaded RAM gray scale bitmap is erased when the printer is turned off.
- Format of the gray scale bitmap: every dot line of gray scale image is indicated by four dot lines of data. The four dot lines of data form different levels of the gray scale bitmaps. The table below shows the gray scale level of a dot and the data of the dot in the four dot lines of data. The corresponding relation is as below:

Actual gray scale level	Data of the first dot line	Data of the second dot line	Data of the third dot line	Data of the fourth dot line
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

ESC c 7 n

[Function] Print downloaded RAM gray scale bitmap, and set print mode.

[Format]

ASCII	ESC	c	7	n
Hex	1B	63	37	n
Decimal	27	99	55	n

[Range] 0 ≤ n ≤ 255

(0 ≤ print mode ≤3, 0 ≤ bitmap number ≤ 7)

[Notes]

- This command is ignored if the downloaded bitmap has not been defined.
- This command is disabled in upside-down print mode.
- If the downloaded bitmap exceeds the print area, the excess data will not be printed.

- This command prints bitmaps downloaded in RAM, but not bitmaps downloaded in Flash, and the corresponding bitmap number is specified by **ESC c**.

Print mode selection			Bitmap number selection		
Hex	**Decimal**	**Mode**	**Hex**	**Decimal**	**Bitmap number**
00	0	Normal	00~07	00~07	00~07
10	16	Double-width			
20	32	Double-height			
30	48	Quadruple			

[Example] 1B 63 37 01 (select bitmap 1, normal mode)
1B 63 37 02 (select bitmap 2, normal mode)
1B 63 37 11 (select bitmap 1, double-width mode)
1B 63 37 21 (select bitmap 1, double-height mode)
1B 63 37 31 (select bitmap 1, quadruple mode)

2.12 Paper control command

ESC c : n

[Function] Choose paper-saving mode and reduce ticket width.

[Format]

ASCII	ESC	c	:	n
Hex	1B	63	3A	n
Decimal	27	99	58	n

[Range] 0 ≤ n ≤ 4

[Notes]

- Paper-saving function refers to vertical compression according to a proportionality factor set by user to reach the goal of saving paper.
- The command only refers to vertical compression.
- The command only works on compressible space. Compressible space includes: space between print data (except space caused by space characters); 1D barcode (The minimal height 1D barcode can be compressed is 30 dots).
- The command compresses compressible space according to a certain proportionality factor, which is set as follows:

m	Proportionality factor setting
0	No compression
1	Compress 25%
2	Compress 50%
3	Compress 75%
4	Compress 100%

- The command only works on the ticket sending this command.
- This command is enabled only in standard mode.

[Default] n = 0

2.13 Double-side relevant command

Function of double-side relevant command:

Printer can be set to the following four modes:

1) Single-side printing mode;
2) Double-side printing mode with single-side command. When the printer receives the cut command, it will automatically divide the print data into two parts, the first part will be printed on the obverse side and the other part will be printed on the reverse side.
3) Double-side printing mode with double-side command. This command will specify the print data to be printed on obverse side or reverse side. The printer will start printing after receiving the cut command or start printing command.
4) Double-side printing mode with pre-defined data on reverse side command. Printer prints the real-time received data on the obverse side and prints the pre-defined data on the reverse side. Printer starts printing after receiving cut command.

US ‘ n

[Function] Select single-side or double-side printing mode.

[Format]

ASCII	US	‘	n
Hex	1F	60	n
Decimal	31	96	n

[Range] 0 ≤n ≤ 3, n = 6

[Notes] The meaning of n is as follows:

N	Printing mode
0	Single-side printing mode, printing on the obverse side.
1	Double-side printing mode with single-side command.
2	Double-side printing mode with double-side command.
3	Double-side printing mode with pre-defined data.
6	Single-side printing mode, printing on the reverse side.

- If two-side printing mode is selected, the printing contents can be printed on the obverse side or reverse side.
- When n = 1, the printing contents will be automatically divided into two parts, one part is printed on the obverse side, and the other part will be printed on the reverse side. The command 1F 61 (choose obverse side / reverse side), 1F 62 (double-side printing) are ignored.
- When n = 2, printing contents will be selected to print on obverse side or reverse side via command 1F 61; printing can be started via sending command 1F 62.
- When n = 3, the received data will be printed on the obverse side, and the pre-defined data will be printed on the reverse side; the pre-defined data printed on the riverse side will be printed only one time on each ticket (cut command is the ticket end command); the ticket length is the length of longer side.

- When n = 0/6, printer will exit double-side printing mode, if there are printing contents in current buffer, the printer will print the contents in buffer and then switch to single-side printing mode.
- When n = 0/6, if the printing position is not at the beginning of line, printer will automatically change the line and switch the printing position to the beginning of line after receiving this command.

[Default] Depending on the printer pre-settings.

[Relative] **US a,US b,**

US a n

[Function] Select thermal printing side (obverse side/reverse side).

[Format] ASCII US a n

Hex 1F 61 n

Decimal 31 97 n

[Range] 0 ,1

[Notes] The meaning of n is as following:

n	Printing mode
0	Select obverse side.
1	Select reverse side.

- This command is only enabled under double-side printing mode with double-side command.
- This command is only enabled when processed at the beginning of the line.
- If either side is larger than the buffer size, printer does print the exceeding parts.

[Default] n = 0

[Relative] **US '**

US b

[Function] Start double-side printing.

[Format] ASCII US b

Hex 1F 62

Decimal 31 98

[Notes] This command is only enabled under double-side printing mode with double-side command, or it will be ignored.

[Relative] **US '**

US c n

[Function] Select or cancel upside-down printing under double-side mode.

[Format] ASCII US c n

Hex 1F 63 n

Decimal 31 99 n

[Range] bit0, bit1

[Notes] The meaning of n is as follows:

Bit	n	Printing mode
bit0	0	Cancel upside-down printing on obverse side.
	1	Select upside-down printing on obverse side.
Bit1	0	Cancel upside-down printing on reverse side.
	1	Select upside-down printing on obverse side.

- This command is only enabled under double-side printing mode.
- The last settings before printing are effective.

[Default] n = 0

[Relative] **US '**

US d n

[Function] Swap the obverse side and reverse side.

[Format] ASCII US d n

Hex 1F 64 n

Decimal 31 100 n

[Range] 0,1

[Notes] The meaning of n is as follows:

n	Printing mode
0	Swap function is disabled.
1	Swap function is enabled.

- After swaping, the paper side, originally should be printed by print head that on obverse side, will be printed by print head on reverse side.
- The last settings before printing are effective.
- This command is only enabled under double side mode.

[Default] n = 0

[Relative] **US '**

US e n k1 d1 d2 … di NUL

[Function] Download one-line/two-line top/bottom message to FLASH memory.

[Format] ASCII US e n k1 d1 d2 … di NUL

Hex 1F 65 n k1 d1 d2 … di 0

Decimal 31 101 n k1 d1 d2 … di 0

[Range] n=0,1,2,3

i <= 60, the parts exceeding one line not be printed.

[Notes]

- Printer download one-line pre-defined message to FLASH memory.
- This command is used in double-side printing mode, user can choose printer to automatically add one-line/two-line message at the end of obverse side/beginning of the reverse side.
- The obverse side/reverse side mentioned in this command is logic side (logic side can be changed via page swap), not physics side (physics side will not be changed by swap command).

- Number of message n:

N	Printing mode
0	Line 1 of message on obverse side.
1	Line 2 of message on obverse side.
2	Line 1 of message on reverse side.
3	Line 2 of message on reverse side.

- Attribute of K1 message:

N	Value	Printing mode
Bit7	0	Cancel italic.
	1	Set italic.
Bit6	0	Cancel white/black reverse.
	1	Set white/black reverse.
Bit5	0	Meaningless.
	1	Meaningless.
Bit4	0	Cancel bold.
	1	Set bold.
Bit3	0	Cancel double width.
	1	Set double width.
Bit2	0	Cancel double height.
	1	Set double height.
Bit0, Bit1	11	Meaningless.
	00	Cancel underline.
	01	Single-dot line for underline.
	10	Double-dot line for underline.

- Only one line of characters can be defined every time when this command is sent, the parts exceeding one line will be ingored.
- If US e n k NUL is sent, the messange in line n will be cleared.
- The character in one line can only use the same attribute settings.

[Default] n = 0

US f n

[Function] Enable top/bottom message.

[Format]

ASCII	US	f	n
Hex	1F	66	n
Decimal	31	102	n

[Range] Bit0~Bit2

[Notes] The meaning of n is as follows:

Bit	Value	Printing mode
Bit0	0	Disable pre-defined bottom message on obverse side.
	1	Enable pre-defined bottom message on obverse side.
Bit1	0	Disable pre-defined top message on reverse side.
	1	Enable pre-defined top message on reverse side.

- If this command is enabled, printer will automatically add one-line/two-line message on the bottom/top of obverse side/reverse side.
- This command is only enabled under double-side mode.

[Default] n = 0

US g n

[Function] Select the n^{th} macro for definition/performance.

[Format]

ASCII	US	g	n
Hex	1F	67	n
Decimal	31	103	n

[Range] 1 ≤n ≤ 25

[Notes]

- The size of macro definition data storage buffer is 50 * 2048 bytes.
- Printer does not detect the validity of macro defined data during the macro definition.
- The macro definitions can be nested, but the nesting depth can only be one macro.

[Default] n = 1

[Example] 1F 67 03 (select the macro ID to be 3)

1D 3A(start to define macro data)

1F 67 01 (select the macro ID to be 1, nest macro)

1D 5E(perform selected macro)

1D 3A(end the definition of macro data)

US h

[Function] Start or end pre-defined data on reverse side.

[Format]

ASCII	US	h
Hex	1F	68
Decimal	31	103

[Notes]

- Start or end pre-defined data on reverse side. And save the pre-defined data to FLASH memory.
- This command is used to start and end the definition of pre-defined data on reverse side.
- The pre-defined data on reverse side is cleared when there isn't defined data (that is, there are two continuous 1F 68).

- Printer will not detect the data validity during the definition of pre-defined data on reverse side.

[Relative] **US ‘**

US i n1 n2

[Function] Define the minimum length for automatically switching to reverse side.

[Format]

ASCII	US	i	n1	n2
Hex	1F	69	n1	n2
Decimal	31	105	n1	n2

[Range] n=n1+n2*256, 0<= n <= 65535

[Notes]

- This command is only enabled under double-side printing mode with single-side command. Under double-side printing mode with single-side command, printer will switch the printing contents to the reverse side if the printing contents exceed a certain length.
- The unit of height is dot, height = n1 + n2 * 256 (dot).
- If printing content height exceeds the print height, the printer will change the next character line to the reverse side. For example, if the character height in line 7 exceeds the printing height during printing line 7, printer will not divide the characters in line 7, but switch the characters at the beginning of line 8 to the reverse side; but if printer can not find the character line for switching in the range of 204 (dot) upward or downward basing on the calculation of newline, printer will divide the character compellingly (printing contents may be divided).

[Default] n = 0

[Relevant command] **US ‘**

US j n [m, o]

[Function] Print variable character strings.

[Format]

ASCII	US	j	n	[m	o]
Hex	1F	6A	n	[m	o]
Decimal	31	106	n	[m	o]

[Range] n: 1 ~ 50(character string ID); when n=0:[m o]

[m o]: Only present when n=0

m: 1 ~ 50(character string ID)

o: Fixed length(1 ~ 57)

[Notes]

- Printer will print the character string with specified ID stored in RAM.
- This command is disabled if character string is not defined,
- If the number of character string with ID is larger than o, then the printer will only print o character strings.

- If the number of character string with ID is smaller than o, space will be automatically supplemented until the number of character reaches o.

[Default] n = 1

[Relevant command] **US ‘**

US k n d1 d2 … di NUL

[Function] Define variable character strings.

[Format]

ASCII	US	k	n	d1	d2	…	di	0
Hex	1F	6B	n	d1	d2	…	di	0
Decimal	31	107	n	d1	d2	…	di	0

[Range] n: 1 ~ 50 (character string ID)
d1~di: Contents of character string, 32≤di≤255

[Notes]

- Define the character string with ID is n.
- Maximum length of each character string is 57 bytes. The character string will be processed as 57 bytes if the byte number exceeds 57, and the data out of 57 bytes will be processed as normal data.
- This character string can be performed in marco definition and pre-defined data on reverse side.
- This character string is stored in RAM and will be lost after turning on the power again.
- The contents of character string can only be characters, and the contained control commands in character string will be ingored.

[Default] n = 1

[Relevant command] **US ‘**

US l n

[Function] Return back to current print mode.

[Format]

ASCII	US	l	n
Hex	1F	6C	n
Decimal	31	108	n

[Range] n=1 query printer mode.

[Notes]

- Definition of printer mode:

Bit	Bit value	Decimal	Function
1,0	00	0	Single-side printing mode.
	01	1	Double-side printing mode with single-side command.
	10	2	Double-side printing mode with double-side command.
	11	3	Double-side printing mode with pre-defined on reverse side command.
2	0	0	Undefined, fixed to 0.

3	0	0	Select obverse side (only valid under double-side printing mode with double-side command)
	1	8	Select reverse side (only valid under double-side printing mode with double-side command)
4, 5	00	0	The paper detection is not finished.
	01	16	The paper detection result matches the set 2ST mode.
	10	32	The paper detection result does not match the set 2ST mode.
	11	48	Undefined.
6	0	0	Undefined, fixed to 0.
7	0	0	Undefined, fixed to 0.

- This command is not a real-time command, printer will execute this command only after executing all the commands before it.

[Default] n = 1

[Relevant command] **US ‘**

2.14 Presenter control command

ESC c 8 n

[Function] Select PRST action mode 1.

[Format]

ASCII	ESC	c	8	n
Hex	1B	63	38	n
Decimal	27	99	56	n

[Range] 0 ≤ n ≤ 3

[Description] Select PRESENTER action mode specified by n as follows:

n	PRST action mode
0	Retraction mode
1	Ejection mode(special model)
2	PRST holding waiting mode
3	PRST disabled mode
4	Throw-out mode

[Notes] Printer operates after receiving command, and then if PRST is enabled, proceeds as follows:

- During printing, the PRST starts and hold the paper head. The PRST will accommodate the printout and hold the end of paper after the printing is finished. After the waiting time, the PRST operates as follows:
  1) In retraction mode, retract the printout into a waste-bin, after reach a setting time. Its main purpose: user doesn’t take it away in specified time, and avoid pollution and keep secrecy for customers.
  2) In ejection mode, discharge the printout after reach a setting time. Its main purpose: If user doesn’t take paper away in specified time, it discharges for next printout
  3) In waiting mode, wait until the user takes the printout away after a setting time.

4) In throw-out mode, no waiting time, and the PRST advances and ejects the printout.

- If the PRST is disabled, it takes no actions during printing and after printing. It is recommended to remove the PRST to prevent from paper jam if the PRST function is not desired.
- **ESC c 9** can set the waiting time before action of PRST. This command is valid only in PRST mode 1.

[Relative] **ESC c 9**

ESC c 9 t

[Function] Set PRST waiting time.

[Format]

ASCII	ESC	c	9	t
Hex	1B	63	39	t
Decimal	27	99	57	t

[Range] 0 ≤ t ≤ 240

[Description] Set the time period (t in seconds) for PRST to wait before it takes any actions set by **ESC c 8**.

[Notes] Actions here mean the PRST action modes specified in **ESC c 8** command.

This command is enabled only under Presenter mode 1 and no throwing out paper and waiting mode.

[Relative] **ESC c 8**

ESC c = m n

[Function] Set thick-paper mode

[Format]

ASCII	ESC	c	=	m
Hex	1B	63	3d	m
Decimal	27	99	61	m

[Range] 0 ≤m≤1

[Description] m=0: disable thick-paper mode, m=1: enable thick-paper mode. m>1: paper ejection distance after cutting paper. The printer does not accommodate paper under this mode.

[Notes]

- Thick-paper mode is recommended when the weight of the paper used exceeds 90g; otherwise paper may not be ejected normally.
- The printer will eject paper directly if m is too large.
- The function will recover to the power-on default configuration if sending command 1b 40 after changing EEPROM.

[Relative] **ESC c 8**

ESC c I

[Function] PRESENTER retract/throw out paper immediately.

[Format]

ASCII	ESC	c	I
Hex	1B	63	49
Decimal	27	99	73

[Description] In retraction mode, send this command to retract the paper immediately; in ejection mode, send this command to eject the paper immediately.

[Notes] This command does not change the waiting time of PRESENTER.

[Relative] **ESC c 8**

GS e n

[Function] Control Presenter(effective in Presenter control mode 2).

[Format]

ASCII	GS	e	n	m	t
Hex	1D	65	n	m	t
Decimal	29	101	n	m	t

[Range] 2 ≤ n ≤ 3, n=5, n=6, n =8, n=18, n=20, n=32; 0 ≤ t ≤ 255

[Description]

n = 2 Execute a ticket retract (only if paper retracting is enabled)

n = 3 Produce a ticket with m steps (1 step = 7.3 mm)

n = 5 Eject ticket

n = 6 Transmit the status byte of the ejector

Bit	Off/On	Hex	Decimal	Function
0	Off	00	0	Paper present in abundance
	On	01	01	Near paper end
1	Off	00	0	Reserved
2	Off	00	0	Paper normal
	On	04	04	Paper end
3	Off	00	0	Ticket not present on the output
	On	08	08	Ticket present on the output
4	Off	00	0	Printer's stepper motor off
	On	10	16	Printer's stepper motor on
5	Off	00	0	Emitter motor off
	On	20	32	Emitter motor on
6	Off	00	0	No error
	On	40	64	Error
7	Off	00	0	Free paper route
	On	80	128	Paper jam

n = 8 Set the length of ticket dispense

n = 18 Disable the dispenser continuous mode: the printed ticket remains in the paper outlet mouth, until a cut command or eject command will be sent.

n = 20 Enable the dispenser continuous mode: the printed ticket doesn't remain in the paper outlet mouth, but be continuously presented.

n = 32 Produce a ticket with m steps (1 step = 7.3 mm) and a timeout t (sec)

[Notes]

- m must be sent with n = 3, n = 8 and n = 32;
- With n=2, n = 5: in IST mode: this command is only effective when there is paper at paper outlet mouth; in 2ST mode, this command is only effective when printer buffer has data. If the ticket has not been cut, the printer will make a full cut before performing the command.
- With n=3, n=32: in IST mode: this command is only effective when printer already printed or fed paper; in 2ST mode, this command is only effective when printer buffer has data. If the ticket has not been cut, the printer will make a full cut before performing the command.
- With n=3, n=8, n=32: the printer executes a check of the ticket produced length: if the m input has a too high value automatically the ticket produced is ejected with the maximum length allowed.
- With n=32: it's necessary to set a timeout that indicate how long the ticket remain presented; when timeout occurs the printer executes a ticket retract or eject in according to printer setup settings.
- This command is enabled only in Presenter mode 2.

[Relative] **ESC I** **FF**

2.15 Other commands

ESC 2

[Function] Select default line spacing.

[Format]

ASCII	ESC	2	
Hex	1B	32	
Decimal	27	50	

[Description] Select 1/6-inch line (approximately 4.23 mm) spacing.

[Notes]

- The line spacing can be set independently in standard mode and in page mode.

[Relative] **ESC 3**

ESC 3 n

[Function] Set line spacing.

[Format]

ASCII	ESC	3	n
Hex	1B	33	n
Decimal	27	51	n

[Range] $0 \le n \le 255$

[Description] Sets the line spacing to [n × vertical or horizontal motion unit] inches.

[Notes]

- The line spacing can be set independently in standard mode and in page mode.
- The horizontal and vertical motion units are specified by **GS P**. Changing the horizontal or vertical motion unit does not affect the current line spacing.
- In standard mode, the vertical motion unit (y) is used.

- In page mode, this command functions as follows, depending on the starting position and direction of the printable area:

1) When the starting position is set to the upper left or lower right of the printable area with **ESC T**, the vertical motion unit (y) is used.
2) When the starting position is set to the upper right or lower left of the print able area with **ESC T**, the horizontal motion unit (x) is used.

The maximum paper feed amount is 600 mm (23.6 inches) for 300 DPI printer and 900mm (35.4 inches) for 203 DPI printer. If a paper feed amount of more than 600 mm (300DPI) or 900mm (203DPI) is set, it is converted to the maximum automatically.

[Default] Line spacing is equivalent to approximately 4.23mm (1/6 inches).

[Relative] **ESC 2**, **GS P**

ESC = n

[Function] Disable or enable peripheral device.

[Format]

ASCII	ESC	=	n
Hex	1B	3D	n
Decimal	27	61	n

[Range] 0 ≤ n ≤ 255

[Description] Disable or enable peripheral device to which host computer sends data.

[Notes]

- The lowest bit of n is effective;
- When the LSB of n is 0, printer is disabled;
- When the LSB of n is 1, printer is enabled'
- When the printer is enabled when power on

When the printer is disabled, it ignores all data except for real-time command **(DLE EOT)** until it is enabled by this command.

[Default] n = 1

ESC @

[Function] Initialize printer.

[Format]

ASCII	ESC	@
Hex	1B	40
Decimal	27	64

[Description]

Clear the data in the print buffer and resets the printer mode to the default mode when the power was turned on.

[Notes]

- The bit-image data or user-defined characters in RAM are cleared.
- The macro definition is not cleared.
- The NV bit-image data is not cleared.

ESC L

[Function] Select page mode.

[Format]

	ASCII	ESC	L
	Hex	1B	4C
	Decimal	27	76

[Description] Switch from standard mode to page mode.

[Notes]

- This command is enabled only when processed at the beginning of a line in standard mode.
- This command has no effect in page mode.
- After printing by **FF** is completed or by **ESC S**, the printer returns to standard mode.
- This command sets the position where data is buffered to the position specified by **ESC T** within the printing area defined by **ESC W**.
- This command switches the settings for the following commands to those for page mode:

1) Set right-side character spacing: **ESC SP**
2) Select line spacing: **ESC 2, ESC 3**

- Only value settings is possible for the following commands in page mode; these commands are not executed until switch to standard mode.

1) Turn 90° clockwise rotation mode on/off: **ESC V**
2) Turn upside-down printing mode on/off: **ESC {**
3) Set left margin: **GS L**
4) Set printable area width: **GS W**

- The printer returns to standard mode, after it is reset, or executes **ESC @**.

[Relative] **FF, CAN, ESC FF, ESC S, ESC T, ESC W, GS $, GS **

ESC S

[Function] Select standard mode.

[Format]

	ASCII	ESC	S
	Hex	1B	53
	Decimal	27	83

[Description] Switch from page mode to standard mode.

[Notes]

- This command is enabled only in page mode.
- Data buffered in page mode are cleared.
- This command sets the print position to the beginning of the line.
- The page mode zone is initialized as default.
- This command switches the settings for the following commands to those for standard mode:

1) Set right-side character spacing: **ESC SP**

2) Select default line spacing: **ESC 2, ESC 3**

- The following commands are enabled only to set in pagemode mode.

1) Set printing area in page mode: **ESC W**

2) Select print direction in page mode: **ESC T**

- The following commands are ignored in standard mode.

1) Set absolute vertical print position in page mode: **GS $**

2) Set relative vertical print position in page mode: **GS **

- Standard mode is selected automatically when the printer is reset, or command **ESC @** is used.

[Relative] **FF, ESC FF, ESC L**

ESC c 0 n

[Function] Select paper type to be used.

[Format]

ASCII	ESC	c	0	n
Hex	1B	63	30	n
Decimal	27	99	48	n

[Range] n =0,1,2

[Description]Select paper type specified by n as follows.

n	Paper type
0	Continuous paper
1,2	Marked paper

[Notes] This command has no effect except n equals 0,1,2.

[Default] n = 0

[Relative] **GS FF**

ESC c 1 nL nH

[Function] Set cutting position

[Format]

ASCII	ESC	c	1	nL	nH
Hex	1B	63	31	nL	nH
Decimal	27	99	49	nL	nH

[Range] 0 ≤ nL ≤ 255; 0 ≤ nH ≤ 255;

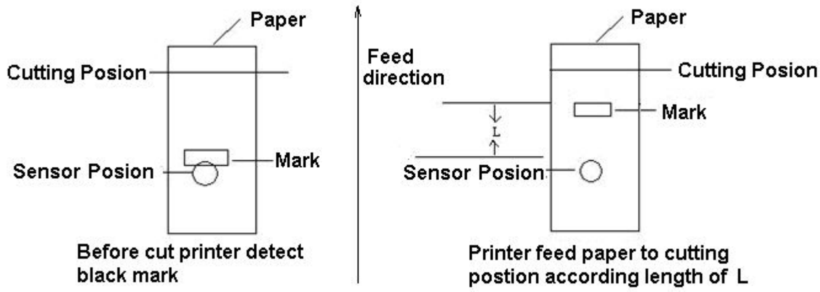
N=(nL + nH x 256)(If n<19,so n = 19)

[Description]Set the cutting position to a place when marked paper is selected.

[Notes]

- When the printer detects the mark (The mark goes to the position of paper sensor.) and feeds paper for a certain distance (Decided by n: for 203DPI model, n/8mm; for 300DPI model, n/12mm), and then starts cutting paper.
- The relationship between N and the cutting position is defined as below.If n increase, the cutting position move backward; If n decrease, the postion move forward : n/8(mm)for 203DPI, n/12(mm)for 300DPI.

- When nL = nH = 255, the N saved to printer will not loss even when printer is turned off/on.
- Refer to the following figure, L is the distance of printer feeding paper corresponding with n.



ESC c 3 n

[Function] Select paper sensor to output paper-end signal.

[Format]

ASCII	ESC	c	3	n
Hex	1B	63	33	n
Decimal	27	99	51	n

[Range] 0 ≤ n ≤ 255

[Description] Select paper sensor to output paper-end signal.

The definition of n bit is as follows:

Bit	0/1	Hex	Decimal	Function
0	0	00	0	undefined
1	0	00	0	Paper-near-end sensor is not available
	1	02	2	Paper-near-end sensor is available
2	0	00	0	undefined
3	0	00	0	Paper end sensor is not available
	1	08	8	Paper end sensor is available
4-7				undefined

[Notes]

- It is possible to select two sensors to output signals. Then, if any of the sensors detects a paper end, the paper end signal is output.
- The command is available only with a parallel interface and is ignored with a serial interface.

[Default] n = 0

ESC c 4 n

[Function] Select paper sensor(s) to stop printing.

[Format] ASCII ESC c 4 n

Hex 1B 63 34 n

Decimal 27 99 52 n

[Range] 0 ≤ n ≤ 255

[Description] Selects the paper sensor(s) used to stop printing when a paper-end is detected, using n as follows:

Bit	0/1	Hex	Decimal	Function
0	0	00	0	Paper roll near end sensor disabled
	1	01	1	Paper roll near end sensor enabled
1	0	00	0	Paper roll near end sensor disabled
	1	02	2	Paper roll near end sensor enabled
2-7				Undefined

[Notes] When either bit 0 or 1 is on, the printer selects the paper roll near-end sensor. When the paper roll sensor detects a paper-end, the printer goes off-line after current printing stops.

[Default] n = 0

ESC c 5 n

[Function] Enable/disable panel buttons.

[Format] ASCII ESC c 5 n

Hex 1B 63 35 n

Decimal 27 99 53 n

[Range] 0 ≤ n ≤ 255

[Description] Enables or disables the panel buttons.

- When the LSB of n is 0, the panel buttons are enabled.
- When the LSB of n is 1, the panel buttons are disabled.

[Notes]

- Only the LSB of n is valid.
- When the panel buttons are disabled, none of them are usable.
- In the macro ready mode, the panel buttons are always enabled.

[Default] n = 0

ESC c @ n

[Function] Enable/disable real-time commands.

[Format] ASCII ESC c @ n

Hex 1B 63 40 n

Decimal 27 99 64 n

[Range] 0 ≤ n ≤ 255

[Description] Enables or disables the real-time commands.

- When the LSB of n is 0, the real-time commands are disabled.
- When the LSB of n is 1, the panel buttons are enabled.

[Notes]

- Only the LSB of n is valid.
- When the printer is turned on, the real-time commands are always enabled.
- When error occurs, the real-time commands are automatically enabled.

[Relative] **DLE EOT n**

ESC i

[Function] Cut paper.

[Format]

ASCII	ESC	i
Hex	1B	69
Decimal	27	105

[Description]Cut paper.

[Notes]

- If the paper feeding length (L1) is less than the minimum page length (Lm) specified when executing this command, the printer feeds the minimum page length and cuts paper. Otherwise, the printer cuts paper directly.
- In marked paper mode, cutting position is specified with marks.
- Under Presenter mode 2, printer does not enter Presenter process after cutting paper; under Presenter mode 1, printer enters Presenter process after cutting paper.

ESC l n X0l X0h Y0l Y0h X1l X1h Y1l Y1h

[Function] Draw a line.

[Format]

ASCII	ESC	l	n	X0l X0h Y0l Y0h X1l X1h Y1l Y1h
Hex	1B	6C	n	X0l X0h Y0l Y0h X1l X1h Y1l Y1h
Decimal	27	108	n	X0l X0h Y0l Y0h X1l X1h Y1l Y1h

[Range] $0 \leq n \leq 255$

[Description] Set coordinates of the starting point and the ending point, and then generate the straight line in the buffer area.

n: Width (dots) of the line.

X0l X0h Y0l Y0h is used to set the starting point coordinate (X0, Y0) of the line, and X1l X1h Y1l Y1h is used to set the ending point coordinate (X1, Y1) of the line.

X0 = X0l + X0h×255, Y0 = Y0l + Y0h×255, X1 = X1l + X1h×255,

Y1 = Y 1l + Y1h×255.

[Notes]

- This command is enabled only in page mode.
- This command can only generate horizontal line or vertical line, and can not generate slanting line. Therefore, when X0 does not equal to X1, Y0 must equal to Y1; also, when Y0 does not equal to Y1, X0 must equal to X1.
- This command is disabled when the starting coordinate and the ending coordinate of the setting line exceed the print area.

ESC v

[Function] Transmit printer status.

[Format]

ASCII	ESC	v
Hex	1B	76
Decimal	27	118

[Description] Transmit the printer status in one-byte data.

[Notes]

- The command is non-real time
- Transmits the status via serial interface and/or USB interface. If printer has both serial and USB interface, data return to serial and USB interface at the same time.

Define returned bytes as follows:

Bit	On/Off	Hex	Device Status
0	0/1	00/01	Paper adequate/Paper near end
1	0/1	00/02	Printhead close/Printhead open
2	0/1	00/04	Paper adequate/Paper end
3	0/1	00/08	Printhead normal/overheat
4	0/1	00/10	Cutter normal/Cutter error

GS FF

[Function] Feed mark to print position.

[Format]

ASCII	GS	FF
Hex	1D	0C
Decimal	29	12

[Description] Feed paper until the next black mark reaches print position.

[Notes]

- This command is enabled only when the paper type is marked paper.
- This command sets the next print position to the beginning of the line.

GS (A pL pH n m

[Function] Execute test print.

[Format]

ASCII	GS	(	A	pL	pH	n	m
Hex	1D	28	41	pL	pH	n	m
Decimal	29	40	65	pL	pH	n	m

[Range] (pL+(pH × 56))=2 (pL=2, pH=0)

$0 \le n \le 2$, $8 \le n \le 50$

$1 \le m \le 3$, $9 \le m \le 51$

[Description]

- Execute the test printing in the mode decided by m.

  m specifies a test pattern.

m	Test pattern
1, 49	Hexadecimal dump
2, 50	Print self-test page

[Notes]

- This command is enabled only when processed at the beginning of a line instandard mode.
- This command is not effective in page mode
- When this command is received during macro definition, the printer ends macro definition and begins performing this command.
- The printer cuts the paper at the end of the test print.

[Example] When m=1, this command is in the hexadecimal dump printing mode. The processing method is as follows:

1) Turn the printer Presenter;
2) Send the command (1d 28 41 02 00 00 01) to enter DUMP mode;

At this moment, the printout is as follows:

Hexadecimal Dump

To terminate hexadecimal dump,
Press FEED button three times

3) Send print data:
 30 31 32 33 34 35 36 37
 31 32 33 34 35 36 37 38
 41 42 43 44 45 46 47 48
 42 43 44 45 46 47 48 49
4) After finishing the printing, press the FEED button three times and retract from DUMP mode. It will show the following information:
 completed

Results:


Hexadecimal Dump

To terminate hexadecimal dump,
Presss FEED button three times

30 31 32 33 34 35 36 37 01234567
31 32 33 34 35 36 37 39 12345679
41 42 43 44 45 46 47 48 ABCDEFGH
42 43 44 45 46 47 48 49 BCDEFGHI

completed

GS C 0 n m

[Function] Select counter print mode.

[Format]

ASCII	GS	C	0	n	m
Hex	1D	43	30	n	m
Decimal	29	67	48	n	m

[Range] $0 \le n \le 5$

$0 \le m \le 2$, $48 \le m \le 50$

[Description] Select a print mode for the serial number counter

n specifies the number of digitals to be printed as follows:

When n = 0, the printer prints the actual digits.

When n = 1 to 5, the printer prints the actual digits indicated by the number value.

m specifies the printing position within the entire range of printer digits as follow:

m	Printing position	Processing of digits less than those specified
0,48	Align right	Adds spaces to the left
1,49	Align right	Adds 0 to the left
2,50	Align left	Adds spaces to the right

[Notes]

- If n or m out of the defined range, the previously set print mode is not changed.
- If n=0, modes not have any meaning.

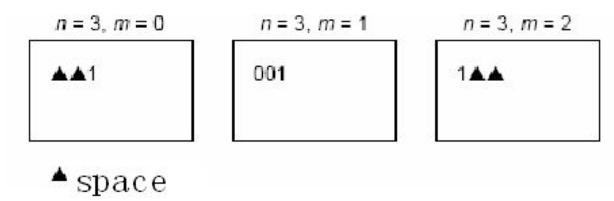
[Default] n = 0, m = 0

[Relative] **GS C 1, GS C 2, GS C ;, GS c**

[Example]

1d 43 30 03 00 (Select count print mode)
1d 43 31 01 00 09 00 01 01 (Select count mode A)
1d 63 (Enter count mode and print)
0a

Send the above sample, and print a number after sending the sample each time. The counter number circles from 1 to 9. Change the parameters of this command, and the format of the counter number is as shown in the following figure.



GS C 1 aL aH bL bH n r

[Function] Select count mode (A).

[Format]

ASCII	GS	C	1	aL	aH	bL	bH	n	r
Hex	1D	43	31	aL	aH	bL	bH	n	r
Decimal	29	67	49	aL	aH	bL	bH	n	r

[Range] 0 ≤ aL ≤ 255

0 ≤ aH ≤ 255

0 ≤ bL ≤ 255

0 ≤ bH ≤ 255

0 ≤ n ≤ 255

0 ≤ r ≤ 255

[Description] Select a count mode for the serial number counter.

- aL, aH, bL, bH specify the counter range.
- n indicates the stepping amount when counting up or down.
- r indicates the repetition number when the counter value is fixed.

[Notes]

- Count-up mode is specified when
  [aL + aH × 256] < [bL + bH × 256], n 0 and r 0.
- Count-down mode is specified when
  [aL + aH × 256] > [bL + bH × 256], n 0 and r 0.
- Count stop when [aL + aH × 256] = [bL + bH × 256], n = 0 or r = 0.
- If setting count-up mode, the minimum value of the counter is [aL + aH × 256], the maximum number is [bL + bH × 256].If counting up reaches a value exceeding the maximum, it is resumed with the minimum value.
- If setting count-down mode, the maximum value of the counter is [aL + aH × 256], the minimum number is [bL + bH × 256].If counting down reaches a value less than the minimum, it is resumed with the maximum value.
- When this command is executed, the internal counter that indicates the repetition number specified by r is cleared.

[Default] aL = 1, aH = 0, bL = 255, bH = 255, n =1, r = 1

[Relative] **GS C 0, GS C 2, GS C ;, GS c**

[Example] Refer to **GS C 0** example

GS C 2 nL nH

[Function] Set the current counter.

[Format]

ASCII	GS	C	2	nL	mH
Hex	1D	43	32	nL	mH
Decimal	29	67	50	nL	mH

[Range] 0 ≤ nL ≤ 255

0 ≤ nH ≤ 255

[Description] Set the serial number counter value.

nL and nH determine the value of the serial number set by [nL + nH ×256].

[Notes]

- In count-up mode, if the counter value specified by this command goes out of the counter operation range specified by **GS C 1** or **GS C**;, it is forced to convert to the minimum value**.**
- In count-down mode, if the counter value specified by this command goes out of the counter operation range specified by **GS C 1** or **GS C**;, it is forced to convert to the maximum value**.**

[Default] nL = 1, nH = 0

[Relative] **GS C 0, GS C 1, GS C ; GS c**

[Example] 1d 43 32 03 00 (Set the current counter as 3)

1d 63 (Enter count mode and print)

0a

Send the above sample, and the counter is 3 after every printing.

GS C : sa ; sb ; sn ; sr ; sc ;

[Function] Select count mode (B).

[Format]

ASCII	GS	C	;	sa	;	sb	;	sn	;	sr	;	sc	;
Hex	1D	43	3B	sa	3B	sb	3B	sn	3B	sr	3B	sc	3B
Decimal	29	67	59	sa	59	sb	59	sn	59	sr	59	sc	59

[Range] “0” ≤ sa ≤ “65535”

“0” ≤ sb ≤ “65535”

“0” ≤ sn “255”

“0” ≤sr ≤“255”

“0” ≤sc ≤“255”

These values are all character strings.

[Description] Select a count mode for the serial number counter.

- sa, sb, sn, sr and scare all displayed in ASCII characters, using the codes for "0" to "9".
- sa and sb set the counter range.
- sn indicates the stepping amount of counting up or down.
- sr indicates the repetition number with the counter value fixed.
- sc indicates the counter value.

[Notes]

- Count-up mode is specified when: sa < sb ,sn ≠ 0 , sr ≠ 0.
- Count-down mode is specified when: sa > sb, sn ≠ 0, sr ≠ 0.
- Count stop when: sa = sb or sn = 0 or sr = 0.
- When count-up mode is specified,sa is the minimum counter value and sb is the maximum counter value.If the counter value set by sc is outside the counter operation range,the counter value is forced to convert to the minimum value.

- When count-down mode is specified,sa is the maximum counter value and sb is the minimum counter value.If the counter value set by sc is outside the counter operation range,the counter value is forced to convert to the maximum value.
- Parameters sa to sc can be omitted. If omitted, these argument values are unchanged.
- Parameters sa to sc must not contain characters, except 0 to 9.
- If an incorrect syntax is used, the corresponding parameter setting has no effect, and the data after that is processed as normal data.

[Default] sa = 1, sb = 65535, sn = 1, sr = 1, sc = 1

[Relative] **GS C 0, GS C 1, GS C 2, GS c**

[Example] **1:**

1D 43 3B 31 3B 39 3B 31 3B 31 3B 32 3B (Select count mode (B))

1D 63 (Enter count mode and print the current counter)

0A

Send the example 1 sample, and the print counter is always 2;

2:

1D 43 3B 31 3B 39 3B 31 3B 31 3B (Select count mode (B))

1D 63 (Enter count mode and print the current counter)

0A

Send the example 2 sample, and the print counter circles from 1 to 9.

GS c

[Function] Print counter value.

[Format]

ASCII	GS	c
Hex	1D	63
Decimal	29	99

[Description] Store a counter value in print buffer, increase or decrease the count value and print it out.

[Notes]

- In count mode, after setting the current counter value in print buffer, the printer will count according to the settings of the count mode. When the printer receives print command or when the buffer is full, the printer prints the counter value;
- Print count mode is set by command **GS C 0**;
- Count mode is set by command **GS C 1 or GS C;**
- In count-up mode, if the counter value specified by this command goes out of the counter operation range specified by **GS C 1 or GS C**, it is forced to convert to the minimum value by **GS c**.
- In count-down mode, if the counter value specified by this command goes out of the counter operation range specified by **GS C 1 or GS C**, it is forced to convert to the maximum value by **GS c**.

[Relative] **GS C 0, GS C 1, GS C 2, GS C;**

GS I n

[Function] Transmit printer configuration.

[Format]

ASCII	GS	I	n
Hex	1D	49	n
Decimal	29	73	n

[Range] n = 68, 69, 153, 80, 81, 82, 83

[Description] Transmit the printer configuration, and the returned information ends with 00.

[Notes]

- n = 68, transmit Firmware version of the printer
  For example, “FV1.030“;
  Hereinto: FV is short for Firmware Version and meaning of the following figures “1.030” are as follows:
  “1”: when hardware changing made the firmware can not compatible afterward, this figure should be changed and thus the firmware version number updated.
  “03”: when faults modified and function updated, this number will be changed and thus the firmware version number updated.
  “0”: firmware branch number, for different customer requirements which are different with standard configurations, this figure will be changed. But as to a specific customer, this number will be fixed.
- n = 69, transmit Boot Loader version of the printer.
- n = 153, Transmit Printer Name of the printer.
- n = 80, transmit Part Number of the printer. The standard printer part number will be fixed to BK-T680.
- n = 81, transmit Printer Serial number of the printer. Serial number of the printer will be wirte into printer when out of factory.
- n = 82, transmit Hardware Version of the printer.
- n = 83, Transmit Printer Production date
- This command is only enabled via serial and/or USB interfaces. If printer has both serial and USB interface, data return to serial and USB interface at the same time.
- `When using USB interface, the transmitted command format please refer to 3.4 Data return format of USB interface.

①GS V m ②GS V m n

[Function] Select cut mode and cut paper.

[Format]

①ASCII	GS	V	m	
Hex	1D	56	m	
Decimal	29	86	m	
②ASCII	GS	V	m	n
Hex	1D	56	m	n
Decimal	29	86	m	n

[Range] ① m = 0, 48

② m = 66, 0 ≤ n ≤255

[Description] Select a mode for cutting paper and executes paper cutting. The value of m selects the mode as follows:

m	Print mode
0,48	Full cut
66	Feeds paper (n × vertical motion unit) inches and cuts the paper fully.

[Notes]

- This command is effective only processed at the beginning of a line.
- When m = 0 or 48, if the paper feed length (L1) is less than the minimum page length (L_m) specified when executing this command, the printer feeds paper (L_m-L1) long and cuts paper. Otherwise, the printer cuts paper directly.
- When m=66, the printer feeds paper (n × vertical motion unit) first. If the totally paper feed length (L2) is less than the minimum page length (L_m) specified when executing this command, the printer feeds paper (L_m-L2) long and cuts paper. Otherwise, the printer cuts paper directly.
- The horizontal and vertical motion units are specified by **GS P**.
- The vertical motion unit (y) is used for calculating the paper feed length.
- In marked paper mode cutting position is set with mark and cut paper.

GS ^ r t m

[Function] Execute macro definition.

[Format]

ASCII	GS	^	r	t	m
Hex	1D	5E	r	t	m
Decimal	29	94	r	t	m

[Range] 0 ≤ r ≤ 255

0 ≤ t ≤ 255

0 ≤ m ≤ 1,

[Notes]

• r specifies times of macro definition.

• t specifies waiting time of macro execution.

• m specifies the mode of macro execution.

• When the LSB of m is 0, the interval time of macro is t × 100 ms, and the macro can be executed r times.

• When the LSB of m is 1, the printer waits for t × 100 ms and does not execute macro definition until user presses the Feed button with LED flashing. The process can continue r times.

• The waiting time is t × 100 ms.

• If receiving this command during macro definition, the printer will stop macro definition and the macro which is defined will be cleared.

• If the macro is undefined or r is 0, the command is disabled.

• In the process of macro execution (m = 1), the printer cannot feed by Feed button.

[Relative] **GS :**

GS

[Function] Start/end macro definition.

[Format]

ASCII	GS	:
Hex	1D	3A
Decimal	29	58

[Notes]

- The printer starts macro definition after receiving this command in normal mode. It ends macro definition after receiving this command in macro definition mode.
- If the printer receives **GS ^** in macro definition mode, the printer will end macro definition and clear it.
- Macro definition is off when powered on.
- **ESC @** cannot clear macro definition, so it can be included in macro definition.
- The data of macro definition can be 2048 bytes. Data out of 2048 bytes will be processed as normal data.

[Relative] **GS ^**

ESC 2

[Function] Set LED flash time.

[Format]

ASCII	ESC	c	F	T1L	T1H	T2L	T2H
Hex	1B	63	46	T1L	T1H	T2L	T2H
Decimal	27	99	70	T1L	T1H	T2L	T2H

[Range] 0 ≤ T1L≤ 255, 0 ≤T1H ≤ 255; 0 ≤ T2L≤ 255, 0 ≤T2H ≤ 255;

[Description] T1 = T1L+(T1H × 256)), T2 = T2L+(T2H × 256));

T1 is the LED on time, T2 is the LED off time, and the unit is 20ms;

[Notes] The settting of line height is independent under standard mode and page mode.

[Relative] **ESC 3**

ESC c

[Function] Set LED flash count.

[Format]

ASCII	ESC	c	H	nL	nH
Hex	1B	63	48	nL	nH
Decimal	27	99	72	nL	nH

[Range] 0 ≤ nL≤ 255, 0 ≤nH ≤ 255;

[Description] The LED flash count n = nL+(nH × 256));

The LED will stop flashing after the LED flash n cycles.

The LED will falsh for ever when n = 0xffff.

3 Programming Instructions

3.1 General Information

1) The instructions are compiled to help users who use commands to control the printer quickly master the command set.
2) All programming examples are given in Hex.
3) Commands cited in the instructions are bolded and underlined. For instance, **1B 40**. Data following the commands are not underlined. For instance, 42 43.
4) Contents in parentheses are interpretations of the commands. Parentheses and contents in them are not parts of the commands transmitted to printer.

3.2 Programming ABC

3.2.1 Print modes

1) Standard mode (Line mode)

Standard mode is the default print mode of Kiosk series. In this mode, the printer prints data and feed paper as long as the line buffer full (data buffered is enough for one print line) or receiving a print command. For instance, **0A**.

Examples:

1B 40 (Initialize the printer)

41 42 43 44 45 46

0A (Print)

Printout: ABCDEF

2) Page mode

In this mode, the printer possesses all data in specified memory and thinks of this as a virtual page. The page is printed when the printer receives print command either **1B 0C** or **0C**.

Examples:

1B 4C(switch into page mode)

1B 57 20 00 00 00 60 02 20 03 (set print area in page mode)

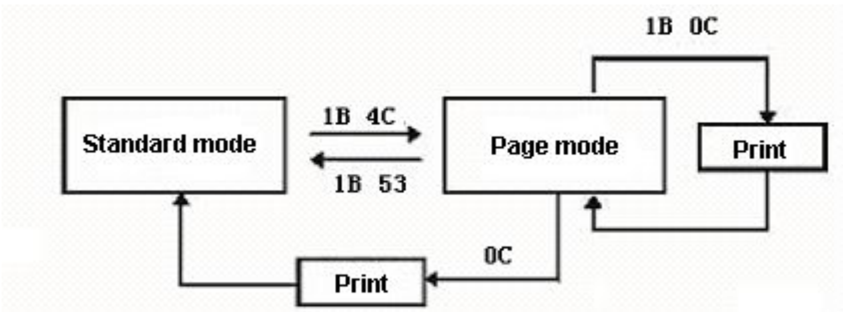
50 72 69 6E 74 20 49 6E 20 50 61 67 65 20 4D 6F 64 65 33 33 33

0C (print)

1D 56 00 (cut paper).

Printout: Print In Page Mode 333

The standard mode is switched with page mode as follows.



3.2.2 Paper supported

Black mark paper is thermal sensitive paper on which black marks are published regularly. The black marks are referenced to set print area and cut position. Refer to relative user's manual for detailed black mark requirements.

Examples:

1B 63 30 02 (select black mark paper)

1D 0C (move the next label to print position)33 33 33 33 42 6C 61 63 6B 20 4D 61 72 6B 20 50 61 70 65 72

0A(print)

1D 56 00(cut paper)

Printout: 3333Black Mark Paper

Continuous paper is normal thermal sensitive paper rolls which is not used for fixing position.

Examples:

1B 63 30 00 (select continuous paper) 33 33 33 33 20 53 65 72 69 65 73 20 50 61 70 65 72

0A (print)

1D 56 00(cut paper)

Printout: 3333 Series Paper

3.2.3 PRST action modes

Refer to commands **1B 63 38 n and 1B 63 39** n for detailed information.

If the PRST is enabled, the PRST will startup to hold the front end of printed paper during printing and accommodate the printed paper. When the printing is finished and the paper is cut, the PRST hands out the printout by holding the back-end of printed paper and waits for a specified time for the user to take it away. After the waiting time, the PRST will

1) In retraction mode, retract the printout into a waste-bin to avoid environmental pollution and keep secret of user-specified information in case the user do not take away the printout within limited time.
2) In ejection mode, discharge the printout to enable the next print (customized model).
3) In waiting mode, wait until the user takes the printout away and discharge it before next printing starts.

If the PRST is disabled, the PRST has no action during printing or after printing. With this selection, the PRST part is disassembled to avoid paper jam.

Examples:

1) Select continuous paper, standard mode, PRST retraction mode and print.

0A (Set the print starting position to the beginning of the line)

1B 40 (initialize printer)

1B 63 30 00 (select continuous paper)

1B 63 39 04 (set PRST waiting time)

1B 63 38 00 (select PRST action mode)

1B 53 (switch to standard mode)

33 33 33 33 33 20 49 6E 20 73 74 61 6E 64 20 6D 6F 64 65 20 61 6E 64 20 74 68 65 20 70 61 70 65 72 20 69 73 20 63 6F 6E 74 69 6E 75 6F 75 73 20 70 61 70 65 72 20 74 68 65 09 70 72 65 73 65 6E 74 65 72 27 73 20 61 63 74 69 6F 6E 20 69 73 20 69 6E 20 72 65 74 72 61 63 74 69 6F 6E 20 6D 6F 64 65 20 21

0A (print)

1D 56 00(cut paper)

After transmits the above command, then

Printout: 33333 In stand mode and the paper is series paper the Presenter's action is receive!

PRST action: After printing and cutting, the PRST presents out the printed paper and waits for the user to take it away. If the printout is not taken away within 4 seconds, the PRST retracts the paper into a waste bin.

2) Select marked paper, page mode and PRST ejection mode.

0A (set the print starting position to the beginning of the line)

1B 40 (initialize printer)

1B 63 30 02 (select black mark paper)

1B 63 39 04 (set PRST waiting time)

1B 63 38 02 (set PRST action mode)

1B 4C (switch into pager mode)

1B 57 00 00 00 00 80 02 00 04 (select print area)

33 33 33 33 33 20 49 6E 20 70 61 67 65 20 6D 6F 64 65 20 61 6E 64 20 74 68 65 20 70 61 70 65 72 20 69 73 20 62 6C 61 63 6B 20 6D 61 72 6B 20 70 61 70 65 72 20 74 68 65 20 70 72 65 73 65 6E 74 65 72 20 69 73 20 69 6E 20 65 6A 65 63 74 69 6F 6E 20 6D 6F 64 65 21

0C (print)

1D 56 00(cut paper)

After transmits the command, then

Printout: 33333 In page mode and the paper is black mark paper the presenter is in ejection mode!

PRST action: After printing and cutting, the PRST presents out the printed paper and wait until the user takes it away.

3.3 Programming examples

In this paragraph, the most commonly used five kinds of commands are outlined below (Refer to 2. command set for detailed usage). The general programming approach follow by programmers is “initialization — >set printing and print data — >print and present printout — >status check”, read descriptions below carefully for quickly mastering this approach.

3.3.1 Initialization

1) Initialize printer: **1B 40**.
2) Select paper type: **1B 63 30 n**, n = 0 (continuous paper); n = 2(black mark paper).
3) Select print mode: **1B 53** switch to standard mode (Default) **1B 4C** switch to page mode.
4) Select PRST action mode: **1B 63 38 n**, **1B 63 39 n**.

3.3.2 Set printing and print data(character /barcode/bit-image)

1) Set print area and print position

a) Horizontal tab(09)

b) Set horizontal absolute print position(1B 24)

c) Set horizontal tab positions(1B 44)

d) Select print direction in page mode(1B 54 n)

e) Set print area in page mode(1B 57)

f) Set relative print position(1B 5C nL nH)

g) Select justification(1B 61 n)

h) Set absolute vertical print position in page mode(1D 24)

i) Set left margin(1D 4C)

j) Set horizontal and vertical motion unit(1D 50)

k) Set print area width(1D 57)

l) Set relative vertical print position in page mode(1D 5C)

Examples:

a) Set print area in page mode (width: 76, height: 100)

0A (set the print starting position to the beginning of the line)

1B 4C (switch to page mode)

1B 57 20 00 00 00 60 02 20 03(set print area)

50 72 69 6E 74 20 49 6E 20 50 61 67 65 20 4D 6F 64 65 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 3333 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

0C (print)

1D 56 00(cut paper)

b) Set horizontal tab position and absolute print position

0A (set the print starting position to the beginning of the line)

1B 40 (Initialize printer)

1B 53(switch to standard mode)

33 33 33 33 33 33

1B 44 08 10 20 00(set horizontal tab position)

09(move the print position to the next horizontal tab position)

33 33 33 33

09(move the print position to the next horizontal tab position)

33 33 33 33

09(move the print position to the next horizontal tab position)

33 33 33 33

0A (print)

33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

0A (print)

33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

1B 24 50 00 (set absolute print position)

4E 65 77 20 61 62 73 6F 6C 75 74 65 20 73 74 61 72 74 20 70 6F 73 69 74 69 6F 6E 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

0A(print)

1D 56 00(cut paper)

c) Select print direction in page mode

1B 4C (switch to page mode)

1B 57 20 00 00 00 60 02 20 03 (set print area in page mode)

1B 54 00(select print direction in page mode)

30 30 30 30 30 30 30 30 30 30 30

0A (print)

1B 54 01(select print direction in page mode)

31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31 31

0A (print)

1B 54 02(select print direction in page mode)

32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32 32

0A(print)

1B 54 03 (select print direction in page mode)

33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33 33

0A(print)

50 72 69 6E 74 20 45 6E 64

0C(print)

1D 56 00(cut paper)

2) Set character property

a) Turn underline mode on/off(1B 2D n)

b) Select print mode(1B 21 n)

c) Turn up-side-down printing mode on/off(1B 7B n)

d) Select character size(1D 21)

e) Turn white/black reverse printing mode on/off(1D 42 n)

Examples:

0A (set the print starting position to the beginning of the line)

1B 40 (Initialize printer)

1B 53(switch to standard mode)

33 33 53 6C 65 63 74 20 75 6E 64 65 72 6C 69 6E 65 20 70 72 69 6E 74 20 6D 6F 64 65 20

1B 2D 02(turn underline mode on)

4E 6F 77 20 69 74 20 77 6F 72 6B 73 20 21 20 53 65 6C 65 63 74 20 20 20 20 20 50 72 69 6E 74 20 6D 6F 64 65 20

1B 21 01(select print mode)

33 33 33 33 33 33 33

1B 21 01(select print mode)

20 4F 70 65 6E 20 69 6E 76 65 72 73 65 20 20 70 72 69 6E 74 20 6D 6F 64 65

0A(print)

1B 7B 01(turn upside-down printing mode on)

33 33 33 33 33 4F 4B 21

0A (print)

1B 7B 00 (turn upside-down printing mode off)

33 33 33 33 33 4F 4B 21 33 33 33 33 33 4E 6F 77 20 49 27 6C 6C 20 73 68 6F 77 20 79 6F 75 20 7A 6F 6F 6D 20 69 6E 20 6D 6F 64 65 20

0A(print)

1D 21 33(select character size)

33 33 33 4F 4B 21 20 77 6F 72 6B 73 21 20

0A (print)

1D 21 00(select character size)

1D 42 01(turn white/black reverse printing mode on)

33 33 33 33 33 33 33

1D 42 00 (turn white/black reverse printing mode off)

33 33 33 33 33

0A (print)

1D 56 00(cut paper)

3) Print character/barcode/bit-image

a) Print characters

To print ASCII characters, send ASCII codes of characters to printer.

Examples:

1B 40(initialize printer)

53 68 61 6E 44 6F 6E 67 20 57 65 69 48 61 69 20 42 65 69 79 61 6E 67 20 43 6F 6D 70 61 6E 79

0A (print)

1D 56 00 (cut paper)

Printout: "ShanDong WeiHai Beiyang Company"

If the character set is Chinese font, send the area codes of wanted Chinese characters to print Chinese characters.

Examples:

1B 40 (Initialize printer)

C9 BD B6 AB D0 C2 B1 B1 D1 F3 D0 C5 CF A2 BC BC CA F5 B9 C9 B7 DD D3 D0 CF DE B9 AB CB BE

0A (print)

1D 56 00 (cut paper)

Printout: "Shandong WeiHai Beiyang Company"

b) Print barcode(Refer to 2.command set for detailed information)

- Select barcode height (1D 68)
- Select barcode width (1D 77)
- Select font for HRI characters (1D 66)
- Select printing position for HRI characters (1D 48)
- Print barcode(1D 6B)

Example 1: **print Code 128 in standard mode**

<u>**0A**</u> (set the print starting position to the beginning of the line)

<u>**1B 40**</u> (Initialize printer)

<u>**1B 53**</u>(switch to standard mode)

33 33 33 50 72 69 6E 74 20 43 6F 64 65

<u>**1D 68 78**</u>(Select barcode height)

<u>**1D 77 03**</u>(Select barcode width)

<u>**1D 66 00**</u>(Select font for HRI characters)

<u>**1D 48 03**</u>(Select printing position for HRI characters)

<u>**1D 6B 49 0B 7B 42 4E 6F 2E 31 32 33 34 35 36**</u> (print barcode)

<u>**1D 56 00**</u>(cut paper)

Example 2: print code128 in page mode

<u>**0A**</u> (set the print starting position to the beginning of the line)

<u>**1B 4C**</u> (switch to page mode)

1B <u>**57 20 00 00 00 60 02 20 03**</u>(set print area in page mode)

<u>**1D 5C 80 00**</u>(set relative vertical print position in page mode)

<u>**1D 68 78**</u>(set barcode height)

<u>**1D 77 03**</u>(set barcode width)

<u>**1D 66 00**</u>(select font for HRI characters)

<u>**1D 48 03**</u>(select printing position for HRI characters)

1D <u>**6B 49 0B 7B 42 4E 6F 2E 31 32 33 34 35 36**</u>(print barcode)

<u>**0C**</u> (print)

<u>**1D 56 00**</u> (cut paper)

Note: In page mode, the value for relative vertical print position shall be large than the selected barcode height. Otherwise, the barcode could not be printed completely.

c) Print bit-images

User may download a bit-image into RAM or Flash and print it (Refer to 2. Command set for details).

Examples: Download a bit-image (bit-image number 0) into RAM and print it.

<u>**0A**</u> (set the print starting position to the beginning of the line)

<u>**1B 53**</u>(switch to standard mode)

<u>**1D 23 00**</u> (specify a number for the bit-image to be downloaded)

<u>**1D 2F 00**</u> (print downloaded bit-image)

<u>**1D 2F 01**</u> (print download bit-image)

<u>**1D 2F 02**</u> (print downloaded bit-image)

<u>**1D 2F 03**</u> (print downloaded bit-image)

1D 56 00(cut paper)

3.3.3 Print and cut paper

To print in line mode, use command **0A**. To print in page mode, use command **0C** or **1B 0C** (do not clear buffer). Cut paper command is **1D 56 00**.

Examples:

1B 4C (switch to page mode)

1B 57 20 00 00 00 60 02 20 03 (set print area in page mode)

50 72 69 6E 74 20 49 6E 20 50 61 67 65 20 4D 6F 64 65 33 33 33

1B 0C (print without clearing buffer)

1D 56 00(cut paper)

1B 0C (print without clearing buffer)

1D 56 00(cut paper)

0C (print and clear buffer)

1D 56 00(cut paper)

3.3.4 Status check

User may check the printer status (normal or error) through either real-time status transmission command or automatic status back function which can transmit status data to host automatically when error (paper end, printhead open, cutter error, paper jam etc.) occurs.

1) Real-time status transmission (**10 04 n**) command

This command is automatically enabled when the printer is turned on or when error occurs after printer is powered on. User may also use real-time command enable/disable command (**1B 63 40 n)** to turn on / off real time commands. When print with printer drivers, the real-time command is automatically disabled by printer driver to avoid rubbish data from the printer. If user wants to check printer status with real-time status transmission command after printing, it is necessary to enable the real-time command first by using command **1B 63 40 n**.

Note: do not check printer status during printing with printer driver due to following reasons.

a) If the status inquiry command is sent to printer in API mode through USB port, the printer will process this command as normal bit-image data and consequently cause abnormal printout.

b) If the status inquiry command is sent to printer through serial interface and USB interface, the printer will also process this command data as normal bit-image data and consequently cause abnormal printout.

2) ASB(**1D 61 n**)

If the ASB command is enabled, the printer will automatically transmit a four-byte status to host when error occurs or specified printer status changes.

How to obtain correct printer status in use of **ASB**?

a) First, confirm if the fourth byte is synchronous data and the frontal three bytes comply with grammar or not.

b) Printer process status is transmitted depending on the third byte of printer return value. The lower four bits of third byte gives information on what process the printer is in and whether there are any errors.

c) The status of the first byte and second byte based on printer process status give exact printer status information.

For example, “01 04 15 F0” is returned by printer.

F0 is synchronous data which cannot be analyzed.

15: i.e. “10＋05” and “05” means the printer is in PRST retraction process and “10” means that there are error occurs in this process.

04: means the PRST paper jam

01: means paper near end.

From information above, the user can learn that the printing has been finished but PRST fails to retract the printout after waiting time and the paper is near end, a new roll is to be prepared.

The detailed information refers to **1D 61 n**

3.3.5 Download bit-images into RAM or Flash

User may download one or more bit-images into RAM (RAM bit-images) or Flash (Flash bit-images). Bit-images in Flash are retained when printer is powered off while those in RAM are cleared when printer is turned off or initialized by using command **1B 40**. Therefore, the users need to download the bit-images into RAM again whenever the printer is restarted or initialized.

Refer to command **1D 2A** (download bit-images into RAM) and **1C 71**(download bit-images into NV) for detailed information on how to perform the download. In both RAM bit-image download and NV bit-image download, a number must be specified for the image to be downloaded. Please read carefully the commands **1D 2A** and **1C 71** **for** the difference between RAM bit-image download and NV bit-image download.

Bit-image data processing by taking an example of BMP format bit-images.

1) The data is arranged in column data type in the bit-image download command. Refer to command **1D 2A** and **1C 71** for details.
2) BMP format bit-image data is arranged in line data type. Normally, a monochromic BMP image is made of sixty two (62) bytes of BMP image attributes description (including bit-image width and height etc.) and normal image data. The BMP image data is arranged in integral multiple of 4 bytes. For example, if the width is 34 dots, 8 bytes instead of 5 bytes are needed to store the data. Therefore, the total amount of bytes a bit-image occupies in the memory is “(Line width in bits+31)/32*4 * line height in bits”

Below is an example written under VC++ environment (for reference only).

Note: A non-monochromic BMP image must be transformed into a monochromic image before downloading it to printer. And, both the height (in pixels) and width (in pixels) of BMP image shall be multiple of eight (8), otherwise the printer may not be able to handle the data correctly.

```
//***********************************************//
//Function: AntiRotateBmp90D                      //
//Utility: Transform bmp format bit-image into printer processable data //
//Parameter: pBmpData---Pointer to source data    //
```

```
//        nPixelsOfWidth----bit-image width(in dots)            //
//        nPixelsOfHeight---bit-image height(in dots)           //
//        pBmpDataRotated---Pinter to target data                //
//Value return: 1: data transformation ok. 0: parameter error.   //
//*********************************************************//
int AntiRotateBmp90D(
    char *pBmpData,
    const int nPixelsOfWidth,
    const int nPixelsOfHeight,
    char *pBmpDataRotated
    )
{
    // Define process variables
    int nBytesOfWidth = 0,nBytesOfHeight = 0;
    int i=0,col=0,row=0,index = 0,colbyte = 0;
    char* midData;
    unsigned char tempdata = 0,colnum = 0,rownum = 0;
    unsigned char temp[8] = {0x80,0x40,0x20,0x10,0x08,0x04,0x02,0x01};

    //Comparing parameters
    if (pBmpData == NULL || pBmpDataRotated == NULL)   return 0;
    if (nPixelsOfWidth <= 0 || nPixelsOfHeight <= 0)   return 0;
    if ((nPixelsOfWidth % 8) != 0 || (nPixelsOfHeight % 8) != 0)   return 0;

    //get real image dimension
    nBytesOfWidth = (nPixelsOfWidth+31)/32*4;
    nBytesOfHeight = nPixelsOfHeight / 8;

    //White/black reverse,the value 1 in BMP data is for white which is contrary to the printer definition.
    midData = (char*)malloc(nBytesOfWidth*nPixelsOfHeight+1);
    for(i=0;i<nBytesOfWidth*nPixelsOfHeight;i++){midData[i] = 0xff-pBmpData[i];}

    //Rotation. BMP data is arranged in line data type while data downloaded to printer are arranged
in column data type.
    for (row = 0; row <nPixelsOfWidth; row++){
        for (colbyte = 0; colbyte < nBytesOfHeight; colbyte++){
            index = row * nBytesOfHeight + colbyte;
            pBmpDataRotated[index] = 0x00;
            for (col = 0; col < 8; col++){
                colnum = col % 8;
```

```
                    rownum = row % 8;
                    if(colnum >= rownum)
                        tempdata = temp[col] & (midData[(nPixelsOfHeight-1-colbyte*8-col) * nBytesOfWidth + row / 8] >> (colnum-rownum));
                    else
                        tempdata = temp[col] & (midData[(nPixelsOfHeight-1-colbyte*8-col) * nBytesOfWidth + row / 8] << (rownum-colnum));
                    pBmpDataRotated[index] |= tempdata;
                }
            }
        }
        // release interim buffer
        free(midData);
        return 1;
}
```

Example: Download a bit-image into RAM and print it.

<u>1D 23 00</u>(Define a number for the image to be downloaded)

<u>1D 2A 1C 08</u> (Defined downloaded bit-image)

00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000003FC0000000
000001FFF000000000007FFF800000000001FC0FC00000000003E003E000000000078003E0200
000000F0001E0200000001C0000E02000000003800000060000000700000006000000070000000E0
000000E0000003E0000000C000000FE0000001C00000FFE000000180000FFFE00000038000FFFF
E00000030807FFFF60000003087FFFFE600000030FFFFFF8600000060FFFFF80600000061FFFF
C00600000061FFF8000600000061FFB0000600000061F830000600000061803000 0E00000060003
8000C000000600078001C000000700078003C0000007000F800780000007803FC00F80000003E0F
FF03F00000003FFF9FFFF00000003FFF9FFFE00000001FFF1FFFC00000000FFE0FFF800000000
7FC07FE0000000003F001F8000000000000000000000000000000000000000000000000000000000
000000FF00000000000007FFC000000000003FFFE000000000007FFFE0000000001FFFFE00000
000003FFFFE0000000000FFF03E0000000001FC601E0000000003E0601E000000000380E01C000
000000701C03C000000000601C038000000000C038078000000000C0780F0000000000E1F01E00
00000000FFF03C0000000000FFE0780000000000FFC0700000000000FF80300000000000 7F00000
0000000003E000000000000000000000000000000000000000000002000000000000000600000
0000000000E0007C0000000001C00FFE0000000003807FFE000000000387FFFE00000000073FFF
FE0000000007FFFFFC000000000FFFFFFC000000000FFFF818007000000FFFC03800F801C00FF
C007000FC03E00FC000E000FC03E0060000C000FC03E00000008000FC03E00040000000FC01C

000C00000007C000001C000000078000001C00000007800000380000000F800000700000000F000
000700000001F000000F80000003E000000FE0000007C000000FF800000F8000000FFF80003F800
00003FFFE00FF00000001FFFFFFFC000000007FFFFFF8000000000FFFFFF00000000001FFFFC0
00000000001FFF8000000000000FE000000000000003FC000000000001FF000000000780FFC00
00000000FFFFF00000000000FFFFC00000000000FFFE000000000000FFF000000000000007F0000
000000000000000000000000000000000000000003F00000000000001FFC000000000007FF
E000000000001FFFE000000000007FFFE00000000000FFFFE00000000001FE03E00000000003F0
01C000000000078001C0000000000F00038000000000001C00038000000000001C0007000000000000380
00E00000000000030001CE00000000070007FE0000000007003FFE000000000703FFFE0000000073
FFFFE0000000007FFFFFC0000000007FFFFFC0000000007FFFE18000000000FFFC03800000000
0FF80070000000000FC000E000000000C0000C000000000000000800000000000000000000000
00000000000000000020000000000000007000000000000000F0000C0000000001E000FC000000
0001C00FFC000000000380FFFC00000000070FFFFC0000000007FFFFFC0000000007FFFFF0000
000000FFFFF00000000000FFFF000000000000FFF000000000000FFC000000000000F38000000
0000000070000000000000000E0000000000000000C000FC0000000001800FFE0000000003807FFE0
00000000303FFFE00000000071FFFFE0000000007FFFFFE000000000FFFFF9C000000000FFFF8
18000000000FFFC03807C000000FFE00700FE000000FF000E01FF00000070000C03FF800000000
00C07FF80000000000007878000000000003F0F01C000000003FF8E01C00000000FFFDC00C000000
03FFFD800C00000007FFFF800C0000001FFFFF000C0000003FE07E000C0000007F001E001C000
000F8001C001C000001E00038003800001C00038007800000380007000F80000070000E003F000
00070001E01FF00000060007FFFFE000000E003FFFFFC000000E03FFFFFF8000000E3FFFFFFF0
000000FFFFFFFFC0000000FFFFFFFE00000000FFFFE00000000001FFFC000000000001FFC0000
000000003FC000000000000003C000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000

1D 23 00 (Select the number of the bit-image to be printed)

1D 2F 00 (Print the RAM bit-image)

3.3.6 Recommendations on applicable flow for programm

Make good use of ASB command during programming, since the ASB command is able to automatically transmit all kinds of printer status. Remember to enable the ASB function by using command **1D 61** after the printer is turned on.

1) Inquiry printer status

   Make sure to send printing data only when the printer is in normal status. We provide all status of printer, at last printer transmits data to print in normal idle status.

2) Transmits the data to be printed (include the command settings before printing)

   If printed data is bit-image, don't transmit status inquiry command when processing printing data.

3) Check printer status after printing. If the ASB is enabled, the status is transmitted automatically.

3.4 Data return format of USB interface

3.4.1 Summary of inquiry commands

Host can inquire BK-T680 printer's status by the following inquiry commands:

- GS I n — Transmit printer configuration
- GS . B E . . > n — Transmit Mandatory and Statistic Data
- DLE EOT n — Real-time status transmission
- GS a n — Enable/Disable Automatic Status Back (ASB)
- GS r n — Transmit status

3.4.2 Data returen format of USB interface

- When printer work under class mode and the ASB function enabled, returned data format are as follows:

  0x05 + ASB + returned data of the inquired commands

  0x05: ASB return data length + 1

  ASB: a four-byte auto status back data, for example (01 00 00 F0). These four bytes will change according to printer's status.

Return data of the Inquired command: return data of the inquired command send to the printer

For example:

When send command 1D 49 99 to inquire printer's name, the returned data will be as follows:

0x05 0x01 0x00 0x00 0xF0 0x42 0x4B 0x2D 0x54 0x30 0x38 0x30 0x49 0x49 0x28 0x55 0x29 0x31

- When printer work under class mode and the ASB function disabled, the printer only return status data of some certain inquiry commands.

For example:

When send command 1D 49 99 to inquire printer's name, the returned data will be as follows:

0x42 0x4B 0x2D 0x54 0x30 0x38 0x30 0x49 0x49 0x28 0x55 0x29 0x31

- When printer works under API mode, returned data format are the same with that the printer work under class mode while the ASB function disabled.

Appendix A: ASCII Table

ASCII	Decimal	HEX	ASCII	Decimal	HEX	ASCII	Decimal	HEX	ASCII	Decimal	HEX
NULL	0	00	**SP**	32	20	@	64	40	`	96	60
SOH	1	01	!	33	21	A	65	41	a	97	61
STX	2	02	“	34	22	B	66	42	b	98	62
EXT	3	03	#	35	23	C	67	43	c	99	63
EOT	4	04	$	36	24	D	68	44	d	100	64
ENQ	5	05	%	37	25	E	69	45	e	101	65
ACK	6	06	&	38	26	F	70	46	f	102	66
BEL	7	07	‘	39	27	G	71	47	g	103	67
BS	8	08	(	40	28	H	72	48	h	104	68
HT	9	09	)	41	29	I	73	49	i	105	69
LF	10	0A	*	42	2A	J	74	4A	j	106	6A
VT	11	0B	+	43	2B	K	75	4B	k	107	6B
FF	12	0C	,	44	2C	L	76	4C	l	108	6C
CR	13	0D	-	45	2D	M	77	4D	m	109	6D
SO	14	0E	.	46	2E	N	78	4E	n	110	6E
SI	15	0F	/	47	2F	O	79	4F	o	111	6F
DLE	16	10	0	48	30	P	80	50	p	112	70
DC1	17	11	1	49	31	Q	81	51	q	113	71
DC2	18	12	2	50	32	R	82	52	r	114	72
DC3	19	13	3	51	33	S	83	53	s	115	73
DC4	20	14	4	52	34	T	84	54	t	116	74
NAK	21	15	5	53	35	U	85	55	u	117	75
SYN	22	16	6	54	36	V	86	56	v	118	76
ETB	23	17	7	55	37	W	87	57	w	119	77
CAN	24	18	8	56	38	X	88	58	x	120	78
EM	25	19	9	57	39	Y	89	59	y	121	79
SUB	26	1A	:	58	3A	Z	90	5A	z	122	7A
ESC	27	1B	;	59	3B	[	91	5B	{	123	7B
FS	28	1C	<	60	3C	\	92	5C	|	124	7C
GS	29	1D	=	61	3D	]	93	5D	}	125	7D
RS	30	1E	>	62	3E	^	94	5E	~	126	7E
US	31	1F	?	63	3F	_	95	5F		127	7F

Appendix B: Code Page Table

Code Page 0 (PC437: USA)

	HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
HEX	BIN	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	NUL 00	DLE 16	SP 32	0 48	@ 64	P 80	` 96	p 112	Ç 128	É 144	á 160	░ 176	└ 192	╨ 208	α 224	240
1	0001	01	XON 17	! 33	1 49	A 65	Q 81	a 97	q 113	ü 129	æ 145	í 161	▒ 177	┴ 193	╤ 209	β 225	± 241
2	0010	02	18	" 34	2 50	B 66	R 82	b 98	r 114	é 130	Æ 146	ó 162	▓ 178	┬ 194	╥ 210	Γ 226	≥ 242
3	0011	03	XOFF 19	# 35	3 51	C 67	S 83	c 99	s 115	â 131	ô 147	ú 163	│ 179	├ 195	╙ 211	π 227	≤ 243
4	0100	EOT 04	DC4 20	$ 36	4 52	D 68	T 84	d 100	t 116	ä 132	ö 148	ñ 164	┤ 180	─ 196	╘ 212	Σ 228	⌠ 244
5	0101	ENQ 05	21	% 37	5 53	E 69	U 85	e 101	u 117	à 133	ò 149	Ñ 165	╡ 181	┼ 197	╒ 213	σ 229	⌡ 245
6	0110	06	22	& 38	6 54	F 70	V 86	f 102	v 118	å 134	û 150	ª 166	╢ 182	╞ 198	╓ 214	µ 230	÷ 246
7	0111	07	23	' 39	7 55	G 71	W 87	g 103	w 119	ç 135	ù 151	º 167	╖ 183	╟ 199	╫ 215	τ 231	≈ 247
8	1000	08	CAN 24	(40	8 56	H 72	X 88	h 104	x 120	ê 136	ÿ 152	¿ 168	╕ 184	╚ 200	╪ 216	Φ 232	° 248
9	1001	HT 09	25	) 41	9 57	I 73	Y 89	i 105	y 121	ë 137	Ö 153	⌐ 169	╣ 185	╔ 201	┘ 217	Θ 233	• 249
A	1010	LF 10	26	* 42	: 58	J 74	Z 90	j 106	z 122	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Ω 234	· 250
B	1011	11	ESC 27	+ 43	; 59	K 75	[91	k 107	{ 123	ï 139	¢ 155	½ 171	╗ 187	╦ 203	█ 219	δ 235	251
C	1100	FF 12	FS 28	, 44	< 60	L 76	\ 92	l 108	¦ 124	î 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	∞ 236	ⁿ 252
D	1101	CR 13	GS 29	- 45	= 61	M 77	] 93	m 109	} 125	ì 141	¥ 157	¡ 173	╜ 189	═ 205	▌ 221	ø 237	² 253
E	1110	14	30	. 46	> 62	N 78	^ 94	n 110	~ 126	Ä 142	₧ 158	« 174	╛ 190	╬ 206	▐ 222	238	■ 254
F	1111	15	31	/ 47	? 63	O 79	_ 95	o 111	SP 127	Å 143	ƒ 159	» 175	┐ 191	╧ 207	▀ 223	239	SP 255

Code Page 1(Katakana)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	▁ 128	┴ 144	SP 160	ｰ 176	ﾀ 192	ﾐ 208	═ 224	× 240
1	0001	▂ 129	┬ 145	｡ 161	ｱ 177	ﾁ 193	ﾑ 209	╞ 225	円 241
2	0010	▃ 130	┤ 146	｢ 162	ｲ 178	ﾂ 194	ﾒ 210	╪ 226	年 242
3	0011	▄ 131	├ 147	｣ 163	ｳ 179	ﾃ 195	ﾓ 211	╡ 227	月 243
4	0100	▅ 132	148	､ 164	ｴ 180	ﾄ 196	ﾔ 212	◢ 228	日 244
5	0101	▆ 133	─ 149	･ 165	ｵ 181	ﾅ 197	ﾕ 213	◣ 229	時 245
6	0110	▇ 134	│ 150	ｦ 166	ｶ 182	ﾆ 198	ﾖ 214	◥ 230	分 246
7	0111	█ 135	│ 151	ｧ 167	ｷ 183	ﾇ 199	ﾗ 215	◤ 231	秒 247
8	1000	▏ 136	┌ 152	ｨ 168	ｸ 184	ﾈ 200	ﾘ 216	♠ 232	〒 248
9	1001	▎ 137	┐ 153	ｩ 169	ｹ 185	ﾉ 201	ﾙ 217	♥ 233	市 249
A	1010	▍ 138	└ 154	ｪ 170	ｺ 186	ﾊ 202	ﾚ 218	♦ 234	区 250
B	1011	▌ 139	┘ 155	ｫ 171	ｻ 187	ﾋ 203	ﾛ 219	♣ 235	町 251
C	1100	▋ 140	╭ 156	ｬ 172	ｼ 188	ﾌ 204	ﾜ 220	● 236	村 252
D	1101	▊ 141	╮ 157	ｭ 173	ｽ 189	ﾍ 205	ﾝ 221	○ 237	人 253
E	1110	▉ 142	╰ 158	ｮ 174	ｾ 190	ﾎ 206	ﾞ 222	╱ 238	▒ 254
F	1111	┼ 143	╯ 159	ｯ 175	ｿ 191	ﾏ 207	ﾟ 223	╲ 239	SP 255

Code Page 2 (PC850: Multilingual)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	Ç 128	É 144	á 160	░ 176	└ 192	ð 208	Ó 224	– 240
1	0001	ü 129	æ 145	í 161	▒ 177	┴ 193	Đ 209	ß 225	± 241
2	0010	é 130	Æ 146	ó 162	▓ 178	┬ 194	Ê 210	Ô 226	‗ 242
3	0011	â 131	ô 147	ú 163	│ 179	├ 195	Ë 211	Ò 227	¾ 243
4	0100	ä 132	ö 148	ñ 164	┤ 180	─ 196	È 212	õ 228	¶ 244
5	0101	à 133	ò 149	Ñ 165	Á 181	┼ 197	ı 213	Õ 229	§ 245
6	0110	å 134	û 150	ª 166	Â 182	ã 198	Í 214	µ 230	÷ 246
7	0111	ç 135	ù 151	º 167	À 183	Ã 199	Î 215	þ 231	¸ 247
8	1000	ê 136	ÿ 152	¿ 168	© 184	╚ 200	Ï 216	Þ 232	° 248
9	1001	ë 137	Ö 153	® 169	╣ 185	╔ 201	┘ 217	Ú 233	¨ 249
A	1010	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Û 234	· 250
B	1011	ï 139	ø 155	½ 171	╗ 187	╦ 203	█ 219	Ù 235	¹ 251
C	1100	î 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	ý 236	³ 252
D	1101	ì 141	Ø 157	¡ 173	¢ 189	═ 205	¦ 221	Ý 237	² 253
E	1110	Ä 142	× 158	« 174	¥ 190	╬ 206	Ì 222	¯ 238	■ 254
F	1111	Å 143	ƒ 159	» 175	┐ 191	¤ 207	▀ 223	´ 239	SP 255

Code Page 3 (PC860: Portuguese)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	Ç 128	É 144	á 160	░ 176	└ 192	╨ 208	α 224	≡ 240
1	0001	ü 129	À 145	í 161	▒ 177	┴ 193	╤ 209	ß 225	± 241
2	0010	é 130	È 146	ó 162	▓ 178	┬ 194	╥ 210	Γ 226	≥ 242
3	0011	â 131	ô 147	ú 163	│ 179	├ 195	╙ 211	π 227	≤ 243
4	0100	ã 132	õ 148	ñ 164	┤ 180	─ 196	╘ 212	Σ 228	⌠ 244
5	0101	à 133	ò 149	Ñ 165	╡ 181	┼ 197	╒ 213	σ 229	⌡ 245
6	0110	Á 134	Ú 150	ª 166	╢ 182	╞ 198	╓ 214	µ 230	÷ 246
7	0111	ç 135	ù 151	º 167	╖ 183	╟ 199	╫ 215	τ 231	≈ 247
8	1000	ê 136	Ì 152	¿ 168	╕ 184	╚ 200	╪ 216	Φ 232	° 248
9	1001	Ê 137	Õ 153	Ò 169	╣ 185	╔ 201	┘ 217	Θ 233	• 249
A	1010	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Ω 234	· 250
B	1011	Í 139	¢ 155	½ 171	╗ 187	╦ 203	█ 219	δ 235	√ 251
C	1100	Ô 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	∞ 236	n 252
D	1101	ì 141	Ù 157	¡ 173	╜ 189	═ 205	▌ 221	ø 237	^{2} 253
E	1110	Ã 142	Pt 158	« 174	╛ 190	╬ 206	▐ 222	∈ 238	■ 254
F	1111	Â 143	Ó 159	» 175	┐ 191	╧ 207	▀ 223	∩ 239	SP 255

Code Page 4 (PC863: Canadian- French)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	Ç 128	É 144	¦ 160	░ 176	└ 192	╨ 208	α 224	≡ 240
1	0001	ü 129	È 145	´ 161	▒ 177	┴ 193	╤ 209	ß 225	± 241
2	0010	é 130	Ê 146	ó 162	▓ 178	┬ 194	╥ 210	Γ 226	≥ 242
3	0011	â 131	ô 147	ú 163	│ 179	├ 195	╙ 211	π 227	≤ 243
4	0100	Â 132	Ë 148	¨ 164	┤ 180	─ 196	╘ 212	Σ 228	⌠ 244
5	0101	à 133	Ï 149	¸ 165	╡ 181	┼ 197	╒ 213	σ 229	⌡ 245
6	0110	¶ 134	û 150	³ 166	╢ 182	╞ 198	╓ 214	µ 230	÷ 246
7	0111	ç 135	ù 151	¯ 167	╖ 183	╟ 199	╫ 215	τ 231	≈ 247
8	1000	ê 136	¤ 152	Î 168	╕ 184	╚ 200	╪ 216	Φ 232	° 248
9	1001	ë 137	Ô 153	⌐ 169	╣ 185	╔ 201	┘ 217	Θ 233	• 249
A	1010	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Ω 234	· 250
B	1011	ï 139	¢ 155	½ 171	╗ 187	╦ 203	█ 219	δ 235	√ 251
C	1100	î 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	∞ 236	ⁿ 252
D	1101	‗ 141	Ù 157	¾ 173	╜ 189	═ 205	▌ 221	ø 237	² 253
E	1110	À 142	Û 158	« 174	╛ 190	╬ 206	▐ 222	∈ 238	■ 254
F	1111	§ 143	ƒ 159	» 175	┐ 191	╧ 207	▀ 223	∩ 239	SP 255

Code Page 5 (PC865: Nordic)

HEX	HEX BIN	8 1000	9 1001	A 1010	B 1011	C 1100	D 1101	E 1110	F 1111
0	0000	Ç 128	É 144	á 160	░ 176	└ 192	╨ 208	α 224	≡ 240
1	0001	ü 129	æ 145	í 161	▒ 177	┴ 193	╤ 209	β 225	± 241
2	0010	é 130	Æ 146	ó 162	▓ 178	┬ 194	╥ 210	Γ 226	≥ 242
3	0011	â 131	ô 147	ú 163	│ 179	├ 195	╙ 211	π 227	≤ 243
4	0100	ä 132	ö 148	ñ 164	┤ 180	─ 196	╘ 212	Σ 228	⌠ 244
5	0101	à 133	ò 149	Ñ 165	╡ 181	┼ 197	╒ 213	σ 229	⌡ 245
6	0110	å 134	û 150	ª 166	╢ 182	╞ 198	╓ 214	µ 230	÷ 246
7	0111	ç 135	ù 151	º 167	╖ 183	╟ 199	╫ 215	τ 231	≈ 247
8	1000	ê 136	ÿ 152	¿ 168	╕ 184	╚ 200	╪ 216	Φ 232	° 248
9	1001	ë 137	Ö 153	⌐ 169	╣ 185	╔ 201	┘ 217	Θ 233	• 249
A	1010	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Ω 234	· 250
B	1011	ï 139	ø 155	½ 171	╗ 187	╦ 203	█ 219	δ 235	√ 251
C	1100	î 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	∞ 236	ⁿ 252
D	1101	ì 141	Ø 157	¡ 173	╜ 189	═ 205	▌ 221	ø 237	² 253
E	1110	Ä 142	Pt 158	« 174	╛ 190	╬ 206	▐ 222	∈ 238	■ 254
F	1111	Å 143	ƒ 159	¤ 175	┐ 191	╧ 207	▀ 223	∩ 239	SP 255

Code Page 6 (PC852 Latin2)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	Ç 128	É 144	á 160	░ 176	└ 192	đ 208	Ó 224	- 240
1	0001	ü 129	Ĺ 145	í 161	▒ 177	┴ 193	Đ 209	ß 225	˝ 241
2	0010	é 130	ĺ 146	ó 162	▓ 178	┬ 194	Ď 210	Ô 226	˛ 242
3	0011	â 131	ô 147	ú 163	│ 179	├ 195	Ë 211	Ń 227	ˇ 243
4	0100	ä 132	ö 148	Ą 164	┤ 180	─ 196	ď 212	ń 228	˘ 244
5	0101	ů 133	Ľ 149	ą 165	Á 181	┼ 197	Ň 213	ň 229	§ 245
6	0110	ć 134	ľ 150	Ž 166	Â 182	Ă 198	Í 214	Š 230	÷ 246
7	0111	ç 135	Ś 151	ž 167	Ě 183	ă 199	Î 215	š 231	¸ 247
8	1000	ł 136	ś 152	Ę 168	Ş 184	╚ 200	ě 216	Ŕ 232	° 248
9	1001	ë 137	Ö 153	ę 169	╣ 185	╔ 201	┘ 217	Ú 233	¨ 249
A	1010	Ő 138	Ü 154	170	║ 186	╩ 202	┌ 218	ŕ 234	˙ 250
B	1011	ő 139	Ť 155	ź 171	╗ 187	╦ 203	█ 219	Ű 235	ű 251
C	1100	î 140	ť 156	Č 172	╝ 188	╠ 204	▄ 220	ý 236	Ř 252
D	1101	Ź 141	Ł 157	ş 173	Ż 189	═ 205	Ţ 221	Ý 237	ř 253
E	1110	Ä 142	× 158	« 174	ż 190	╬ 206	Ů 222	ţ 238	■ 254
F	1111	Ć 143	č 159	» 175	┐ 191	¤ 207	▀ 223	´ 239	SP 255

Code Page 7 (PC858)

HEX	HEX: 8	9	A	B	C	D	E	F
HEX / BIN	1000	1001	1010	1011	1100	1101	1110	1111
0 / 0000	Ç 128	É 144	á 160	░ 176	└ 192	ð 208	Ó 224	— 240
1 / 0001	ü 129	æ 145	í 161	▒ 177	┴ 193	Đ 209	ß 225	± 241
2 / 0010	é 130	Æ 146	ó 162	▓ 178	┬ 194	Ê 210	Ô 226	‗ 242
3 / 0011	â 131	ô 147	ú 163	│ 179	├ 195	Ë 211	Ò 227	¾ 243
4 / 0100	ä 132	ö 148	ñ 164	┤ 180	─ 196	È 212	õ 228	¶ 244
5 / 0101	à 133	ò 149	Ñ 165	Á 181	┼ 197	€ 213	Õ 229	§ 245
6 / 0110	å 134	û 150	ª 166	Â 182	ã 198	Í 214	µ 230	÷ 246
7 / 0111	ç 135	ù 151	º 167	À 183	Ã 199	Î 215	þ 231	¸ 247
8 / 1000	ê 136	ÿ 152	¿ 168	© 184	╚ 200	Ï 216	Þ 232	° 248
9 / 1001	ë 137	Ö 153	® 169	╣ 185	╔ 201	┘ 217	Ú 233	¨ 249
A / 1010	è 138	Ü 154	¬ 170	║ 186	╩ 202	┌ 218	Û 234	· 250
B / 1011	ï 139	ø 155	½ 171	╗ 187	╦ 203	█ 219	Ù 235	¹ 251
C / 1100	î 140	£ 156	¼ 172	╝ 188	╠ 204	▄ 220	ý 236	³ 252
D / 1101	ì 141	Ø 157	¡ 173	¢ 189	═ 205	¦ 221	Ý 237	² 253
E / 1110	Ä 142	× 158	« 174	¥ 190	╬ 206	Ì 222	¯ 238	■ 254
F / 1111	Å 143	ƒ 159	» 175	┐ 191	¤ 207	▀ 223	´ 239	SP 255

Code Page 8 (PC866: Cyrillic #2)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	А 128	Р 144	а 160	░ 176	└ 192	╨ 208	р 224	Ё 240
1	0001	Б 129	С 145	б 161	▒ 177	┴ 193	╤ 209	с 225	ё 241
2	0010	В 130	Т 146	в 162	▓ 178	┬ 194	╥ 210	т 226	Є 242
3	0011	Г 131	У 147	г 163	│ 179	├ 195	╙ 211	у 227	є 243
4	0100	Д 132	Ф 148	д 164	┤ 180	─ 196	╘ 212	ф 228	Ї 244
5	0101	Е 133	Х 149	е 165	╡ 181	┼ 197	╒ 213	х 229	ї 245
6	0110	Ж 134	Ц 150	ж 166	╢ 182	╞ 198	╓ 214	ц 230	Ў 246
7	0111	З 135	Ч 151	з 167	╖ 183	╟ 199	╫ 215	ч 231	ў 247
8	1000	И 136	Ш 152	и 168	╕ 184	╚ 200	╪ 216	ш 232	° 248
9	1001	Й 137	Щ 153	й 169	╣ 185	╔ 201	┘ 217	щ 233	∙ 249
A	1010	К 138	Ъ 154	к 170	║ 186	╩ 202	┌ 218	ъ 234	· 250
B	1011	Л 139	Ы 155	л 171	╗ 187	╦ 203	█ 219	ы 235	√ 251
C	1100	М 140	Ь 156	м 172	╝ 188	╠ 204	▄ 220	ь 236	№ 252
D	1101	Н 141	Э 157	н 173	╜ 189	═ 205	▌ 221	э 237	¤ 253
E	1110	О 142	Ю 158	о 174	╛ 190	╬ 206	▐ 222	ю 238	■ 254
F	1111	П 143	Я 159	п 175	┐ 191	╧ 207	▀ 223	я 239	SP 255

Code Page 9 (WPC1252)

	HEX	8	9	A	B	C	D	E	F
HEX	BIN	1000	1001	1010	1011	1100	1101	1110	1111
0	0000	€ 128	SP 144	SP 160	° 176	À 192	Ð 208	à 224	ð 240
1	0001	SP 129	‘ 145	¡ 161	± 177	Á 193	Ñ 209	á 225	ñ 241
2	0010	‚ 130	’ 146	¢ 162	² 178	Â 194	Ò 210	â 226	ò 242
3	0011	ƒ 131	“ 147	£ 163	³ 179	Ã 195	Ó 211	ã 227	ó 243
4	0100	„ 132	” 148	¤ 164	´ 180	Ä 196	Ô 212	ä 228	ô 244
5	0101	… 133	• 149	¥ 165	µ 181	Å 197	Õ 213	å 229	õ 245
6	0110	† 134	– 150	¦ 166	¶ 182	Æ 198	Ö 214	æ 230	ö 246
7	0111	‡ 135	— 151	§ 167	· 183	Ç 199	× 215	ç 231	÷ 247
8	1000	ˆ 136	˜ 152	¨ 168	¸ 184	È 200	Ø 216	è 232	ø 248
9	1001	‰ 137	™ 153	© 169	¹ 185	É 201	Ù 217	é 233	ù 249
A	1010	Š 138	š 154	ª 170	º 186	Ê 202	Ú 218	ê 234	ú 250
B	1011	‹ 139	› 155	« 171	» 187	Ë 203	Û 219	ë 235	û 251
C	1100	Œ 140	œ 156	¬ 172	¼ 188	Ì 204	Ü 220	ì 236	ü 252
D	1101	SP 141	SP 157	- 173	½ 189	Í 205	Ý 221	í 237	ý 253
E	1110	Ž 142	ž 158	® 174	¾ 190	Î 206	Þ 222	î 238	þ 254
F	1111	SP 143	Ÿ 159	¯ 175	¿ 191	Ï 207	ß 223	ï 239	ÿ 255

Appendix C: Code 128 Barcode

1. Description of the CODE128 Barcode

In code 128 barcode system, it is possible to represent 128 ASCII characters, 2-digit numerals and some special characters in using one barcode character that is defined by combining one of code set A, code set B and code set C. Each code set is used for representing the following characters:

- Code set A: ASCII characters 00H to 5FH
- Code set B: ASCII characters 20H to 7FH
- Code set C: 2-digit numeral characters using one character (100 numerals from 00 to 99)

The following special characters are also available in CODE128:

- SHIFT characters

  In code set A, the character just after SHIFT is processed as a character for code set B. In code set B, the character just after SHIFT is processed as a character for code set A. The character recovers the previous code set before SHIFT when starting from the second character. SHIFT characters are used only for Switch between code set A and code set B, but cannot make the current code set enter or exit code set c status.
- Code set selection character (CODE A, CODE B, CODE C).

  This character switches the following code set to code set A, B, or C.
- Function character (FNC1, FNC2, FNC3, FNC4)

  The usage of function characters depends on the application software. In code set C, only FNC1 is available.

2. Character Set

Characters in Code Set A:

Character	Transmit data		Character	Transmit data		Character	Transmit data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
NULL	00	0	(	28	40	P	50	80
SOH	01	1	)	29	41	Q	51	81
STX	02	2	*	2A	42	R	52	82
ETX	03	3	+	2B	43	S	53	83
EOT	04	4	,	2C	44	T	54	84
ENQ	05	5	-	2D	45	U	55	85
ACK	06	6	.	2E	46	V	56	86
BEL	07	7	/	2F	47	W	57	87
BS	08	8	0	30	48	X	58	88
HT	09	9	1	31	49	Y	59	89
LF	0A	10	2	32	50	Z	5A	90
VT	0B	11	3	33	51	[	5B	91
FF	0C	12	4	34	52	\	5C	92
CR	0D	13	5	35	53	]	5D	93
SO	0E	14	6	36	54	^	5E	94
SI	0F	15	7	37	55	_	5F	95
DLE	10	16	8	38	56	FNC1	7B,31	123,49
DC1	11	17	9	39	57	FNC2	7B,32	123,50
DC2	12	18	:	3A	58	FNC3	7B,33	123,51
DC3	13	19	;	3B	59	FNC4	7B,34	123,52
DC4	14	20	<	3C	60	SHIFT	7B,53	123,83
NAK	15	21	=	3D	61	CODEB	7B,42	123,66
SYN	16	22	>	3E	62	CODEC	7B,43	123,67
ETB	17	23	?	3F	63			
CAN	18	24	@	40	64			
EM	19	25	A	41	65			
SUB	1A	26	B	42	66			
ESC	1B	27	C	43	67			
FS	1C	28	D	44	68			
GS	1D	29	E	45	69			
RS	1E	30	F	46	70			
US	1F	31	G	47	71			
SP	20	32	H	48	72			
!	21	33	I	49	73			
"	22	34	J	4A	74			
#	23	35	K	4B	75			
$	24	36	L	4C	76			
%	25	37	M	4D	77			
&	26	38	N	4E	78			
'	27	39	O	4F	79			

Characters in Code Set B:

Character	Transmit data		Character	Transmit data		Character	Transmit data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
SP	20	32	H	48	72	p	70	112
!	21	33	I	49	73	q	71	113
"	22	34	J	4A	74	r	72	114
#	23	35	K	4B	75	s	73	115
$	24	36	L	4C	76	t	74	116
%	25	37	M	4D	77	u	75	117
&	26	38	N	4E	78	v	76	118
'	27	39	O	4F	79	w	77	119
(	28	40	P	50	80	x	78	120
)	29	41	Q	51	81	y	79	121
*	2A	42	R	52	82	z	7A	122
+	2B	43	S	53	83	{	7B,7B	123,123
,	2C	44	T	54	84	|	7C	124
-	2D	45	U	55	85	}	7D	125
.	2E	46	V	56	86	—	7E	126
/	2F	47	W	57	87	DEL	7F	127
0	30	48	X	58	88	FNC1	7B,31	123,49
1	31	49	Y	59	89	FNC2	7B,32	123,50
2	32	50	Z	5A	90	FNC3	7B,33	123,51
3	33	51	[	5B	91	FNC4	7B,34	123,52
4	34	52	\	5C	92	SHIFT	7B,53	123,83
5	35	53	]	5D	93	CODEA	7B,41	123,65
6	36	54	^	5E	94	CODEC	7B,43	123,67
7	37	55	_	5F	95			
8	38	56	`	60	96			
9	39	57	a	61	97			
:	3A	58	b	62	98			
;	3B	59	c	63	99			
<	3C	60	d	64	100			
=	3D	61	e	65	101			
>	3E	62	f	66	102			
?	3F	63	g	67	103			
@	40	64	h	68	104			
A	41	65	i	69	105			
B	42	66	j	6A	106			
C	43	67	k	6B	107			
D	44	68	l	6C	108			
E	45	69	m	6D	109			
F	46	70	n	6E	110			
G	47	71	o	6F	111			

Characters in Code Set C:

Character	Transmit data		Character	Transmit data		Character	Transmit data	
	Hex	Decimal		Hex	Decimal		Hex	Decimal
0	00	0	40	28	40	80	50	80
1	01	1	41	29	41	81	51	81
2	02	2	42	2A	42	82	52	82
3	03	3	43	2B	43	83	53	83
4	04	4	44	2C	44	84	54	84
5	05	5	45	2D	45	85	55	85
6	06	6	46	2E	46	86	56	86
7	07	7	47	2F	47	87	57	87
8	08	8	48	30	48	88	58	88
9	09	9	49	31	49	89	59	89
10	0A	10	50	32	50	90	5A	90
11	0B	11	51	33	51	91	5B	91
12	0C	12	52	34	52	92	5C	92
13	0D	13	53	35	53	93	5D	93
14	0E	14	54	36	54	94	5E	94
15	0F	15	55	37	55	95	5F	95
16	10	16	56	38	56	96	60	96
17	11	17	57	39	57	97	61	97
18	12	18	58	3A	58	98	62	98
19	13	19	59	3B	59	99	63	99
20	14	20	60	3C	60	FNC1	7B,31	123,49
21	15	21	61	3D	61	CODEA	7B,41	123,65
22	16	22	62	3E	62	CODEB	7B,42	123,66
23	17	23	63	3F	63			
24	18	24	64	40	64			
25	19	25	65	41	65			
26	1A	26	66	42	66			
27	1B	27	67	43	67			
28	1C	28	68	44	68			
29	1D	29	69	45	69			
30	1E	30	70	46	70			
31	1F	31	71	47	71			
32	20	32	72	48	72			
33	21	33	73	49	73			
34	22	34	74	4A	74			
35	23	35	75	4B	75			
36	24	36	76	4C	76			
37	25	37	77	4D	77			
38	26	38	78	4E	78			
39	27	39	79	4F	79			